Natalyx MVP Product Flow Reshape — consolidated programme plan

> Planning-only. Do not implement, deploy, migrate, change cloud resources, alter existing PRs or clean protected worktrees from this document. Alan should review/sign off this consolidated plan before proposed assignments become expected work.

Inspection snapshot: 2026-07-10 20:12 America/New_York. Planning branch: `plan/mvp-product-flow-reshape`. GitHub Project: MVP Product Flow Reshape. Proposed PRs: 50.

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Executive summary

The programme is a controlled rebuild around free Node 0 workspace → explicit entitlement/commitment/payment → typed promotion authorization → enrollment-pending paid case, not a patch stack on the current case-shaped Node 0 branch. The live dev endpoint is currently 503, so A0 recovery and current CI/PostgreSQL evidence come first. PR #122's update materially improves invitation and case-surface safety and should be salvaged; PR #124 remains valuable but structurally unsafe unchanged. Alan's unfinished review is an explicit blocking checkpoint, not an assumed approval.

The canonical package proposes 50 focused PRs across app, infra, intelligence, Learn, participant design system and organisation planning. It keeps free education/safety capability, payment and truth state separate; uses one processor-neutral promotion authorization; preserves authorship/provenance; and gates real PII/PHI behind independent evidence rather than deploy success.

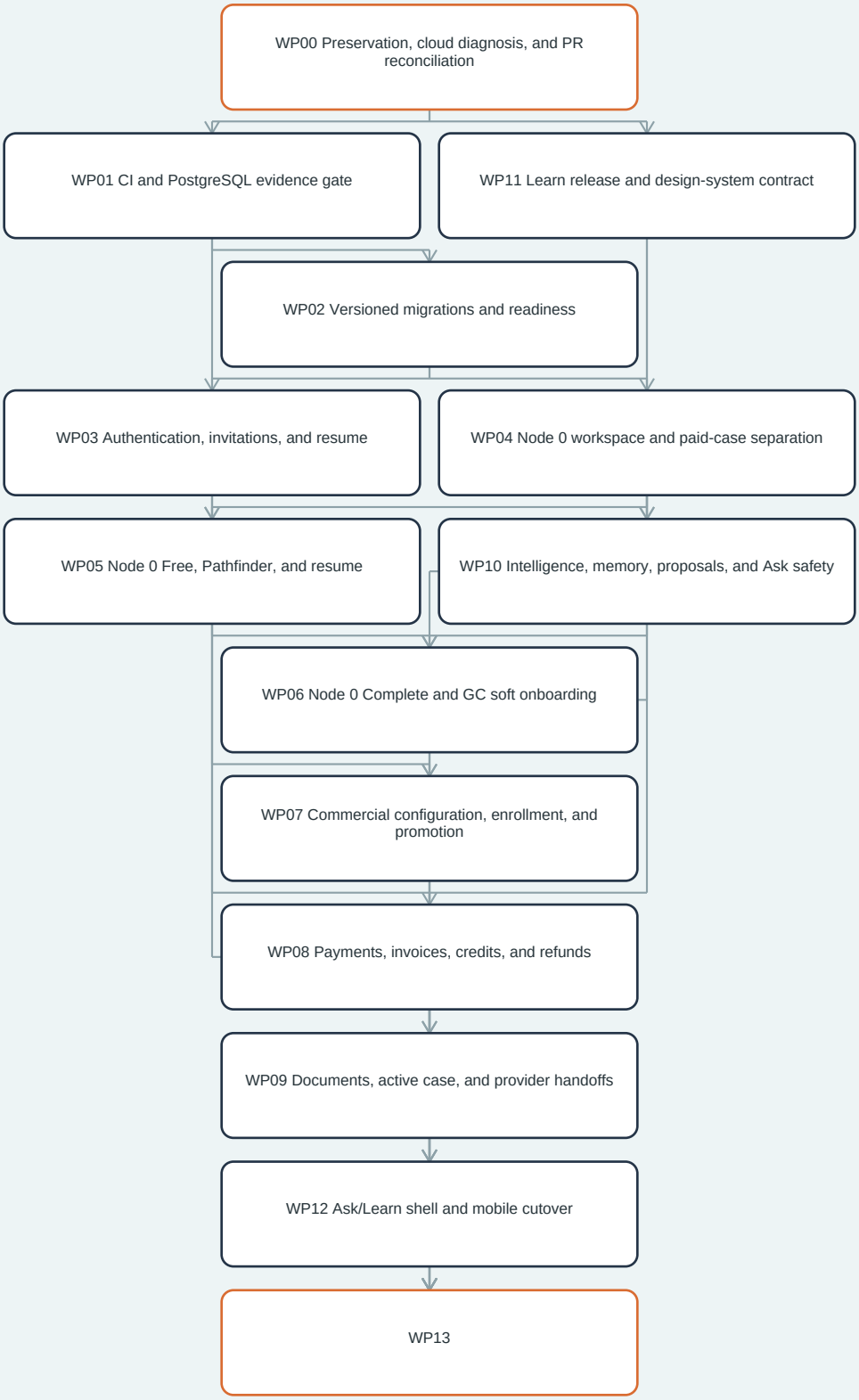
Current verified verdict

Area	Verdict	Evidence / consequence
Live cloud	Not usable now	dev.app.natalyx.health returned ALB 503 at the inspection snapshot; A0 is open.
AWS detail	Partially unverified	Local natalyx-dev SSO expired; service counts/task revision must be captured during A0.
CI/Postgres	Insufficient on main	Container build + contract smoke exist; no main Python/PostgreSQL evidence job.
PR #122	Materially improved, not final architecture	Latest e853821 fixes return_to/TTL/CAS/lens/CI and should be selectively retained.
PR #124	Do not merge unchanged	Free Node 0 creates a canonical case; unaddressed link can lead to case membership; no commercial promotion/provenance contract.
Payments/documents/PHI	Not implemented/ready	No money ledger; documents metadata-only; app-layer authorization and synthetic-first PHI gate remain.

Architecture principles

- natalyx-app owns routing, validation, permissions, participant/workspace/case/commercial truth and PHI policy.
- Workspace, entitlement, commitment, payment, case, invitation/claim, document and journey-stage state machines remain separate.
- PostgreSQL is production authority; established migrations are preferred pending ADR; runtime identities lose DDL authority.
- Only `byos.workspace_promotion_authorized.v1` authorizes workspace-to-case promotion; redirects and raw webhooks never do.
- Commercial/money events are append-only; balances/statuses are projections.
- IP observations/provisional assertions remain attributed; GC/provider confirmation appends revisions and never silently erases/elevates source.
- Provider/intelligence/Learn projections are purpose-, role-, consent-, provenance- and version-bound.
- Security/accessibility/responsive interaction can block release; aesthetic/token purity alone generally cannot.
- Synthetic only through A2, approved PII scope at B1, PHI only after formal B2.

Dependency graph



The graph is package-level; the PR table below is the executable dependency authority.

Global PR table and execution order

P00-01

ID	P00-01
Title	Approve programme baseline and extraction ledger
Repository	.github
Work package	WP00 / Preservation, cloud diagnosis, and PR reconciliation
DRI	Luke
Implementer	Luke
Reviewer	Alan
Final approver	Luke
Dependencies	P00-01 plan sign-off
Gate	Pre-A0
Size	Medium
Readiness	Needs plan approval
Blocking decision	None
Replacement/source PR	All listed draft PRs and protected worktrees
Project item	P00-01

P01-01

ID	P01-01
Title	Establish current-test and PostgreSQL CI evidence
Repository	app
Work package	WP01 / CI and PostgreSQL evidence gate
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P00-01
Gate	A0
Size	Medium
Readiness	Dependencies unmet
Blocking decision	None
Replacement/source PR	PR #122 unittest workflow and backend-neutral fixes after reconciliation
Project item	P01-01

P11-01

ID	P11-01
Title	Publish validated Learn artifact and provenance contract
Repository	learn
Work package	WP11 / Learn release and design-system contract
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P00-01
Gate	A2
Size	Large
Readiness	Dependencies unmet
Blocking decision	None
Replacement/source PR	Protected feature/learn-resource-rendering-contract, feature/node0-beta-learn-resources
Project item	P11-01

P11-03

ID	P11-03
Title	Release participant design-system accessibility and interaction contract
Repository	design-system
Work package	WP11 / Learn release and design-system contract
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P00-01
Gate	A1
Size	Medium
Readiness	Dependencies unmet
Blocking decision	None
Replacement/source PR	Design-system PRs #7 and #8, dirty local brand assets
Project item	P11-03

P02-01

ID	P02-01
Title	Decide migration architecture and compatibility contract
Repository	app
Work package	WP02 / Versioned migrations and readiness
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P01-01
Gate	A0
Size	Small
Readiness	Blocked by decision
Blocking decision	D-MIG-01
Replacement/source PR	app/storage/schema.py, app/storage/connection.py
Project item	P02-01

P03-01

ID	P03-01
Title	Harden auth return-to and participant resume
Repository	app
Work package	WP03 / Authentication, invitations, and resume
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P01-01
Gate	A1
Size	Medium
Readiness	Dependencies unmet
Blocking decision	None
Replacement/source PR	PR #122 return_to fixes
Project item	P03-01

P13-01

ID	P13-01
Title	Expose known-revision health, readiness, and dependency diagnostics
Repository	app
Work package	WP13 / Deployment, cloud gates, and PHI readiness
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P01-01
Gate	A0
Size	Large
Readiness	Dependencies unmet
Blocking decision	None
Replacement/source PR	Current /healthz, direct 503 evidence
Project item	P13-01

P11-02

ID	P11-02
Title	Serve approved Learn catalog versions from intelligence
Repository	intelligence
Work package	WP11 / Learn release and design-system contract
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P11-01
Gate	A2
Size	Large
Readiness	Dependencies unmet
Blocking decision	None
Replacement/source PR	Intelligence issue #14, protected dirty learn_resources.py
Project item	P11-02

P02-02

ID	P02-02
Title	Introduce versioned migration runner and readiness checks
Repository	app
Work package	WP02 / Versioned migrations and readiness
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P02-01
Gate	A0
Size	Large
Readiness	Dependencies unmet
Blocking decision	D-MIG-01
Replacement/source PR	Current init_db startup path
Project item	P02-02

P09-01

ID	P09-01
Title	Define document lifecycle, authorization, and retention contract
Repository	app
Work package	WP09 / Documents, active case, and provider handoffs
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P00-01, P02-01
Gate	A2
Size	Medium
Readiness	Blocked by decision
Blocking decision	D-DOC-01, D-DOC-02, D-PHI-01
Replacement/source PR	Current metadata-only documents.py, provider branch
Project item	P09-01

P13-02

ID	P13-02
Title	Harden guarded deployment, rollback, and evidence capture
Repository	infra
Work package	WP13 / Deployment, cloud gates, and PHI readiness
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P01-01, P13-01
Gate	A0
Size	Large
Readiness	Dependencies unmet
Blocking decision	D-OPS-01
Replacement/source PR	Existing guarded app/infra workflows
Project item	P13-02

P11-04

ID	P11-04
Title	Adopt approved Learn and source-card recipes in app
Repository	app
Work package	WP11 / Learn release and design-system contract
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P11-02, P11-03
Gate	A2
Size	Large
Readiness	Dependencies unmet
Blocking decision	None
Replacement/source PR	PR #122 Learn slot, PR #124 dependency note
Project item	P11-04

P02-03

ID	P02-03
Title	Baseline canonical schema and prove expand-contract repair
Repository	app
Work package	WP02 / Versioned migrations and readiness
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P02-02
Gate	A1
Size	Large
Readiness	Dependencies unmet
Blocking decision	None
Replacement/source PR	app/storage/schema.py current definitions, PR #122 forward-migration tests after reconciliation
Project item	P02-03

P02-04

ID	P02-04
Title	Move intelligence persistence to a revisioned migration namespace
Repository	intelligence
Work package	WP02 / Versioned migrations and readiness
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P02-01, P02-02
Gate	A1
Size	Large
Readiness	Dependencies unmet
Blocking decision	D-MIG-01
Replacement/source PR	src/natalyx_intelligence/shared/persistence.py runtime _ensure_schema
Project item	P02-04

P09-02

ID	P09-02
Title	Provision private encrypted document and quarantine storage
Repository	infra
Work package	WP09 / Documents, active case, and provider handoffs
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P09-01, P13-02
Gate	A2
Size	Large
Readiness	Dependencies unmet
Blocking decision	D-DOC-02, D-PHI-01
Replacement/source PR	Existing infra platform stack patterns
Project item	P09-02

P04-01

ID	P04-01
Title	Separate Node 0 workspace storage from paid cases
Repository	app
Work package	WP04 / Node 0 workspace and paid-case separation
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P02-03
Gate	A1
Size	Exceptional
Readiness	Dependencies unmet
Blocking decision	CHECK-ALAN-122-124
Replacement/source PR	PR #124 _ensure_node0_case as behavior to replace, candidate worktrees
Project item	P04-01

P07-01

ID	P07-01
Title	Add versioned commercial configuration and terms acceptance
Repository	app
Work package	WP07 / Commercial configuration, enrollment, and promotion
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P02-03
Gate	A2
Size	Large
Readiness	Dependencies unmet
Blocking decision	D-COM-01, D-COM-02, D-COM-03
Replacement/source PR	Payment/business-model DOCX
Project item	P07-01

P09-03

ID	P09-03
Title	Implement resilient document upload, scan, and download
Repository	app
Work package	WP09 / Documents, active case, and provider handoffs
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P09-02, P02-03
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet
Blocking decision	D-DOC-01, D-DOC-02
Replacement/source PR	Current metadata scaffold
Project item	P09-03

P03-02

ID	P03-02
Title	Add low-risk workspace claim lifecycle
Repository	app
Work package	WP03 / Authentication, invitations, and resume
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P03-01, P04-01
Gate	A1
Size	Large
Readiness	Dependencies unmet
Blocking decision	D-AUTH-01, CHECK-ALAN-122-124
Replacement/source PR	PRs #123/#124, dirty onboarding-link worktree
Project item	P03-02

P03-03

ID	P03-03
Title	Bind paid-case invitations to identity and consent
Repository	app
Work package	WP03 / Authentication, invitations, and resume
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P03-01, P04-01
Gate	A2
Size	Large
Readiness	Dependencies unmet
Blocking decision	D-AUTH-01, CHECK-ALAN-122-124
Replacement/source PR	PR #122 CAS/TTL/self-invite fixes, current invitations.py
Project item	P03-03

P04-02

ID	P04-02
Title	Add attributable workspace revision and projection contracts
Repository	app
Work package	WP04 / Node 0 workspace and paid-case separation
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P04-01
Gate	A1
Size	Large
Readiness	Dependencies unmet
Blocking decision	D-DATA-01, D-DATA-02, CHECK-ALAN-122-124
Replacement/source PR	Current facts/memory models, GC authorship audit
Project item	P04-02

P05-01

ID	P05-01
Title	Replace Pathfinder routing with deterministic entry decisions
Repository	app
Work package	WP05 / Node 0 Free, Pathfinder, and resume
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P03-01, P04-01
Gate	A1
Size	Medium
Readiness	Dependencies unmet
Blocking decision	None
Replacement/source PR	Current pathfinder_destinations.py, PR #119 product-alignment doc
Project item	P05-01

P08-01

ID	P08-01
Title	Add append-only commercial and money ledgers
Repository	app
Work package	WP08 / Payments, invoices, credits, and refunds
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P02-03, P07-01
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet
Blocking decision	D-COM-01, D-COM-02, D-COM-03
Replacement/source PR	Payment/business-model DOCX
Project item	P08-01

P03-04

ID	P03-04
Title	Replace shared signup admission with claim-aware Cognito admission
Repository	infra
Work package	WP03 / Authentication, invitations, and resume
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P03-02, P03-03
Gate	A1
Size	Large
Readiness	Blocked by decision
Blocking decision	D-AUTH-02
Replacement/source PR	infra platform-stack.ts current allowlist + shared invite code
Project item	P03-04

P04-03

ID	P04-03
Title	Adapt legacy Node 0 case-shaped data without semantic promotion
Repository	app
Work package	WP04 / Node 0 workspace and paid-case separation
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P04-02
Gate	A1
Size	Large
Readiness	Dependencies unmet
Blocking decision	None
Replacement/source PR	PRs #117/#121/#123/#124 and dirty candidate worktrees
Project item	P04-03

P10-01

ID	P10-01
Title	Version the scoped intelligence context envelope
Repository	intelligence
Work package	WP10 / Intelligence, memory, proposals, and Ask safety
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P01-01, P04-02
Gate	A2
Size	Large
Readiness	Dependencies unmet
Blocking decision	D-DATA-01, D-AI-01
Replacement/source PR	Current case_context advisory stub
Project item	P10-01

P05-02

ID	P05-02
Title	Rebuild Node 0 Free candidate and outreach domain
Repository	app
Work package	WP05 / Node 0 Free, Pathfinder, and resume
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P04-02, P05-01
Gate	A1
Size	Exceptional
Readiness	Dependencies unmet
Blocking decision	CHECK-ALAN-122-124
Replacement/source PR	PR #117 direct algorithm/test salvage, PR #124, quiz rewrite worktree
Project item	P05-02

P06-01

ID	P06-01
Title	Add Node 0 Complete entitlement and versioned feature gates
Repository	app
Work package	WP06 / Node 0 Complete and GC soft onboarding
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P04-01, P07-01, P08-01
Gate	A2
Size	Large
Readiness	Dependencies unmet
Blocking decision	D-COM-01
Replacement/source PR	Business-model Node 0 feature table
Project item	P06-01

P07-02

ID	P07-02
Title	Implement commitment and enrollment-pending lifecycle
Repository	app
Work package	WP07 / Commercial configuration, enrollment, and promotion
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P07-01, P08-01
Gate	A2
Size	Large
Readiness	Dependencies unmet
Blocking decision	D-COM-04
Replacement/source PR	Business-model conversion trigger
Project item	P07-02

P10-02

ID	P10-02
Title	Enforce private/shared memory and evidence-bound proposal approval
Repository	app
Work package	WP10 / Intelligence, memory, proposals, and Ask safety
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P04-02, P10-01
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet
Blocking decision	D-DATA-01, D-AI-01
Replacement/source PR	Current facts/memory/proposals scaffolds, GC authorship audit
Project item	P10-02

P10-03

ID	P10-03
Title	Harden Ask and Learn grounding and degraded-mode behavior
Repository	intelligence
Work package	WP10 / Intelligence, memory, proposals, and Ask safety
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P10-01, P11-02
Gate	A2
Size	Large
Readiness	Dependencies unmet
Blocking decision	D-AI-01
Replacement/source PR	Current ask_natalyx.py and PR #124 Ask integration
Project item	P10-03

P05-03

ID	P05-03
Title	Ship resumable Node 0 Free participant surface
Repository	app
Work package	WP05 / Node 0 Free, Pathfinder, and resume
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P03-01, P05-02, P11-03
Gate	A1
Size	Large
Readiness	Dependencies unmet
Blocking decision	CHECK-ALAN-122-124
Replacement/source PR	PRs #121/#124 selective UI port, DS PR #8
Project item	P05-03

P06-02

ID	P06-02
Title	Add GC soft onboarding and attributable confirmation
Repository	app
Work package	WP06 / Node 0 Complete and GC soft onboarding
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P03-02, P04-02, P06-01
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet
Blocking decision	D-DATA-01, D-DATA-02, CHECK-ALAN-122-124
Replacement/source PR	PRs #123/#124 UX only, GC authorship audit
Project item	P06-02

P08-02

ID	P08-02
Title	Add processor-neutral payment command and webhook inbox
Repository	app
Work package	WP08 / Payments, invoices, credits, and refunds
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P08-01, P07-02
Gate	A2
Size	Exceptional
Readiness	Blocked by decision
Blocking decision	D-PAY-01, D-COM-05
Replacement/source PR	Business-model processor-neutral requirement
Project item	P08-02

P10-04

ID	P10-04
Title	Add intelligence safety evaluations and cross-service observability
Repository	intelligence
Work package	WP10 / Intelligence, memory, proposals, and Ask safety
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P10-02, P10-03
Gate	A2
Size	Large
Readiness	Dependencies unmet
Blocking decision	None
Replacement/source PR	Current intelligence auth/telemetry tests
Project item	P10-04

P13-03

ID	P13-03
Title	Automate synthetic A1 core participant journey verification
Repository	app
Work package	WP13 / Deployment, cloud gates, and PHI readiness
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P03-02, P04-03, P05-03, P11-03, P13-02
Gate	A1
Size	Large
Readiness	Dependencies unmet
Blocking decision	None
Replacement/source PR	Existing synthetic sandbox/scenario utilities
Project item	P13-03

P06-03

ID	P06-03
Title	Add readiness report and role-scoped progress surfaces
Repository	app
Work package	WP06 / Node 0 Complete and GC soft onboarding
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P06-02, P10-02, P11-04
Gate	A2
Size	Large
Readiness	Dependencies unmet
Blocking decision	D-DATA-01
Replacement/source PR	PR #124 score/report concepts
Project item	P06-03

P07-03

ID	P07-03
Title	Emit processor-neutral workspace promotion authorization
Repository	app
Work package	WP07 / Commercial configuration, enrollment, and promotion
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P04-02, P07-02, P08-02
Gate	A2
Size	Large
Readiness	Dependencies unmet
Blocking decision	D-COM-04, D-DATA-01
Replacement/source PR	Business-model promotion invariants
Project item	P07-03

P07-04

ID	P07-04
Title	Promote workspace exactly once from authorization event
Repository	app
Work package	WP07 / Commercial configuration, enrollment, and promotion
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P03-03, P04-02, P07-03
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet
Blocking decision	D-COM-04, D-DATA-01, CHECK-ALAN-122-124
Replacement/source PR	byos.workspace_promotion_authorized.v1 contract, PR #124 source linkage concepts
Project item	P07-04

P09-04

ID	P09-04
Title	Complete active-case participant document and progress surfaces
Repository	app
Work package	WP09 / Documents, active case, and provider handoffs
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P03-03, P07-04, P09-03, P11-03
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet
Blocking decision	D-COM-04, CHECK-ALAN-122-124
Replacement/source PR	PR #122 selective/direct reuse after checkpoint, PR #124 case UI
Project item	P09-04

P08-03

ID	P08-03
Title	Implement milestone invoicing, credits, and deterministic refunds
Repository	app
Work package	WP08 / Payments, invoices, credits, and refunds
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P08-02, P07-04, P09-04
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet
Blocking decision	D-COM-04, D-COM-06, D-COM-07
Replacement/source PR	Payment/business-model milestone/refund tables
Project item	P08-03

P09-05

ID	P09-05
Title	Implement consent-scoped provider handoff snapshots and operations
Repository	app
Work package	WP09 / Documents, active case, and provider handoffs
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P04-02, P09-03, P09-04, P10-02
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet
Blocking decision	D-PROV-01, D-PHI-01
Replacement/source PR	Protected provider-exchange worktree: selective port only after provenance fix
Project item	P09-05

P12-01

ID	P12-01
Title	Unify Ask and Learn shell with explicit context scopes
Repository	app
Work package	WP12 / Ask/Learn shell and mobile cutover
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P10-03, P11-04, P09-04
Gate	A2
Size	Large
Readiness	Dependencies unmet
Blocking decision	D-AI-01
Replacement/source PR	Current Ask widget/Learn workspace, PR #124
Project item	P12-01

P08-04

ID	P08-04
Title	Add payment reconciliation and safe service-pause operations
Repository	app
Work package	WP08 / Payments, invoices, credits, and refunds
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P08-03
Gate	A2
Size	Large
Readiness	Dependencies unmet
Blocking decision	D-COM-08
Replacement/source PR	Business-model failure handling
Project item	P08-04

P13-05

ID	P13-05
Title	Enforce case RLS and a non-owner PostgreSQL runtime
Repository	app
Work package	WP13 / Deployment, cloud gates, and PHI readiness
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P02-03, P03-03, P04-03, P09-05, P10-02
Gate	B2
Size	Exceptional
Readiness	Dependencies unmet
Blocking decision	D-PHI-01
Replacement/source PR	case-shell.md current app-layer-only/RLS gap
Project item	P13-05

P12-02

ID	P12-02
Title	Complete mobile participant interaction and accessibility cutover
Repository	app
Work package	WP12 / Ask/Learn shell and mobile cutover
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P05-03, P06-03, P09-04, P12-01
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet
Blocking decision	None
Replacement/source PR	PR #122 mobile journey, PR #124 Node 0, DS PR #8
Project item	P12-02

P13-06

ID	P13-06
Title	Harden B1/B2 backups, audit, alarms, and cohort controls
Repository	infra
Work package	WP13 / Deployment, cloud gates, and PHI readiness
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P09-02, P13-02, P13-05
Gate	B2
Size	Exceptional
Readiness	Dependencies unmet
Blocking decision	D-PHI-01, D-PHI-02
Replacement/source PR	Existing audit/backup/platform stack constructs
Project item	P13-06

P13-04

ID	P13-04
Title	Automate synthetic A2 commercial and document journey verification
Repository	app
Work package	WP13 / Deployment, cloud gates, and PHI readiness
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P06-03, P07-04, P08-04, P09-05, P10-04, P12-02, P13-03
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet
Blocking decision	None
Replacement/source PR	Payment document smoke paths, A1 harness
Project item	P13-04

P12-03

ID	P12-03
Title	Retire legacy participant routes behind verified redirects
Repository	app
Work package	WP12 / Ask/Learn shell and mobile cutover
DRI	Alan
Implementer	Alan
Reviewer	Luke
Final approver	Luke
Dependencies	P12-02, P13-04
Gate	B1
Size	Large
Readiness	Dependencies unmet
Blocking decision	None
Replacement/source PR	Current route inventory, PRs #112/#119 planning maps
Project item	P12-03

P13-07

ID	P13-07
Title	Publish B1 and B2 privacy, PHI, and operational go-no-go runbooks
Repository	.github
Work package	WP13 / Deployment, cloud gates, and PHI readiness
DRI	Luke
Implementer	Joint
Reviewer	External/TBD
Final approver	Luke
Dependencies	P13-04, P13-05, P13-06
Gate	B2
Size	Large
Readiness	Blocked by decision
Blocking decision	D-PHI-01, D-PHI-02
Replacement/source PR	Natalyx compliance/company records by reference, all gate evidence
Project item	P13-07

Topological execution order:

Order	ID	Title	Repository	Dependencies	Gate
1	P00-01	Approve programme baseline and extraction ledger	.github	P00-01	Pre-A0
2	P01-01	Establish current-test and PostgreSQL CI evidence	app	P00-01	A0
3	P11-01	Publish validated Learn artifact and provenance contract	learn	P00-01	A2
4	P11-03	Release participant design-system accessibility and interaction contract	design-system	P00-01	A1
5	P02-01	Decide migration architecture and compatibility contract	app	P01-01	A0
6	P03-01	Harden auth return-to and participant resume	app	P01-01	A1

Order	ID	Title	Repository	Dependencies	Gate
7	P13-01	Expose known-revision health, readiness, and dependency diagnostics	app	P01-01	A0
8	P11-02	Serve approved Learn catalog versions from intelligence	intelligence	P11-01	A2
9	P02-02	Introduce versioned migration runner and readiness checks	app	P02-01	A0
10	P09-01	Define document lifecycle, authorization, and retention contract	app	P00-01, P02-01	A2
11	P13-02	Harden guarded deployment, rollback, and evidence capture	infra	P01-01, P13-01	A0
12	P11-04	Adopt approved Learn and source-card recipes in app	app	P11-02, P11-03	A2
13	P02-03	Baseline canonical schema and prove expand-contract repair	app	P02-02	A1
14	P02-04	Move intelligence persistence to a revisioned migration namespace	intelligence	P02-01, P02-02	A1
15	P09-02	Provision private encrypted document and quarantine storage	infra	P09-01, P13-02	A2
16	P04-01	Separate Node 0 workspace storage from paid cases	app	P02-03	A1
17	P07-01	Add versioned commercial configuration and terms acceptance	app	P02-03	A2
18	P09-03	Implement resilient document upload, scan, and download	app	P09-02, P02-03	A2
19	P03-02	Add low-risk workspace claim lifecycle	app	P03-01, P04-01	A1
20	P03-03	Bind paid-case invitations to identity and consent	app	P03-01, P04-01	A2
21	P04-02	Add attributable workspace revision and projection contracts	app	P04-01	A1
22	P05-01	Replace Pathfinder routing with deterministic entry decisions	app	P03-01, P04-01	A1
23	P08-01	Add append-only commercial and money ledgers	app	P02-03, P07-01	A2
24	P03-04	Replace shared signup admission with claim-aware Cognito admission	infra	P03-02, P03-03	A1
25	P04-03	Adapt legacy Node 0 case-shaped data without semantic promotion	app	P04-02	A1
26	P10-01	Version the scoped intelligence context envelope	intelligence	P01-01, P04-02	A2
27	P05-02	Rebuild Node 0 Free candidate and outreach domain	app	P04-02, P05-01	A1
28	P06-01	Add Node 0 Complete entitlement and versioned feature gates	app	P04-01, P07-01, P08-01	A2
29	P07-02	Implement commitment and enrollment-pending lifecycle	app	P07-01, P08-01	A2

Order	ID	Title	Repository	Dependencies	Gate
30	P10-02	Enforce private/shared memory and evidence-boun d proposal approval	app	P04-02, P10-01	A2
31	P10-03	Harden Ask and Learn grounding and degraded-mo de behavior	intelligence	P10-01, P11-02	A2
32	P05-03	Ship resumable Node 0 Free participant surface	app	P03-01, P05-02, P11-03	A1
33	P06-02	Add GC soft onboarding and attributable confirmation	app	P03-02, P04-02, P06-01	A2
34	P08-02	Add processor-neutral payment command and webhook inbox	app	P08-01, P07-02	A2
35	P10-04	Add intelligence safety evaluations and cross-service observability	intelligence	P10-02, P10-03	A2
36	P13-03	Automate synthetic A1 core participant journey verification	app	P03-02, P04-03, P05-03, P11-03, P13-02	A1
37	P06-03	Add readiness report and role-scoped progress surfaces	app	P06-02, P10-02, P11-04	A2
38	P07-03	Emit processor-neutral workspace promotion authorization	app	P04-02, P07-02, P08-02	A2
39	P07-04	Promote workspace exactly once from authorization event	app	P03-03, P04-02, P07-03	A2
40	P09-04	Complete active-case participant document and progress surfaces	app	P03-03, P07-04, P09-03, P11-03	A2
41	P08-03	Implement milestone invoicing, credits, and deterministic refunds	app	P08-02, P07-04, P09-04	A2
42	P09-05	Implement consent-scoped provider handoff snapshots and operations	app	P04-02, P09-03, P09-04, P10-02	A2
43	P12-01	Unify Ask and Learn shell with explicit context scopes	app	P10-03, P11-04, P09-04	A2
44	P08-04	Add payment reconciliation and safe service-pause operations	app	P08-03	A2
45	P13-05	Enforce case RLS and a non-owner PostgreSQL runtime	app	P02-03, P03-03, P04-03, P09-05, P10-02	B2
46	P12-02	Complete mobile participant interaction and accessibility cutover	app	P05-03, P06-03, P09-04, P12-01	A2
47	P13-06	Harden B1/B2 backups, audit, alarms, and cohort controls	infra	P09-02, P13-02, P13-05	B2
48	P13-04	Automate synthetic A2 commercial and document journey verification	app	P06-03, P07-04, P08-04, P09-05, P10-04, P12-02, P13-03	A2
49	P12-03	Retire legacy participant routes behind verified redirects	app	P12-02, P13-04	B1
50	P13-07	Publish B1 and B2 privacy, PHI, and operational go-no-go runbooks	.github	P13-04, P13-05, P13-06	B2

Parallel lanes and critical path

Critical path:

`P00-01` → `P01-01` → `P02-01` → `P02-02` → `P02-03` → `P07-01` → `P08-01` → `P07-02` → `P08-02` → `P07-03` → `P07-04` → `P09-04` → `P08-03` → `P08-04` → `P13-04` → `P13-07`

Required dependency joins:

- P07-03 additionally waits for P04-02 (workspace/provenance); P07-04 additionally waits for P03-03 (paid-case invitation).
- P09-04 additionally waits for P09-03 and P11-03; P13-04 additionally waits for P06-03, P09-05, P10-04, P12-02 and A1 runner P13-03.
- P13-07 additionally waits for P13-05 RLS/non-owner runtime and P13-06 backup/audit/control evidence before B2.

First three implementation PRs:

1. `P01-01` — Establish current-test and PostgreSQL CI evidence
2. `P02-01` — Decide migration architecture and compatibility contract
3. `P13-01` — Expose known-revision health, readiness, and dependency diagnostics

Lane	Lead	Items	Coordination
Foundation and live recovery	Alan	P01-01, P02-01, P02-02, P02-03, P02-04, P13-01, P13-02	Luke approves ADR and A0; no replacement participant surface required.
Alan reconciliation	Alan	CHECK-ALAN-122-124, CLEAN-01	Runs beside CI/A0; blocks replacement auth/Node0/case surfaces, not evidence work.
Identity and workspace	Alan	P03-01, P03-02, P03-03, P03-04, P04-01, P04-02, P04-03	Luke decides role/privacy/authorship contract; begins after checkpoint.
Participant Free and Complete	Alan	P05-01, P05-02, P05-03, P06-01, P06-02, P06-03	Depends workspace/provenance; Luke accepts complete-the-match and projections.
Commercial and payment	Alan	P07-01, P07-02, P07-03, P07-04, P08-01, P08-02, P08-03, P08-04	Domain/config work can proceed beside participant UI once Luke/finance/legal decisions are explicit.
Documents and provider exchange	Alan	P09-01, P09-02, P09-03, P09-04, P09-05	Infra storage can follow ADR while UI waits on case/provenance and checkpoint.
Intelligence, Learn and DS	Alan	P10-01, P10-02, P10-03, P10-04, P11-01, P11-02, P11-03, P11-04, P12-01	Learn and DS repositories can progress without blocking A0; app adoption waits on pinned releases.
Cutover and formal gates	Joint	P12-02, P12-03, P13-03, P13-04, P13-05, P13-06, P13-07	Alan verifies technical evidence; Luke and external reviewers accept B1/B2.

Early sequencing:

1. P00-01
2. P01-01 + P02-01 + P13-01 + CHECK-ALAN-122-124
3. P02-02 + P13-02 + P11-01 + P11-03
4. P02-03 + P02-04 + P03-01 + P04-01

State transition contract

Common envelope:

```
event_id
entity_type
entity_id
event_type
from_state
to_state
expected_revision
result_revision
actor_type
actor_subject_or_service
authority_scope
evidence_refs[]
idempotency_key
```

```
correlation_id
causation_id
policy_version
schema_version
commercial_config_version (nullable)
reason_code
occurred_at
recorded_at
safe_metadata
```

- Reusing an idempotency key with the same canonical payload returns the original result; a different payload returns 409 idempotency_conflict.
- A wrong from_state or expected revision returns 409 stale_transition; no generic setter silently coerces state.
- The state write, append-only audit event and outbox entry commit in one transaction.
- Retry resumes from committed event/projection; recovery uses an explicit compensating or forward-repair event.
- Audit safe_metadata excludes raw PHI, PII beyond essential identifiers, invitation tokens, processor payloads and document contents.

Participant entry and routing (`entry_session`)

Owner: natalyx-app. States: `unstarted`, `education`, `identity_required`, `authenticated`, `routed`, `handed_off`, `expired`.

unstarted

From	unstarted
To	education
Actor	visitor
Evidence	GET/explicit entry choice
Idempotency key	entry:{session}:start
Invalid transitions	No participant/case write
Audit event	entry_started
Retry	Return current session
Recovery	Create fresh anonymous session

education

From	education
To	identity_required
Actor	visitor/policy
Evidence	route requires saved workspace
Idempotency key	entry:{session}:identity
Invalid transitions	Cannot grant workspace/case
Audit event	identity_required
Retry	Preserve safe return intent
Recovery	Restart at public choice

identity_required

From	identity_required
To	authenticated
Actor	Cognito/auth provider
Evidence	verified subject/session
Idempotency key	auth:{subject}:{nonce}
Invalid transitions	Invalid/expired token
Audit event	identity_bound
Retry	Replay returns authenticated session
Recovery	Rechallenge; no partial membership

education|authenticated

From	education|authenticated
To	routed
Actor	participant + deterministic policy
Evidence	validated route answers + policy version
Idempotency key	route:{session}:{answer_hash}
Invalid transitions	Unknown/unsafe route goes review/education
Audit event	entry_routed
Retry	Same input returns same decision
Recovery	Allow corrected answers/new decision revision

routed

From	routed
To	handed_off
Actor	participant
Evidence	accepted destination and authorized target
Idempotency key	handoff:{session}:{route_version}
Invalid transitions	Target not authorized/available
Audit event	entry_handed_off
Retry	Resume same safe destination
Recovery	Return to route result with explanation

any nonterminal

From	any nonterminal
To	expired
Actor	system
Evidence	TTL elapsed/revoked session
Idempotency key	entry:{session}:expire
Invalid transitions	No transition from handed_off needed
Audit event	entry_expired
Retry	Idempotent
Recovery	New session; no case/access change

Node 0 workspace (`node0_workspace`)

Owner: natalyx-app. States: `draft`, `active`, `match_ready`, `promotion_pending`, `promoted`, `archived`.

none

From	none
To	draft
Actor	authenticated IP
Evidence	first explicit Node 0 start
Idempotency key	workspace:{owner}:create
Invalid transitions	Existing active workspace returns existing
Audit event	workspace_created
Retry	Unique owner/idempotency key
Recovery	Forward repair orphan membership

draft

From	draft
To	active
Actor	IP
Evidence	minimum workspace consent/profile
Idempotency key	workspace:{id}:activate:{revision}
Invalid transitions	Missing consent/owner
Audit event	workspace_activated
Retry	Return current active revision
Recovery	Correct inputs and retry

active

From	active
To	match_ready
Actor	IP + selected GC evidence policy
Evidence	selected candidate + required provisional/GC-authored fields
Idempotency key	workspace:{id}:ready:{revision}
Invalid transitions	Critical unknown/decline/revoked claim
Audit event	workspace_match_ready
Retry	No duplicate event
Recovery	Reopen active with reason revision

match_ready

From	match_ready
To	promotion_pending
Actor	promotion event consumer
Evidence	valid byos.workspace_promotion_authorized.v1
Idempotency key	promotion:{commitment}:v1
Invalid transitions	No authorization; already promoted; stale config
Audit event	workspace_promotion_requested
Retry	Replay resumes/finds same promotion
Recovery	Forward-repair incomplete transaction

promotion_pending

From	promotion_pending
To	promoted
Actor	promotion service
Evidence	paid case + manifest committed
Idempotency key	promotion:{commitment}:v1
Invalid transitions	Projection invariant/privacy failure
Audit event	workspace_promoted
Retry	Return same case ID
Recovery	Repair from immutable manifest; never duplicate

draft|active|match_ready

From	draft|active|match_ready
To	archived
Actor	IP/operator per policy
Evidence	explicit archive + retention policy
Idempotency key	workspace:{id}:archive:{request}
Invalid transitions	Promotion pending/promoted
Audit event	workspace_archived
Retry	Idempotent
Recovery	Restore only through new audited revision if policy permits

Entitlement (`workspace_entitlement`)

Owner: natalyx-app commercial domain. States: `free`, `complete_pending`, `complete_active`, `suspended`, `revoked`.

none

From	none
To	free
Actor	system
Evidence	active Node 0 workspace
Idempotency key	entitlement:{workspace}:free
Invalid transitions	No workspace
Audit event	entitlement_granted
Retry	Return current
Recovery	Repair projection from event

free

From	free
To	complete_pending
Actor	IP/commercial service
Evidence	accepted Complete offer/config
Idempotency key	entitlement:{workspace}:{commitment}
Invalid transitions	Unknown price/config/terms
Audit event	complete_requested
Retry	Return pending
Recovery	Cancel/expire commitment

complete_pending

From	complete_pending
To	complete_active
Actor	payment domain
Evidence	settled Complete ledger entry
Idempotency key	entitlement:{payment}:activate
Invalid transitions	Amount/currency/config mismatch
Audit event	complete_activated
Retry	Replay no-op
Recovery	Reconcile and retry from inbox

complete_active

From	complete_active
To	suspended
Actor	authorized commercial policy
Evidence	refund/reversal or approved pause reason
Idempotency key	entitlement:{id}:suspend:{event}
Invalid transitions	Cannot remove free/safety capability
Audit event	entitlement_suspended
Retry	No-op if same event
Recovery	Reinstate or revoke through new event

complete_active|suspended

From	complete_active|suspended
To	revoked
Actor	authorized operator/policy
Evidence	fraud/refund/terms revocation evidence
Idempotency key	entitlement:{id}:revoke:{event}
Invalid transitions	No silent case cancellation
Audit event	entitlement_revoked
Retry	Idempotent
Recovery	Manual reviewed regrant creates new entitlement event

Commercial commitment (`journey_commitment`)

Owner: natalyx-app commercial domain. States: `draft`, `accepted`, `funding_pending`, `funded`, `cancelled`, `expired`.

none

From	none
To	draft
Actor	IP
Evidence	active workspace + displayed config/terms
Idempotency key	commitment:{workspace}:{config}
Invalid transitions	No active config/workspace
Audit event	commitment_drafted
Retry	Return existing draft
Recovery	Create new draft after expiry/version change

draft

From	draft
To	accepted
Actor	IP
Evidence	terms/privacy consent + payment path
Idempotency key	commitment:{id}:accept:{terms_hash}
Invalid transitions	Stale/missing acceptance
Audit event	commitment_accepted
Retry	Return accepted
Recovery	New acceptance revision if version changes

accepted

From	accepted
To	funding_pending
Actor	commercial service
Evidence	enrollment invoice/payment intent created
Idempotency key	commitment:{id}:funding
Invalid transitions	No invoice or invalid amount
Audit event	funding_requested
Retry	Resume same invoice
Recovery	Cancel/reissue under audited adjustment

funding_pending

From	funding_pending
To	funded
Actor	payment domain
Evidence	net settled enrollment funding meets versioned requirement
Idempotency key	enrollment-funding:{id}:v1
Invalid transitions	Redirect/pending/failed/reversed evidence
Audit event	commercial.enrollment_funding_confirmed.v1
Retry	Exactly-once event/outbox replay
Recovery	Reconcile inbox/ledger; no browser repair

draft|accepted|funding_pending

From	draft|accepted|funding_pending
To	cancelled
Actor	IP/authorized operator
Evidence	explicit cancellation and refund policy
Idempotency key	commitment:{id}:cancel:{request}
Invalid transitions	Already funded/promoted uses case cancellation rail
Audit event	commitment_cancelled
Retry	Idempotent
Recovery	New commitment required

draft|accepted

From	draft|accepted
To	expired
Actor	system
Evidence	offer/terms TTL or config deactivation
Idempotency key	commitment:{id}:expire
Invalid transitions	No expiry after funding
Audit event	commitment_expired
Retry	Idempotent
Recovery	Draft new version

Payment (`payment_obligation + append-only entries`)

Owner: natalyx-app payment domain. States: `created`, `processing`, `settled`, `failed`, `partially_refunded`, `refunded`, `reversed`.

none

From	none
To	created
Actor	billing policy
Evidence	valid invoice/obligation
Idempotency key	payment:{invoice}:create
Invalid transitions	Duplicate or invalid invoice
Audit event	payment_created
Retry	Return current intent
Recovery	Void/reissue through adjustment

created|failed

From	created|failed
To	processing
Actor	processor adapter
Evidence	signed provider acknowledgement
Idempotency key	processor:{provider_event}
Invalid transitions	Amount/currency/customer mismatch
Audit event	payment_processing
Retry	Inbox replay no-op
Recovery	Retry same intent or audited replacement

processing|created

From	processing|created
To	settled
Actor	payment domain
Evidence	verified settlement event
Idempotency key	processor:{provider_event}
Invalid transitions	Browser redirect or unverified event
Audit event	payment_settled
Retry	Exactly-once append
Recovery	Reconcile provider and append repair event

created|processing

From	created|processing
To	failed
Actor	payment domain
Evidence	verified failure/timeout policy
Idempotency key	processor:{provider_event}
Invalid transitions	Cannot overwrite settled
Audit event	payment_failed
Retry	Replay no-op
Recovery	Retry/new attempt linked to obligation

settled|partially_refunded

From	settled|partially_refunded
To	partially_refunded|refunded
Actor	authorized refund service
Evidence	approved refund calculation + processor confirmation
Idempotency key	refund:{refund_request}
Invalid transitions	Exceeds settled less prior refunds
Audit event	refund_settled
Retry	Replay no-op
Recovery	Reconcile/adjust; never mutate original

settled

From	settled
To	reversed
Actor	payment domain
Evidence	verified chargeback/reversal
Idempotency key	processor:{provider_event}
Invalid transitions	Unsupported/manual assertion
Audit event	payment_reversed
Retry	Replay no-op
Recovery	Open discrepancy and apply policy; case is not deleted

Paid case lifecycle (`case`)

Owner: natalyx-app. States: `enrollment_pending`, `active`, `paused`, `closed`, `cancelled`.

none

From	none
To	enrollment_pending
Actor	promotion service
Evidence	byos.workspace_promotion_authorized.v1 + projection manifest
Idempotency key	promotion:{commitment}:v1
Invalid transitions	No authorization/duplicate workspace
Audit event	paid_case_created
Retry	Return same case
Recovery	Forward-repair same promotion

enrollment_pending

From	enrollment_pending
To	active
Actor	authorized coordinator/domain service
Evidence	approved onboarding/compatibility/intake/education evidence
Idempotency key	case:{id}:activate:{evidence_hash}
Invalid transitions	Missing/revoked evidence or GC consent
Audit event	case_activated
Retry	No duplicate activation/invoice
Recovery	Return pending with unmet checklist

active

From	active
To	paused
Actor	authorized commercial/operations policy
Evidence	audited non-critical pause reason
Idempotency key	case:{id}:pause:{event}
Invalid transitions	Safety-critical access cannot be paused
Audit event	case_paused
Retry	Idempotent
Recovery	Resume after evidence; preserve entitled docs

paused

From	paused
To	active
Actor	authorized operator/domain service
Evidence	pause reason resolved
Idempotency key	case:{id}:resume:{event}
Invalid transitions	Unresolved safety/authorization block
Audit event	case_resumed
Retry	Idempotent
Recovery	Escalate manual review

active|paused

From	active|paused
To	closed
Actor	authorized coordinator
Evidence	birth/close or completed termination handoff evidence
Idempotency key	case:{id}:close:{evidence_hash}
Invalid transitions	Open critical obligations
Audit event	case_closed
Retry	No-op if same evidence
Recovery	Reopen only by exceptional audited event

enrollment_pending|active|paused

From	enrollment_pending|active|paused
To	cancelled
Actor	authorized operator under policy
Evidence	withdrawal/non-clearance/cancellation + financial disposition
Idempotency key	case:{id}:cancel:{request}
Invalid transitions	Cannot erase records/refund ledger
Audit event	case_cancelled
Retry	Idempotent
Recovery	Replacement is a new case/workspace policy event

Low-risk workspace identity claim (`workspace_claim`)

Owner: natalyx-app. States: `issued`, `pending_identity`, `identity_bound`, `consented`, `claimed`, `mismatch_pending`, `revoked`, `expired`.

none

From	none
To	issued
Actor	workspace IP
Evidence	selected candidate + allowed claim scope
Idempotency key	claim:{workspace}:{candidate}:{generation}
Invalid transitions	Existing live claim superseded
Audit event	workspace_claim_issued
Retry	Return current token only at issuance
Recovery	Issue new generation; old revoked

issued

From	issued
To	pending_identity
Actor	link holder
Evidence	valid token landing
Idempotency key	claim:{id}:land
Invalid transitions	Expired/revoked token
Audit event	claim_opened
Retry	Safe repeat
Recovery	Authenticate or exit; no data grant

pending_identity

From	pending_identity
To	identity_bound|mismatch_pending
Actor	auth provider
Evidence	verified subject/email and binding policy
Idempotency key	claim:{id}:bind:{subject}
Invalid transitions	Already claimed/different subject
Audit event	claim_identity_bound|claim_mismatch
Retry	CAS no-op or mismatch state
Recovery	Issuer review/new claim

identity_bound

From	identity_bound
To	consented
Actor	GC claimant
Evidence	workspace consent version
Idempotency key	claim:{id}:consent:{version}
Invalid transitions	Stale/missing consent
Audit event	claim_consented
Retry	Idempotent
Recovery	Display current consent

consented

From	consented
To	claimed
Actor	claim service
Evidence	authorized low-risk workspace membership transaction
Idempotency key	claim:{id}:complete
Invalid transitions	Case membership requested or conflicting claimant
Audit event	workspace_claimed
Retry	Return same membership
Recovery	Transaction rollback/forward repair

issued|pending_identity|identity_bound|consented|mismatch_pending

From	issued|pending_identity|identity_bound|consented|mismatch_pending
To	revoked|expired
Actor	issuer/system
Evidence	revocation or TTL
Idempotency key	claim:{id}:terminal:{event}
Invalid transitions	No terminal overwrite
Audit event	claim_revoked|claim_expired
Retry	Idempotent
Recovery	Fresh claim only

Paid-case membership invitation (`case_invitation`)

Owner: natalyx-app. States: `issued`, `pending_identity`, `identity_bound`, `mismatch_pending`, `issuer_confirmed`, `consented`, `redeemed`, `declined`, `revoked`, `expired`, `superseded`.

none

From	none
To	issued
Actor	authorized IP case member
Evidence	enrollment_pending/active case + recipient binding
Idempotency key	case-invite:{case}:{role}:{generation}
Invalid transitions	Self/member/active-GC/unauthorized
Audit event	case_invite_issued
Retry	Return current; supersede prior live GC invite
Recovery	Issue corrected invite

issued

From	issued
To	pending_identity
Actor	recipient
Evidence	valid addressed token landing
Idempotency key	case-invite:{id}:land
Invalid transitions	Expired/revoked
Audit event	case_invite_opened
Retry	Safe repeat
Recovery	Authenticate with return_to

pending_identity

From	pending_identity
To	identity_bound|mismatch_pending
Actor	auth provider
Evidence	verified subject/email
Idempotency key	case-invite:{id}:bind:{subject}
Invalid transitions	Terminal/duplicate/different subject
Audit event	case_invite_bound|mismatch
Retry	CAS protects terminal
Recovery	Issuer confirmation or corrected invite

mismatch_pending

From	mismatch_pending
To	issuer_confirmed
Actor	authorized issuer
Evidence	explicit reviewed mismatch confirmation
Idempotency key	case-invite:{id}:confirm:{issuer}
Invalid transitions	Issuer unauthorized/recipient changed
Audit event	case_invite_mismatch_confirmed
Retry	CAS no-op
Recovery	Revoke and reissue

identity_bound|issuer_confirmed

From	identity_bound|issuer_confirmed
To	consented
Actor	invitee
Evidence	paid-case role/privacy consent version
Idempotency key	case-invite:{id}:consent:{version}
Invalid transitions	Stale/missing consent
Audit event	case_invite_consented
Retry	Idempotent
Recovery	Present current consent

consented

From	consented
To	redeemed
Actor	invitation service
Evidence	single transaction creates person/party/membership
Idempotency key	case-invite:{id}:redeem
Invalid transitions	Existing member/active GC/expired/revoked
Audit event	case_invite_redeemed
Retry	Return same membership
Recovery	Rollback entire transaction; forward repair from invite

issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented

From	issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented
To	declined|revoked|expired|superseded
Actor	invitee/issuer/system
Evidence	explicit decline/revoke/TTL/new generation
Idempotency key	case-invite:{id}:terminal:{event}
Invalid transitions	No terminal overwrite
Audit event	case_invite_terminal
Retry	Idempotent CAS
Recovery	Fresh invitation only

Document lifecycle (`document + document_version + upload`)

Owner: natalyx-app; storage in natalyx-infra. States: `upload_requested`, `uploading`, `quarantined`, `scanning`, `available`, `review`, `approved`, `replaced`, `deleted`, `failed`.

none

From	none
To	upload_requested
Actor	authorized case member
Evidence	document purpose/type/size intent
Idempotency key	upload:{case}:{client_request}
Invalid transitions	Unauthorized/type/size/retention block
Audit event	upload_requested
Retry	Return same intent
Recovery	Expire/orphan intent

upload_requested

From	upload_requested
To	uploading
Actor	client/storage service
Evidence	authorized upload starts
Idempotency key	upload:{id}:start
Invalid transitions	Expired intent/wrong key
Audit event	upload_started
Retry	Multipart resume
Recovery	Abort and mark failed/orphan cleanup

uploading

From	uploading
To	quarantined
Actor	finalize service
Evidence	object exists + size/checksum/content sniff
Idempotency key	upload:{id}:finalize:{checksum}
Invalid transitions	Mismatch/missing object
Audit event	upload_quarantined
Retry	Repeat verify
Recovery	Delete bad object; request new upload

quarantined

From	quarantined
To	scanning
Actor	scan worker
Evidence	durable queued task
Idempotency key	scan:{version}:{engine_version}
Invalid transitions	Already available/deleted
Audit event	scan_started
Retry	Leased retry
Recovery	Lease expiry requeues after ECS replacement

scanning

From	scanning
To	available|failed
Actor	scan service
Evidence	signed clean/malicious/error result
Idempotency key	scan:{version}:{result_id}
Invalid transitions	Unknown result/engine
Audit event	scan_completed|scan_failed
Retry	Inbox replay no-op
Recovery	Remain quarantined on timeout/error; security review

available

From	available
To	review|approved
Actor	authorized participant/reviewer
Evidence	purpose-specific review/evidence
Idempotency key	document:{version}:review:{event}
Invalid transitions	Wrong role/stale version
Audit event	document_reviewed
Retry	Idempotent
Recovery	New review revision

available|review|approved

From	available|review|approved
To	replaced
Actor	authorized owner
Evidence	new clean version linked
Idempotency key	document:{id}:replace:{new_version}
Invalid transitions	New version not available
Audit event	document_replaced
Retry	Return current version
Recovery	Keep prior per retention; roll current pointer forward

available|review|approved|replaced|failed

From	available|review|approved|replaced|failed
To	deleted
Actor	authorized policy job/operator
Evidence	retention/delete request and no hold
Idempotency key	document:{version}:delete:{request}
Invalid transitions	Legal hold/provider obligation
Audit event	document_deleted
Retry	Tombstone idempotent
Recovery	Object-delete retry/orphan alarm; metadata/audit retained

Programme gates

Gate A0 — Infrastructure recovery and known-revision health

Field	Contract
Prerequisite PRs/gates	P01-01, P02-02, P13-01, P13-02
Prerequisite decisions	D-MIG-01, D-OPS-01
Responsible owner	Alan
Verifier	Luke

Acceptance matrix:

- □ CloudFormation/ECS/ALB state is captured from refreshed dev credentials and is stable.
- □ The running app and intelligence image SHAs/task definitions are known and map to reviewed commits.
- □ Liveness and readiness are both healthy; PostgreSQL, migrations and required private intelligence dependencies pass their bounded checks.
- □ A synthetic unauthenticated health request and authenticated minimal route succeed without PII/PHI.
- □ Rollback or forward-repair path is rehearsed and the prior compatible revision is identified.

Required evidence:

- AWS account/region and stack/service snapshots
- ALB status and target health
- image SHA/task definition
- readiness JSON/log correlation
- CI/PostgreSQL artifacts
- rollback rehearsal

Failure and rollback response. Do not proceed to participant validation. Freeze live updates, retain evidence, restore the known compatible image/schema or forward-repair under the approved runbook, and keep the Project gate blocked.

Gate A1 — Synthetic core participant journey

Field	Contract
Prerequisite PRs/gates	P02-04, P03-01, P03-02, P03-04, P04-01, P04-02, P04-03, P05-01, P05-02, P05-03, P11-03, P13-03
Prerequisite decisions	D-AUTH-01, D-AUTH-02, D-DATA-01, D-DATA-02, D-DATA-03
Responsible owner	Alan
Verifier	Luke

Acceptance matrix:

- □ A new synthetic IP signs up, is deterministically routed, enters Node 0 Free and no paid case is created.

- □ Candidate/outreach data remains IP-attributed and provisional; no score or answer becomes GC/provider truth.
- □ A synthetic GC begins a low-risk claim, binds identity/consent and sees only the allowed workspace projection.
- □ Cross-role negative tests prove class-2 IP assertions and class-4 GC-private answers do not leak.
- □ Both roles log out/in/resume and complete the mobile journey at tested widths.
- □ Known revision, audit events, correlation IDs and sanitized evidence are attached to the gate item.

Required evidence:

- A1 synthetic run artifact
- separate IP/GC subject IDs (synthetic only)
- authorization matrix results
- mobile screenshots/accessibility output
- audit event trace
- database invariant queries

Failure and rollback response. Stop before Complete, commercial or paid-case validation. Revert/disable the failing participant feature flag or deploy the known A0 revision; preserve synthetic data only for defect diagnosis, then purge under the runbook.

Gate A2 — Synthetic commercial, promotion, billing and document journey

Field	Contract
Prerequisite PRs/gates	P03-03, P06-01, P06-02, P06-03, P07-01, P07-02, P07-03, P07-04, P08-01, P08-02, P08-03, P08-04, P09-01, P09-02, P09-03, P09-04, P09-05, P10-01, P10-02, P10-03, P10-04, P11-01, P11-02, P11-04, P12-01, P12-02, P13-04
Prerequisite decisions	D-COM-01, D-COM-02, D-COM-03, D-COM-04, D-PAY-01, D-COM-05, D-COM-06, D-COM-07, D-COM-08, D-DOC-01, D-DOC-02, D-PROV-01, D-AI-01
Responsible owner	Alan
Verifier	Luke

Acceptance matrix:

- □ Synthetic Complete entitlement does not itself create a case and records its commercial configuration version/credit.
- □ Only byos.workspace_promotion_authorized.v1 authorizes exactly one enrollment_pending case; callbacks/redirects/replays create none/duplicates.
- □ Recipient-bound paid-case invitation and consent grant correct membership; activation waits for approved evidence.
- □ Enrollment/activation/milestone/credit/refund ledgers reconcile, including duplicate and out-of-order provider events.
- □ Synthetic files follow upload/quarantine/scan/available/download/replacement/revocation/recovery and never become public.
- □ Provider packet contains only purpose-allowed consented sources; intelligence scopes and proposals do not upgrade truth.

Required evidence:

- A2 synthetic run artifact
- processor simulator event ledger
- promotion idempotency query
- invoice/refund explanations
- S3/KMS/IAM/quarantine assertions
- provider snapshot checksum
- Ask/proposal safety evals

Failure and rollback response. Disable commercial/document/provider features, retain free/safety content, prevent live money or real documents, reconcile synthetic ledger inconsistencies, and restore the A1-known revision or forward-repair as approved.

Gate B1 — Real-user and PII beta readiness

Field	Contract
Prerequisite PRs/gates	A2, P12-03, P13-07
Prerequisite decisions	D-PHI-01, D-PHI-02
Responsible owner	Luke
Verifier	External/TBD

Acceptance matrix:

- □ A written B1 data-classification boundary permits named PII and expressly prohibits PHI/clinical documents.
- □ Named beta users, support/on-call, access review, incident response, backup/restore, retention/deletion and complaint routes are ready.
- □ Security/privacy review has no unresolved critical finding; vendors handling PII are approved.
- □ A2 evidence is current for the exact candidate revision and rollback/feature-disable controls are operational.
- □ Consent/privacy/terms copy is approved and data-subject correction/deletion workflows are testable.

Required evidence:

- Signed B1 checklist
- data inventory/classification
- access review
- incident tabletop
- backup/restore result
- vendor register
- current A2 bundle

Failure and rollback response. No real-user invitation. Continue synthetic-only, close control gaps, invalidate stale evidence and re-run the exact affected reviews/tests before a new go/no-go.

Gate B2 — PHI readiness and formal go/no-go

Field	Contract
Prerequisite PRs/gates	B1, P13-05, P13-06, P13-07
Prerequisite decisions	D-PHI-01, D-PHI-02
Responsible owner	Luke
Verifier	External security/privacy/compliance reviewers

Acceptance matrix:

- □ Formal PHI data flows, risk assessment, policies, BAAs/vendor controls, least privilege, audit/log retention, encryption/key management, incident/breach response and backup/restore are approved.
- □ Application, infrastructure, document, provider and intelligence threat models have current evidence and no unresolved critical/high risk without accepted exception.
- □ Operational staff training, access provisioning/deprovisioning, monitoring, support and emergency stop are complete.
- □ The exact release passed A2 and B1 revalidation; a signed human go/no-go records scope, date, evidence versions and approvers.
- □ The gate states explicitly that testing/deployment alone did not establish HIPAA/PHI readiness.

Required evidence:

- Signed go/no-go
- risk/security/privacy assessments
- BAA/vendor evidence
- access/log/incident/backup controls

- penetration/security testing
- training/operations evidence
- current technical gate bundles

Failure and rollback response. PHI remains prohibited. Keep B1 PII scope only if still independently valid; disable any PHI-capable upload/context path, remediate, and repeat formal review.

GitHub Project operating model

Project: MVP Product Flow Reshape.

Pre-change inspection

Direct GraphQL inspection found Project 3 open with only GitHub's default Title, Assignees, Status, Labels, Linked PRs, Milestone, Repository, Reviewers, Parent issue, Sub-issues progress, Created, Updated and Closed fields, and zero items. The programme sync therefore does not duplicate prior programme items.

Operating model

Convention	Rule
Project	MVP Product Flow Reshape
Number	3
Status	Reuse GitHub's Todo / In Progress / Done field for execution state.
Hierarchy	Draft items cannot be true GitHub sub-issues. Work-package parents are identifiable draft epics; child PR/decision/gate items link by Work package and Dependencies fields and exact IDs. Convert only when real implementation begins.
Readiness	Readiness is independent from Status. A Todo item may be blocked, awaiting reconciliation, plan-ready or ready to implement.
Assignment	Alan is a proposed implementer pending his review/sign-off; Project fields describe the recommendation, not an imposed assignment.
Reviewer Field	GitHub reserves the literal custom-field name Reviewer, so Programme reviewer holds the requested planning reviewer role without duplicating the built-in Reviewers metadata.
Source Of Truth	programme_data.py is the package generator source; rendered Markdown is review material; the Project is operational control plane after sync. Changes must update data, regenerate, validate and re-sync together.

Fields

Field	Type	Options / use
Item type	single	Work package, Proposed PR, Existing PR, Existing issue, Decision, Review checkpoint, Programme gate, Cleanup task
Work package	single	Programme, WP00, WP01, WP02, WP03, WP04, WP05, WP06, WP07, WP08, WP09, WP10, WP11, WP12, WP13
Target repository	single	.github, natalyx-app, natalyx-infra, natalyx-intelligence, natalyx-learn-resources, natalyx-design-system, Multiple / external
Proposed PR ID	text	Free text
DRI	single	Luke, Alan, Joint, External / TBD, N/A
Implementer	single	Luke, Alan, Joint, External / TBD, N/A
Programme reviewer	single	Luke, Alan, Joint, External / TBD, N/A
Final approver	single	Luke, Alan, Joint, External / TBD, N/A
Dependencies	text	Free text
Programme gate	single	Pre-A0, A0, A1, A2, B1, B2, Post-B2, N/A
Size	single	Small, Medium, Large, Exceptional, N/A
Blocking decision	text	Free text
Source / salvage PR	text	Free text
Plan document path	text	Free text
Readiness	single	Needs plan approval, Awaiting reconciliation, Blocked by decision, Dependencies unmet, Ready to implement, In implementation, Verification, Complete, Retained for salvage
Disposition	single	Programme baseline, Proposed, Retain pending review, Direct reuse, Selective port, Rewrite, Supersede after extraction, Defer, Gate criterion, Decision needed, Checkpoint, Cleanup only

GitHub's existing Status field is reused; no duplicate status field is created. Custom fields are the minimum set needed to hold the fields explicitly required by the programme brief for draft items as well as real PRs/issues.

Item catalogue

Item type	Expected count	Naming convention
Work packages	14	[WPxx] title
Proposed PRs	50	[Pxx-yy] title
Existing PRs	9	Actual linked PR
Existing issues	2	Actual linked issue
Decisions	22	[DECISION D-*] title
Review checkpoints	1	[CHECKPOINT] title
Programme gates	5	[GATE Ax] title
Cleanup tasks	3	[CLEANUP C-*] title

Live post-sync validation

Check	Verified result
Validated At	2026-07-10 21:59 America/New_York
Project Id	PVT_kwDOEAWZDM4BYbai
Items	106
Draft Items	95
Linked Items	11
Total Fields	29
Custom Fields	16
Statuses	All 106 items are Todo
Required Values	Complete on every item; Proposed PR ID complete on all 50 proposed PR items
Type Counts	14 WP; 50 proposed PR; 9 existing PR; 2 existing issue; 22 decision; 1 checkpoint; 5 gate; 3 cleanup
Result	Passed: exact counts, unique stable IDs/URLs, no missing fields or links

The synchronizer read these values back from Project 3 after the final update; it did not infer them from the local catalogue. All items remain planning-only and no proposed PR is marked `Ready to implement`.

Dependency and hierarchy representation

Draft items cannot be true GitHub sub-issues. Work-package parents are identifiable draft epics; child PR/decision/gate items link by Work package and Dependencies fields and exact IDs. Convert only when real implementation begins.

Each proposed PR carries a Work package value and exact Dependencies text. Work-package draft bodies list child PR IDs, decisions, gates and plan path. Decision/gate/checkpoint bodies list what they block. This provides deterministic hierarchy without creating empty issues/PRs.

Status, readiness and disposition

Dimension	Meaning
Status	Todo / In Progress / Done: execution state only.
Readiness	Whether plan/decision/dependency/reconciliation conditions permit implementation.
Disposition	How existing/proposed/decision/gate/cleanup material should be treated.
Ownership	DRI/implementer/reviewer/approver recommendation; Alan remains proposed until sign-off.

Project sync and maintenance

Dry-run/read current state:

```
/Users/lukerhodes/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 scripts/sync_proje
ct.py
```

Apply idempotent updates:

```
/Users/lukerhodes/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 scripts/sync_proje
ct.py --apply
```

After any canonical change, regenerate Markdown, validate, re-render the PDF, then re-run the idempotent Project sync. Do not hand-edit IDs/titles/dependencies in only one surface.

Final Project validation checklist

- □ One identifiable item for every WP00-WP13.
- □ One proposed-PR draft item for all 50 PR IDs.
- □ Actual linked PRs #112/#117/#119/#121/#122/#123/#124 and DS #7/#8.
- □ Actual linked Learn #3 and Intelligence #14 dependency issues.

- □ Alan reconciliation checkpoint present and blocking named replacement work.
- □ Every blocking decision and A0/A1/A2/B1/B2 gate present.
- □ Ownership, dependencies, readiness, plan path, source/salvage and disposition populated.
- □ No duplicate titles/IDs and Project/document parity validator passes.

Ownership, delegation and review

Work package	DRI	Implementer	Reviewer	Approver	Split
WP00	Luke	Luke (proposed)	Alan	Luke	Programme baseline; Alan sign-off precedes mandatory assignments.
WP01	Alan	Alan (proposed)	Luke	Luke	Technical evidence foundation.
WP02	Alan	Alan (proposed)	Luke + higher-capability	Luke	Alan proposes framework; Luke approves compatibility/risk.
WP03	Alan	Alan (proposed)	Luke + security	Luke	Luke decides admission/access policy.
WP04	Alan	Alan (proposed)	Luke + privacy	Luke	Joint domain/provenance design.
WP05	Alan	Alan (proposed)	Luke	Luke	Luke owns Free boundary and sensitive question/score policy.
WP06	Alan	Alan (proposed)	Luke + privacy	Luke	Luke owns cross-role projection and Complete contract.
WP07	Luke	Alan (proposed)	Luke + finance/legal	Luke	Luke decides commercial semantics; Alan implements event/state contract.
WP08	Luke	Alan (proposed)	payment/security/finance	Luke	Mandatory external review before real money.
WP09	Alan	Alan (proposed)	Luke + security/privacy/provider	Luke	Luke approves retention and provider purpose.
WP10	Alan	Alan (proposed)	Luke + AI safety/privacy	Luke	Luke approves truth/scope policy.
WP11	Alan	Alan (proposed)	Luke + accessibility/content	Luke	Separate repo releases; aesthetic purity is not universal gate.
WP12	Alan	Alan (proposed)	Luke + accessibility	Luke	Luke owns product acceptance; participant surfaces only.
WP13	Joint	Alan (proposed)	Luke + external security/privacy/compliance	Luke	Alan verifies technical gates; Luke accepts; B1/B2 need independent review.

Checkpoint	When	Owner	Evidence
Architecture review	Before any package PR plan is marked ready	Luke	Approved entities/contracts/decision items
PR-plan approval	Before implementation branch	Luke + proposed implementer	Scope, non-goals, dependencies, tests, rollout/repair
Alan PR reconciliation	Before replacement auth/Node0/case surfaces	Alan; Luke accepts	CHECK-ALAN-122-124 complete
Pre-merge review	Every PR	Named reviewer/approver	CI, objective acceptance, risk review
Cloud verification	Gate-related PRs	Alan	Known revision and redacted live/synthetic artifacts
Gate acceptance	A0/A1/A2/B1/B2	Luke	Gate matrix; external sign-off where required

Delegation rule: Luke decides product/lifecycle/commercial/privacy/PHI/provider boundaries and accepts gates; Alan proposes implementation and owns technical delivery/validation after sign-off. Joint planning is mandatory for migration, invitation admission,

authorship/projection, promotion, activation, document plane and B1/B2. External review is mandatory for payment/refund, auth security, document/PHI, provider sharing and AI truth boundaries.

Existing-PR extraction and Alan checkpoint

> Read-only reconciliation record. No PR, branch or worktree may be closed, retargeted, rebased, cleaned or deleted from this ledger alone.

Inspection snapshot: 2026-07-10 20:12 America/New_York.

Preservation and extraction sequence

1. Preserve dirty worktrees and identify active branches/PRs.
2. Reinspect latest heads.
3. Complete this extraction ledger.
4. Reconcile Alan's findings.
5. Approve replacement architecture/PR boundaries.
6. Create/ready Project draft items.
7. Open replacement drafts only when implementation starts.
8. Extract/port/rewrite value.
9. Close superseded PRs only after accounting.
10. Delete branches/worktrees only after closure and team confirmation.

Summary

PR	Latest head	Useful capability	Destination	Status	Disposition
app#112	9605d88e31bd7ea5ccb0ddc524c5f2412efb242e	Source-first 81-node/12-cluster surface map, participant-renderer boundaries, wave plan and cross-repo seams.	P00-01, P09-04, P11-03, P11-04, P12-03	Catalogued; content remains to be reconciled into replacement PR acceptance tests.	Selective port, then supersede after extraction
app#117	5ac09f0cdd9299667fd70e845e606b1892f98ece	Deterministic candidate questionnaire, scoring/status calculation, conversation-guide resource, routes and extensive Node 0 tests.	P04-01, P04-02, P05-02, P05-03	Unextracted; algorithm/test review required after Alan checkpoint and Luke field/score decision.	Selective port and rewrite
app#119	a8cd2f428535807749 0f02089ac8f9e6f1e8bed3	Concise product-leverage framing and boundaries for the MVP flow.	P00-01, P04-01, P05-01, P06-01, P10-01, P12-03	Catalogued; reconcile terminology into approved programme.	Retain as source or merge separately after plan decision
app#121	529d377e23376268b44c95ba9974aa0cd133275f	Primary Node 0 participant model/UI, PDF/guide presentation and focused surface tests.	P05-03, P11-03, P12-02	Unextracted; #124 may contain later copies but equivalence must be proved by file/commit comparison.	Selective UI/test port, then supersede
app#122	e85382150853cf4774948c55b91e330e7e69617e	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P01-01, P02-03, P03-01, P03-03, P04-02, P09-04, P10-02, P11-03, P12-02	Reinspected at latest head. It is exactly e853821 (no newer delta); retention strengthened, but final extraction waits for Alan checkpoint.	Retain pending Alan reconciliation; selective port or rebase only after architecture approval
app#123	34078f810b634aa6ddce5954763c03cb46b574d8	GC onboarding form/copy, hosted auth return, invite handoff and role-specific surface work.	P03-01, P03-02, P06-02, P06-03	Unextracted; likely partially copied into #124 but not proven complete.	Selective UX/test port and rewrite
app#124	84a5a847fea71213b6257549e5a620f654836cb1	Current consolidation of Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitations, DS assets and large focused test suites.	P04-01, P04-02, P04-03, P05-02, P05-03, P06-02, P06-03, P09-04, P10-02, P11-04, P12-02	Unextracted; Alan review explicitly incomplete and unresolved.	Retain unchanged as evidence; selective port/rewrite after checkpoint
ds#7	cedfcc489b1f60842aed3a9701348c85964cfd96	Documents an escaping/attrs hardening follow-up relevant to participant recipe security.	P11-03	Direct-reuse candidate; not yet merged.	Retain for direct review/merge or fold into P11-03
ds#8	4656426e493e243785f8490589bf44fc0e05b8d6	Consumer parity and bounded app-owned layout guidance for Node 0/case/mobile adoption.	P11-03, P12-02	Direct-reuse candidate; local dirty asset delta separately protected.	Prefer direct review/merge if still correct

Detailed PR records

app#112 — BYOS journey surface migration: surface map + workplan (spec handoff)

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-app/pull/112
Latest inspected head	9605d88e31bd7ea5ccb0ddc524c5f2412efb242e
Inspection time	2026-07-11T00:59:11Z (2026-07-10 20:59:11 America/New_York)
Useful capability	Source-first 81-node/12-cluster surface map, participant-renderer boundaries, wave plan and cross-repo seams.
Reusable commits/files	docs/plans/byos-journey-surface-map.md; docs/plans/byos-journey-surface-workplan.md; single documentation commit 9605d88.
Tests worth preserving	Docs-only; preserve its stated acceptance/test map, not CI results as product evidence.
Obsolete/unsafe assumptions	Predates the explicit free-workspace/paid-case/commercial/provenance split and cannot be the implementation sequence by itself.
Replacement destination	P00-01, P09-04, P11-03, P11-04, P12-03
Extraction status	Catalogued; content remains to be reconciled into replacement PR acceptance tests.
Outstanding reviewer input	No GitHub review recorded.
Recommended disposition	Selective port, then supersede after extraction
Exact close condition	Every unique architecture/product decision is either adopted in the programme or rejected with rationale; Alan confirms no needed review input remains; replacement plan is approved.
Exact branch/worktree delete condition	PR is closed, commit is reachable from retained history or archived reference, no attached dirty worktree/unique commit remains, and Luke/Alan explicitly confirm cleanup.

app#117 — Node 0 beta: Outreach Readiness Quiz and deterministic Conversation Guide

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-app/pull/117
Latest inspected head	5ac09f0cdd9299667fd70e845e606b1892f98ece
Inspection time	2026-07-11T00:59:11Z (2026-07-10 20:59:11 America/New_York)
Useful capability	Deterministic candidate questionnaire, scoring/status calculation, conversation-guide resource, routes and extensive Node 0 tests.
Reusable commits/files	app/node0.py; app/node0_conversation_guide.py; resource JSON; focused tests; only the non-ranking/safety-safe algorithms that pass D-DATA-03.
Tests worth preserving	tests/python/test_node0.py contains useful deterministic and authorization cases.
Obsolete/unsafe assumptions	Large mixed UI/style/schema PR; case-shaped persistence, 0-100 ranking and unverified third-party answers conflict with workspace/provenance contract.
Replacement destination	P04-01, P04-02, P05-02, P05-03
Extraction status	Unextracted; algorithm/test review required after Alan checkpoint and Luke field/score decision.
Outstanding reviewer input	No GitHub review recorded.
Recommended disposition	Selective port and rewrite
Exact close condition	Approved questionnaire/guide behavior and every valuable test are mapped to P05-02/P05-03 or deliberately rejected; replacement PRs contain provenance and Free-tier contract.
Exact branch/worktree delete condition	Close conditions met, stacked PRs #121/#123 are reconciled, branch has no unique retained worktree content, and team confirms deletion.

app#119 — Docs: align MVP app flows to process leverage points

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-app/pull/119
Latest inspected head	a8cd2f4285358077490f02089ac8f9e6f1e8bed3
Inspection time	2026-07-11T00:59:11Z (2026-07-10 20:59:11 America/New_York)
Useful capability	Concise product-leverage framing and boundaries for the MVP flow.
Reusable commits/files	docs/plans/mvp-process-leverage-alignment.md and its two documentation commits.
Tests worth preserving	Docs-only; no executable behavior.
Obsolete/unsafe assumptions	Does not define implementation state machines, funding, provenance or gate evidence; title similarity must not make it the programme authority.
Replacement destination	P00-01, P04-01, P05-01, P06-01, P10-01, P12-03
Extraction status	Catalogued; reconcile terminology into approved programme.
Outstanding reviewer input	No GitHub review recorded.
Recommended disposition	Retain as source or merge separately after plan decision
Exact close condition	Luke decides whether the source document remains useful independently; if superseded, every unique principle is cited/adopted or rejected in the approved programme.
Exact branch/worktree delete condition	Only after close/merge, source commit remains reachable, no worktree depends on branch and team confirms.

app#121 — Node 0 beta: primary participant surface

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-app/pull/121
Latest inspected head	529d377e23376268b44c95ba9974aa0cd133275f
Inspection time	2026-07-11T00:59:11Z (2026-07-10 20:59:11 America/New_York)
Useful capability	Primary Node 0 participant model/UI, PDF/guide presentation and focused surface tests.
Reusable commits/files	app/node0_surface.py; node0_pdf.py; route/template/asset/test slices after comparison with #124.
Tests worth preserving	Node 0 surface/rendering tests worth porting to P05-03.
Obsolete/unsafe assumptions	Stacked on #117 rather than main, inherits case-shaped candidate model, and includes broad Wave 0/DS content already landed through #115; only tip 529d377 is genuinely unique. It predates the authorship/workspace split.
Replacement destination	P05-03, P11-03, P12-02
Extraction status	Unextracted; #124 may contain later copies but equivalence must be proved by file/commit comparison.
Outstanding reviewer input	No GitHub review recorded.
Recommended disposition	Selective UI/test port, then supersede
Exact close condition	All unique surface/PDF/test value is proven present in an approved replacement or explicitly rejected; stacked dependency is no longer needed.
Exact branch/worktree delete condition	After #117/#123/#124 reconciliation, PR closure, no dirty worktree/unique commits, and team confirmation.

app#122 — Wave 1 participant case shell surfaces

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-app/pull/122
Latest inspected head	e85382150853cf4774948c55b91e330e7e69617e
Inspection time	2026-07-11T00:59:11Z (2026-07-10 20:59:11 America/New_York)
Useful capability	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.
Reusable commits/files	Invitation commits/files and tests; case_journey/case_home presenter and paired-lens/mobile tests; .github/workflows/python-tests.yml; DS vendoring/escaping contract where still applicable.
Tests worth preserving	Reported 499-test suite; green unittest/build/contract checks observed at e853821. Preserve concurrency, logged-out invite, paired IP/GC, Learn timeout and render tests.
Obsolete/unsafe assumptions	No Node 0 workspace target, payment/promotion or separate claim model; facts/proposals lack strong authorship/approval authority; generic invitation status setter remains; case lifecycle is not the whole programme state model.
Replacement destination	P01-01, P02-03, P03-01, P03-03, P04-02, P09-04, P10-02, P11-03, P12-02
Extraction status	Reinspected at latest head. It is exactly e853821 (no newer delta); retention strengthened, but final extraction waits for Alan checkpoint.
Outstanding reviewer input	Alan's 2026-07-10 review-round comment records fixes from earlier 8df63dd to e853821. No formal GitHub review object; current planning must not imply all architecture is approved.
Recommended disposition	Retain pending Alan reconciliation; selective port or rebase only after architecture approval
Exact close condition	CHECK-ALAN-122-124 is complete; every fix/finding/test has a destination; replacement contracts cover workspace/case/provenance; the relevant replacements have merged or the team explicitly elects to merge a narrowed #122.
Exact branch/worktree delete condition	Only after final disposition/closure, all commits/tests are reachable or intentionally discarded, no worktree depends on branch, and Alan/Luke confirm cleanup.

app#123 — Node 0 beta: GC onboarding and hosted handoff

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-app/pull/123
Latest inspected head	34078f810b634aa6ddce5954763c03cb46b574d8
Inspection time	2026-07-11T00:59:11Z (2026-07-10 20:59:11 America/New_York)
Useful capability	GC onboarding form/copy, hosted auth return, invite handoff and role-specific surface work.
Reusable commits/files	Onboarding templates/routes/tests after field-level privacy and claim separation review.
Tests worth preserving	Preserve hosted-return, invalid/decline and GC lens interaction cases after comparison with #124/dirty onboarding branch.
Obsolete/unsafe assumptions	Stacked on #121, predates PR #122 hardening, and conflates soft onboarding with case-shaped data/membership; GC answers do not revise IP assertions. It also mixes participant work with a large internal/operator redesign and vendored internal DS, which must stay out of this participant programme.
Replacement destination	P03-01, P03-02, P06-02, P06-03
Extraction status	Unextracted; likely partially copied into #124 but not proven complete.
Outstanding reviewer input	No GitHub review recorded.
Recommended disposition	Selective UX/test port and rewrite
Exact close condition	Hosted return and onboarding value is mapped to claim-aware, attributable GC replacement PRs and no unique behavior remains.
Exact branch/worktree delete condition	After stacked PR/worktree reconciliation, closure, retained commits/tests are reachable, and team confirms.

app#124 — Consolidate BYOS Node 0 handoff

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-app/pull/124
Latest inspected head	84a5a847fea71213b6257549e5a620f654836cb1
Inspection time	2026-07-11T00:59:11Z (2026-07-10 20:59:11 America/New_York)
Useful capability	Current consolidation of Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitations, DS assets and large focused test suites.
Reusable commits/files	Deterministic guide/resource/tests, selected Node 0/GC/case presentation, source-aware Learn dependency notes, subject to file-level extraction.
Tests worth preserving	test_node0.py and test_case_home.py contain broad behavior/render coverage; green build/contract-smoke observed, but no main Python/PostgreSQL proof.
Obsolete/unsafe assumptions	Node 0 Free creates a synthetic canonical case; unaddressed link can lead to case membership; IP third-party assertions drive score/invite; notes become shared memory; no commercial event, payment ledger or promotion. It contains older #122 equivalents but not the later hardening commits through e853821; its checks omit the unittest workflow.
Replacement destination	P04-01, P04-02, P04-03, P05-02, P05-03, P06-02, P06-03, P09-04, P10-02, P11-04, P12-02
Extraction status	Unextracted; Alan review explicitly incomplete and unresolved.
Outstanding reviewer input	Alan's review is explicitly incomplete and unresolved (halfway through per team update). No completion is assumed; CHECK-ALAN-122-124 blocks final disposition and replacement work.
Recommended disposition	Retain unchanged as evidence; selective port/rewrite after checkpoint
Exact close condition	Alan completes and reconciles findings; every valuable commit/file/test has a destination; replacement architecture is approved; extracted replacements cover value; Luke and Alan confirm supersession.
Exact branch/worktree delete condition	Only after closure, replacement merges/extraction verification, no protected worktree depends on branch and explicit team cleanup confirmation.

ds#7 — Document Jinja attrs hardening follow-up

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-design-system/pull/7
Latest inspected head	cedfcc489b1f60842aed3a9701348c85964cfd96
Inspection time	2026-07-11T00:59:11Z (2026-07-10 20:59:11 America/New_York)
Useful capability	Documents an escaping/attrs hardening follow-up relevant to participant recipe security.
Reusable commits/files	Documentation commit/contract as direct source for P11-03.
Tests worth preserving	Preserve attrs/escaping contract tests in the eventual DS release.
Obsolete/unsafe assumptions	Draft guidance alone does not prove consumer safety and should not block cloud recovery for aesthetic reasons.
Replacement destination	P11-03
Extraction status	Direct-reuse candidate; not yet merged.
Outstanding reviewer input	No recorded review in this inspection.
Recommended disposition	Retain for direct review/merge or fold into P11-03
Exact close condition	Merged as-is after review or content/tests are present in an approved DS replacement.
Exact branch/worktree delete condition	After merge/closure, no unique dirty asset changes on the branch and team confirmation.

ds#8 — Add consumer visual parity guidance

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-design-system/pull/8
Latest inspected head	4656426e493e243785f8490589bf44fc0e05b8d6
Inspection time	2026-07-11T00:59:11Z (2026-07-10 20:59:11 America/New_York)
Useful capability	Consumer parity and bounded app-owned layout guidance for Node 0/case/mobile adoption.
Reusable commits/files	Guidance and validation expectations for P11-03/P12-02.
Tests worth preserving	Use concrete responsive/a11y/security requirements; do not turn every token mismatch into a blocker.
Obsolete/unsafe assumptions	Local worktree contains extra dirty brand assets not represented by remote PR; do not overwrite or assume included.
Replacement destination	P11-03, P12-02
Extraction status	Direct-reuse candidate; local dirty asset delta separately protected.
Outstanding reviewer input	Open, not draft; no recorded formal review in this inspection.
Recommended disposition	Prefer direct review/merge if still correct
Exact close condition	Merged or equivalent guidance/tests are released and app pins that release.
Exact branch/worktree delete condition	After closure/merge and explicit confirmation that local dirty assets are committed elsewhere or intentionally discarded.

Latest PR #122 reinspection

Latest inspected head: `e85382150853cf4774948c55b91e330e7e69617e` at 2026-07-11T00:59:11Z (2026-07-10 20:59:11 America/New_York). The earlier reviewed workflow run/comment baseline was `8df63dd`; Alan's current head is `e853821`.

Fixed or materially improved since the earlier snapshot:

- Logged-out invite delivery/resume and validated internal return_to across login/signup/confirm/resend.
- Paired IP/GC lens semantics and render-time action labels.
- CAS redemption/expiry/mismatch transitions, self-invite refusal and one-live-GC constraints.
- 14-day expiry ceiling, portable integrity handling and duplicate-live-invite forward migration.
- Blocking Python CI workflow, fail-closed Learn approval, bounded Learn timeout, SQL-limited timeline and safer cookie/dead-button behavior.

Remaining conflicts/gaps:

- No distinct Node 0 workspace or workspace claim target; it cannot be stacked unchanged with #124.
- No commercial commitment/funding/promotion or separate entitlement/payment/document lifecycle.
- Generic fact/proposal/invitation rails still need explicit author/authority/evidence transition commands.
- The case projection is useful case-stage presentation, not a universal programme state machine.

Decomposition impact: retain more of #122's CI, invitation and case-surface tests/implementation, but do not collapse WP03/WP04/WP07/WP09. Workspace/case separation and Alan reconciliation remain blocking.

Alan reconciliation checkpoint

Field	Contract
Name	Alan review reconciliation: PRs #122 and #124
Status	Unresolved input
Blocks	P03-01, P03-02, P03-03, P03-04, P04-01, P04-02, P04-03, P05-01, P05-02, P05-03, P06-01, P06-02, P06-03, P09-04
May amend	WP01, WP03, WP04, WP05, WP06, WP09, WP10, WP11, WP12, WP13
Avoid wasted review	Do not ask Alan to polish code already known to be architecturally obsolete. Complete the safety/defect review at the contract/test level, identify reusable commits/tests, and stop once findings are captured. Obsolete code can be superseded while its findings become replacement acceptance tests.

Finding recording:

- Record each finding with PR/head SHA, file/line or diff anchor, severity, reproducible behavior, architecture/product implication, proposed acceptance test and destination PR ID.
- Mark whether the finding concerns reusable behavior, obsolete architecture, security/privacy, accessibility, migration, concurrency, test coverage or product semantics.
- Update this extraction ledger first; then update the canonical programme data, regenerate all plans/manifest and re-sync the Project so no source diverges.

Resolution rule:

- Evidence beats plan text: a verified defect or missing contract amends the package.
- Alan proposes technical alternatives and records a concise dissent where needed; Luke resolves product, lifecycle, privacy, commercial and cross-repository architecture decisions.
- Security/privacy/payment/PHI disagreements that exceed internal authority go to the named external reviewer; implementation remains blocked.

Complete only when:

- □ Alan confirms his #122 changes and unfinished #124 findings are fully captured.
- □ Every finding has a disposition and destination or explicit no-action rationale.
- □ Luke and Alan accept the workspace/case, invitation, authorship and PR-boundary contract, or an explicit unresolved decision remains blocked.
- □ No replacement implementation item is marked Ready to implement until both the checkpoint and its own dependencies/decisions are clear.

Protected local worktrees

Repository	Path	Branch/head	Protected content	Destination	Safety rule
app	`/Users/lukerhodes/Natalyx/natalyx-app`	feature/provider-exchange-layer-v0 / 84a5a84	substantial uncommitted provider handoff/ops/document work	P09-01/P09-03/P09-05 after provenance/security review	Do not stage, reset, stash, switch, commit, clean or overwrite.
app	`/Users/lukerhodes/.codex/worktrees/1b3f/natalyx-app`	feature/node0-surrogate-onboarding-link / 529d377	substantial uncommitted onboarding/invitation work	P03-02/P03-03/P06-02	Preserve until file/test extraction is signed off.
app	`/Users/lukerhodes/.codex/worktrees/39b2/natalyx-app`	detached candidate rewrite / 9605d88	substantial uncommitted Node 0 quiz rewrite	P04-02/P05-02	Preserve; compare with predecessor and #117/#124 before any discard.
app	`/Users/lukerhodes/.codex/worktrees/6233/natalyx-app`	detached predecessor candidate model / dc6a2f0	substantial uncommitted predecessor model	P04-02/P05-02 if unique	Preserve until rewrite diff/extraction ledger is complete.
learn	`/Users/lukerhodes/Natalyx/natalyx-learn-resources`	feature/learn-resource-rendering-contract / 9c9a9c2	substantial uncommitted builder/schema/templates/resource work	P11-01	Do not update/switch/clean; create a future isolated extraction worktree.
learn	`/Users/lukerhodes/.codex/worktrees/9dfc/natalyx-learn-resources`	feature/node0-beta-learn-resources / e1fb2d6	uncommitted Node 0 Learn implementation	P11-01 after provenance/no-fake-content review	Preserve until compared with rendering-contract branch.
intelligence	`/Users/lukerhodes/Natalyx/natalyx-intelligence`	main (behind origin/main) / 9d824e8	dirty Learn registry/corpus plus uv.lock	P11-02/P10-03 as appropriate	Do not pull/reset/stash; extract in isolated worktree later.
design-system	`/Users/lukerhodes/Natalyx/Natalyx Design System`	feature/ds-consumer-parity-guidance / 4656426	dirty brand assets beyond PR #8	P11-03 after asset review	Remote PR and local asset delta are separate evidence; do not overwrite.
internal-design-system	`/Users/lukerhodes/Natalyx/Natalyx Internal View Design System`	main / current local	substantial uncommitted operator shell/design-system work	Not part of participant DS; review only if P09-05 operator UI needs approved internal recipes	Preserve repo boundary and dirty worktree.

Cleanup tasks

ID	Task	Dependencies	Owner
CLEAN-01	Verify extraction completeness before any PR closure	CHECK-ALAN-122-124	Joint
CLEAN-02	Close superseded PRs only after replacement evidence is reachable	CLEAN-01	Luke
CLEAN-03	Delete remote branches/worktrees only after closure and team confirmation	CLEAN-02	Alan

Data authorship and provenance

Class definitions:

Class	Meaning
1	IP-owned observation or relationship context
2	Provisional IP-supplied candidate information
3	GC-authored or GC-confirmed information
4	GC-only private information
5	Provider- or reviewer-confirmed information

Pathfinder known-carrier/route answers

Element	Pathfinder known-carrier/route answers
Class	1
Subject	IP journey
Current author	IP
Storage	participants.intake_data and pathfinder_context
Provenance	participant settings; inconsistent per-field source
Visibility	IP/app routing; legacy reviewer surfaces
Verification	Not applicable as GC truth
Correction/replacement	IP appends/amends route-answer revision
GC sees it?	No by default
IP sees GC answer?	N/A
Downstream influence	Entry routing only
Promotion behavior	Route/source reference only
Recommended contract	Never create a GC fact, entitlement, membership or case.

IP preferences about a GC/journey (location, communication, involvement, reduction/twins, relationship)

Element	IP preferences about a GC/journey (location, communication, involvement, reduction/twins, relationship)
Class	1
Subject	IP preferences
Current author	IP
Storage	legacy participant intake_data
Provenance	participant-supplied, no field history
Visibility	IP; some selected axes in frozen matching UI
Verification	IP preference, not eligibility
Correction/replacement	IP revision
GC sees it?	Only an approved purpose-limited projection
IP sees GC answer?	Only separate GC preference projection
Downstream influence	Routing/compatibility discussion
Promotion behavior	As IP preference with author
Recommended contract	Race-based preference is separately blocked for Luke/external review.

Candidate nickname/name and pronouns

Element	Candidate nickname/name and pronouns
Class	2
Subject	Prospective GC
Current author	IP/co-IP
Storage	node0_candidates columns
Provenance	row creator only; no field history
Visibility	IP/co-IP only
Verification	Unverified
Correction/replacement	IP revision; GC identity later supersedes reference
GC sees it?	No current path
IP sees GC answer?	GC-authored identity after explicit share
Downstream influence	Candidate list/guide
Promotion behavior	Link selected candidate reference only
Recommended contract	Paid identity comes from authenticated GC/person record.

Relationship to prospective GC

Element	Relationship to prospective GC
Class	1
Subject	IP-GC relationship
Current author	IP
Storage	node0_candidates.responses_json q1_2
Provenance	Implicit IP
Visibility	IP private
Verification	Observation
Correction/replacement	IP revision; GC may provide separate relationship answer
GC sees it?	No by default
IP sees GC answer?	Approved relationship projection
Downstream influence	Outreach route
Promotion behavior	Optional IP-attributed context
Recommended contract	Conflicting answers coexist; neither silently overwrites.

Pregnancy history, family-complete, citizenship and age

Element	Pregnancy history, family-complete, citizenship and age
Class	2
Subject	Prospective GC
Current author	IP
Storage	responses_json q2_1-q2_4
Provenance	Implicit IP; copied to guide packet
Visibility	IP private
Verification	Unverified third-party assertion
Correction/replacement	GC class-3/4 revision; reviewer class-5 later
GC sees it?	Only if Luke approves labelled confirmation prompt
IP sees GC answer?	Only purpose-limited status
Downstream influence	Current score/status/invite gate
Promotion behavior	Exclude raw values by default
Recommended contract	Cannot gate account/claim/onboarding/eligibility from IP answer.

Gestational-health or systemic-health assertions

Element	Gestational-health or systemic-health assertions
Class	2
Subject	Prospective GC health
Current author	IP
Storage	responses_json q3_1-q3_2 and guide packet
Provenance	Implicit IP
Visibility	IP private
Verification	Unverified/sensitive
Correction/replacement	GC private self-report; provider evidence class 5
GC sees it?	Prefer remove before consent; otherwise labelled prompt
IP sees GC answer?	Only reviewed coordination outcome
Downstream influence	Current critical/status/guide
Promotion behavior	Do not promote raw assertion
Recommended contract	Remove or make optional/provisional; never provider material.

Substance, schedule, support and stability assertions

Element	Substance, schedule, support and stability assertions
Class	2
Subject	Prospective GC circumstances
Current author	IP
Storage	responses_json q4_1/q4_3/q4_4/q5_5
Provenance	Implicit IP
Visibility	IP private
Verification	Unverified
Correction/replacement	GC self-report revision
GC sees it?	No by default
IP sees GC answer?	Only explicit progress/status projection
Downstream influence	Current score/things-to-uncover
Promotion behavior	Exclude raw; optional private question list
Recommended contract	May inform low-pressure questions only.

Candidate state and derived proximity

Element	Candidate state and derived proximity
Class	2
Subject	Prospective GC location
Current author	IP/system derivative
Storage	responses_json q4_2 plus IP profile state
Provenance	IP and derived-from missing
Visibility	IP private
Verification	Unverified
Correction/replacement	GC class-3 state revision
GC sees it?	Optional labelled confirmation
IP sees GC answer?	GC-shared state as separate revision
Downstream influence	Score/route
Promotion behavior	GC-authored state only when permitted
Recommended contract	Derived value inherits weakest source status.

Worldview, communication, power, tested relationship and privacy observations

Element	Worldview, communication, power, tested relationship and privacy observations
Class	1
Subject	Relationship
Current author	IP
Storage	responses_json q5_1-q5_6
Provenance	Implicit IP
Visibility	IP private
Verification	Personal observation
Correction/replacement	IP revision; GC separate answer
GC sees it?	No automatic visibility
IP sees GC answer?	Only GC-approved projection
Downstream influence	Private outreach guide
Promotion behavior	Exclude unless expressly shared
Recommended contract	Suitable for private reflection, not eligibility.

GC spouse/household assertion

Element	GC spouse/household assertion
Class	2
Subject	GC/third party
Current author	IP
Storage	responses_json q5_7
Provenance	Implicit IP
Visibility	IP private
Verification	Unverified
Correction/replacement	GC private revision
GC sees it?	No default
IP sees GC answer?	Purpose-limited status only
Downstream influence	Conversation prompt
Promotion behavior	Exclude
Recommended contract	No provider/case truth.

Observed openness/signals

Element	Observed openness/signals
Class	1
Subject	IP-GC interaction
Current author	IP
Storage	responses_json q5_8
Provenance	Implicit IP
Visibility	IP private
Verification	Observation
Correction/replacement	IP revision
GC sees it?	No default
IP sees GC answer?	N/A
Downstream influence	Low-pressure outreach route
Promotion behavior	Exclude or retain IP-private
Recommended contract	Never imply GC consent/willingness.

Archetype, score, status, score suppression, things-to-uncover and guide route

Element	Archetype, score, status, score suppression, things-to-uncover and guide route
Class	2 derived
Subject	Prospective GC
Current author	System from IP input
Storage	node0_candidates derived columns/JSON
Provenance	No derived_from/source version
Visibility	IP private
Verification	Unverified derivative
Correction/replacement	Recompute from revision; retain old derivation
GC sees it?	No by default
IP sees GC answer?	N/A
Downstream influence	Ordering, readiness, guide and invite gating
Promotion behavior	Do not promote as truth
Recommended contract	Recommendation is remove ranking; use non-ranking questions-to-clarify.

Guide evidence/scaffold/snapshot/PDF

Element	Guide evidence/scaffold/snapshot/PDF
Class	2 derived
Subject	Outreach about GC
Current author	System for IP
Storage	candidate guide JSON/PDF
Provenance	Raw IP responses copied
Visibility	IP private
Verification	Unverified
Correction/replacement	New immutable version on source change
GC sees it?	No automatic visibility
IP sees GC answer?	N/A
Downstream influence	Conversation preparation/Ask
Promotion behavior	Exclude by default
Recommended contract	Workspace-private; never provider/canonical truth.

Candidate ordering/outreach state/decline

Element	Candidate ordering/outreach state/decline
Class	1/2
Subject	IP workflow and GC response
Current author	IP, GC or system
Storage	candidate row plus timeline
Provenance	Single enum obscures actor/reason
Visibility	IP; GC only own flow
Verification	Workflow evidence
Correction/replacement	Append actor/reason transition
GC sees it?	Own invitation status
IP sees GC answer?	Purpose-limited response status
Downstream influence	Outreach workflow
Promotion behavior	Selected relationship history only
Recommended contract	Decline is not eligibility; retain append-only actor.

Invited email and IP message

Element	Invited email and IP message
Class	2/1
Subject	Prospective GC contact / IP message
Current author	IP
Storage	invitations
Provenance	Created-by subject
Visibility	Issuer/internal; token landing
Verification	Email unverified until identity bind
Correction/replacement	Revoke/reissue; mismatch confirmation
GC sees it?	Own invitation/message
IP sees GC answer?	Claim status, not private answers
Downstream influence	Identity claim
Promotion behavior	Membership audit reference only
Recommended contract	Original recipient assertion remains in audit; do not silently rewrite.

Redeemed subject/email and GC decline

Element	Redeemed subject/email and GC decline
Class	3
Subject	GC/account
Current author	Authenticated GC
Storage	invitation claim/redemption
Provenance	Verified account event
Visibility	Issuer sees bounded status; operator audit
Verification	Identity-bound, not professional truth
Correction/replacement	New claim/identity-review event
GC sees it?	Own status
IP sees GC answer?	Bounded status/email per policy
Downstream influence	Workspace/case membership
Promotion behavior	Purpose/membership event only
Recommended contract	No clinical/eligibility meaning.

GC name, preferred name and state

Element	GC name, preferred name and state
Class	3
Subject	GC
Current author	Authenticated GC
Storage	people plus gc_onboarding:first_run_profile
Provenance	Broad user-supplied
Visibility	Currently shared fact
Verification	Self-attested
Correction/replacement	GC revision/history UI
GC sees it?	Yes
IP sees GC answer?	Approved identity/state projection
Downstream influence	Profile/routing/onboarding
Promotion behavior	Preserve GC author and self-attested status
Recommended contract	Revises, never erases, IP assertion.

GC relationship/familiarity/readiness

Element	GC relationship/familiarity/readiness
Class	3
Subject	GC/relationship
Current author	GC
Storage	shared first_run_profile
Provenance	User-supplied
Visibility	Shared latent; no clear current UI
Verification	Self-attested familiarity, not eligibility
Correction/replacement	GC revision
GC sees it?	Yes
IP sees GC answer?	Only approved progress/relationship projection
Downstream influence	Onboarding/readiness presentation
Promotion behavior	Preserve source/meaning
Recommended contract	Do not label as medical/overall readiness.

GC email, phone, best time and contact preference

Element	GC email, phone, best time and contact preference
Class	4
Subject	GC
Current author	GC/account
Storage	private contact_preferences; preference duplicated shared
Provenance	User/account supplied
Visibility	GC owner/internal purpose
Verification	Self-attested; email account-bound
Correction/replacement	GC revision
GC sees it?	Yes
IP sees GC answer?	Minimal operational projection only
Downstream influence	Contact coordination
Promotion behavior	Private; project only explicit contact channel
Recommended contract	Remove shared duplication and audit every projection.

GC case-profile consent

Element	GC case-profile consent
Class	3
Subject	GC
Current author	GC
Storage	case_profile_consent/shared fact in PR #124
Provenance	User-supplied
Visibility	Status may be shared; evidence private/audit
Verification	Authenticated acceptance
Correction/replacement	Revoke/new consent version
GC sees it?	Yes
IP sees GC answer?	Purpose/status only
Downstream influence	Projection/membership authorization
Promotion behavior	Carry consent ID/version/status, not form content
Recommended contract	Consent is purpose/version specific and revocable.

GC direct clinical/social screening answers

Element	GC direct clinical/social screening answers
Class	4 then 5
Subject	GC
Current author	GC; later reviewer/provider
Storage	legacy GC intake_data
Provenance	No per-field provenance/history
Visibility	GC-private/reviewer; some legacy summaries
Verification	Self-report until evidence review
Correction/replacement	GC revision; dedicated class-5 evidence revision
GC sees it?	Yes
IP sees GC answer?	Reviewed coordination status only
Downstream influence	Eligibility/readiness/provider review
Promotion behavior	Private self-report with separate verified revisions
Recommended contract	Never collected/answered authoritatively by IP or upgraded by generic proposal.

GC preferences and boundaries

Element	GC preferences and boundaries
Class	3 or 4
Subject	GC
Current author	GC
Storage	legacy intake_data / future authored assertions
Provenance	User-supplied
Visibility	Per field/purpose
Verification	Self-attested
Correction/replacement	GC revision
GC sees it?	Yes
IP sees GC answer?	Only explicit compatibility/coordination projection
Downstream influence	Relationship discussion/routing
Promotion behavior	Preserve field-level visibility
Recommended contract	No blanket sharing.

GC insurance identifiers and answers

Element	GC insurance identifiers and answers
Class	4 then 5
Subject	GC
Current author	GC; reviewer/provider confirms
Storage	legacy intake_data / future document-fact rail
Provenance	Currently broad
Visibility	GC/private reviewer
Verification	Self-report then reviewed
Correction/replacement	GC/provider evidence revision
GC sees it?	Yes
IP sees GC answer?	Coverage/review status only
Downstream influence	Insurance coordination/provider packet
Promotion behavior	Raw identifiers stay private
Recommended contract	Never promise coverage; class 5 evidence linked.

IP Conversation Guide notes

Element	IP Conversation Guide notes
Class	1
Subject	IP relationship intent
Current author	IP
Storage	currently memory_records.scope=shared
Provenance	IP but scope unsafe
Visibility	Future all-case-member risk
Verification	Private note
Correction/replacement	IP revision/delete per retention
GC sees it?	No by default
IP sees GC answer?	N/A
Downstream influence	Ask/guide
Promotion behavior	Off by default
Recommended contract	Change default to IP-private; explicit share creates a new projection.

Ask context and AI proposals about GC

Element	Ask context and AI proposals about GC
Class	2 until 3/5 evidence
Subject	GC/case
Current author	AI from mixed sources
Storage	known_intake_state, proposals, memory/facts
Provenance	Insufficiently typed
Visibility	Depends caller/payload
Verification	Proposal only
Correction/replacement	Reject or evidence-linked revision
GC sees it?	Only scope-authorized output
IP sees GC answer?	Only scope-authorized output
Downstream influence	Guidance, case facts, provider prep
Promotion behavior	Never upgrades class
Recommended contract	Every datum carries subject/author/class/source/allowed use; generic approval cannot create class 5.

GC self-authored case facts

Element	GC self-authored case facts
Class	3/4
Subject	GC
Current author	GC
Storage	facts with user-supplied provenance
Provenance	Owner/created_by but no author role/supersedes
Visibility	Private/shared broad
Verification	Self-attested
Correction/replacement	Append superseding revision
GC sees it?	Yes
IP sees GC answer?	Only shared purpose projection
Downstream influence	Case/Ask/readiness
Promotion behavior	Preserve visibility/verification
Recommended contract	Add field-level author, revision and verification.

Reviewer/provider conclusions

Element	Reviewer/provider conclusions
Class	5
Subject	GC/case
Current author	Authorized reviewer/provider
Storage	future evidence rail; current broad fact provenance allows it
Provenance	Must include evidence/source/authority
Visibility	Purpose/role scoped
Verification	Confirmed
Correction/replacement	New evidence-linked revision
GC sees it?	Per policy
IP sees GC answer?	Approved coordination status
Downstream influence	Eligibility, routing, provider and milestone evidence
Promotion behavior	Retain class/evidence
Recommended contract	Only dedicated role-gated evidence rail can create class 5.

Provider packet facts/documents

Element	Provider packet facts/documents
Class	3/5 plus consent
Subject	GC/provider purpose
Current author	GC/operator/provider sources
Storage	dirty handoff packet/snapshot work
Provenance	Current selector too broad
Visibility	Named recipient/purpose only
Verification	Per included item
Correction/replacement	New snapshot/revision/revocation
GC sees it?	Preview/consent
IP sees GC answer?	No raw packet unless policy permits
Downstream influence	Provider handoff
Promotion behavior	Immutable purpose-bound snapshot
Recommended contract	Class 1/2 excluded; class 4 explicit GC selection; exact source/evidence required.

Global contract:

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.
- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

Concise answer for Alan

The intended parent answers today because Node 0 was built as a private pre-contact reflection tool: relationship context and observed signals can tailor a low-pressure conversation guide before the prospective GC has an account. That limited use is reasonable. What is not reasonable is treating those answers as if the GC made or confirmed them, especially health, age, citizenship, family status, insurance, consent, preferences or eligibility. Those must be GC-authored, with reviewer/provider confirmation where required. When the GC joins, her answer appends an attributable revision; the earlier IP statement remains labelled as IP-reported/unconfirmed, GC-private data stays private, and only an explicit purpose-limited projection reaches the IP. The current score/invite gating and missing reconciliation should be replaced.

Migration architecture

Options:

1. Alembic with reviewed operations/raw SQL
2. Custom ordered Python migrations with a demonstrated reason not to use an established framework
3. A standalone established runner such as Flyway/dbmate if its runtime/deployment burden fits

Provisional recommendation: Favor Alembic unless the ADR proves it cannot safely serve the current raw-SQL/SQLite compatibility architecture.

- PostgreSQL is production authority; SQLite is a supported local/test projection only if the ADR retains it.
- Fingerprint existing databases before baseline stamping; never blindly stamp unknown schema.
- App and intelligence use separate revision namespaces and PostgreSQL advisory-lock keys.

- Migration identities hold DDL; app/intelligence runtime identities do not after adoption.
- Run migrations in a deploy-controlled one-off task before app rollout, not opportunistically from web startup.
- Use expand in release N, bounded/backpressure-aware backfill, dual-compatible N+1, then contract only when N cannot run.
- CI covers fresh, current, one-revision-behind, N-1 app on expanded schema, concurrent runner, checksum drift and interrupted backfill.
- Irreversible changes require backup evidence, explicit approval and forward-repair plan. Schema rollback is not automatic.

Promotion, commercial and payment contract

Sole authorization: `byos.workspace_promotion_authorized.v1`. Producer: natalyx-app commitment/commercial domain after deterministic validation. Consumer: natalyx-app promotion service.

Payload:

```
event_id
idempotency_key
workspace_id
commitment_id
owner_person_id
owner_subject
commercial_config_version
terms_acceptance_ids[]
payment_plan_id
funding_evidence.kind
funding_evidence.commercial_event_id
funding_evidence.settled_amount_minor
funding_evidence.currency
credit_allocation_ids[]
initial_case_state=enrollment_pending
occurred_at
correlation_id
causation_id
schema_version
```

Conditions:

- Workspace is active/match-ready, unarchived and not already promoted.
- Commitment references the same workspace, has current terms/privacy acceptances and is in funding_pending/funded per the approved state contract.
- Payment path and activated commercial configuration are pinned.
- Required enrollment funding has a verified settled ledger entry or separately approved financing/manual-funding evidence; a processor webhook is first normalized and cannot itself authorize promotion.
- Credit allocation is non-negative, reproducible and cannot exceed the applicable obligation.
- No cancellation, compliance hold, reversal that invalidates funding, or existing promotion is present.

Constraints:

- one promotion per workspace
- one case per promotion event
- one result per idempotency key
- one primary owner membership per created case

No browser success/return route and no raw processor callback can emit or substitute for this event.

Append-only ledgers:

- `commercial_events`
- `processor_events`
- `invoice_events`
- `payment_ledger_entries`
- `credit_ledger_entries`
- `refund_ledger_entries`

- `adjustment_ledger_entries`
- `performance_obligation_events`
- `promotion_events`

Rules:

- Every record carries currency in integer minor units, commercial configuration version, source/idempotency/correlation identifiers and immutable timestamps.
- Mutable balances/statuses are projections only; reversals, chargebacks, credits, refunds and adjustments append entries.
- Processor events are signature-verified, stored once and normalized before domain use.
- Node 0 Complete credit cannot create a negative JMF obligation and stays distinct from cash/refund entries.
- Pass-through clinic/legal/psych/background/escrow/insurance/GC funds never enter Natalyx revenue ledgers.
- Clinical-linked invoices require durable evidence plus dignity-of-timing delay and can never be the first clinical notification.
- Refundable cash = max(0, settled JMF cash - earned JMF - prior cash refunds - chargebacks/reversals), with policy hooks for unresolved cases.

Invitation safety

Retain from #122:

- hashed tokens
- purpose-specific TTL with a 14-day ceiling
- CAS terminal transitions
- atomic membership creation
- one-live-GC constraint
- self/existing-member refusal
- safe return_to
- portable integrity/concurrency tests

Required changes:

- Exactly one target: workspace_id or case_id; include purpose, role, access_class, recipient_binding_mode and policy version.
- Email-less possession may only open pending low-risk workspace claim; it cannot reveal sensitive data or grant paid-case membership.
- Paid-case invitation is addressed, authenticated, consented and atomically redeemed; mismatch requires explicit policy/issuer handling and preferably fresh bound token.
- Cognito admission and app membership are separate decisions; account creation never equals workspace/case access.
- Verified identity comes from auth claims, not form-entered email; duplicate claims return the original result.
- Never put raw token/email/PHI in general participant timeline payloads; revoke/expire/supersede fail closed.
- Remove generic unconditional status mutation; expose authorized transition commands only.

Documents and S3

Storage

- Private versioned quarantine bucket and clean vault with block-public-access, SSL-only and KMS encryption.
- App can create short-lived unique upload targets in quarantine but cannot read unscanned content; scanner moves/copies only clean versions to vault.
- Object key: env/{org_id}/{case_id}/{document_id}/{version_id}/{generated_object_id}; no names, emails, filenames or clinical labels.
- Authorization is always a Postgres membership/role/visibility/consent decision independent of the key.
- CloudTrail S3 data events and app document_access_event records provide purpose-aware access evidence.

Upload

- Configurable maximum size; provisional 25 MiB awaits D-DOC-02.
- Initial proposed allowlist PDF/JPEG/PNG; DOCX only after scanner/archive review; no macro-enabled Office formats.
- Unicode-normalize/sanitize display filename; byte-sniff MIME, expected length and required SHA-256 checksum.
- Never reuse a key. Finalize verifies object version, HeadObject size/checksum/type and upload-session ownership.
- S3 event -> durable queue -> scanner -> signed clean/rejected/error result; unavailable/timed-out scan remains quarantined.

Download

- App authenticates and authorizes exact version/purpose, records access event, then returns a very short-lived clean-vault GET with safe content disposition.
- App proxy is reserved for justified transformation/audit cases, not the default durable path.

Recovery

- Upload sessions/tasks live in Postgres/queue so ECS replacement does not lose them; leases expire and retry idempotently.
- Replacement creates a new immutable version and supersedes the prior current pointer.
- Deletion is an audited tombstone followed by retention/legal-hold controlled purge; object-delete failures alert and retry.
- Orphan reconciliation handles expired intents and objects without document versions; lifecycle aborts incomplete multipart uploads.
- Provider share uses a purpose-bound snapshot/checksum/expiry, not direct object-key authority.

Design-system release rubric

Class	Items
Release-blocking	XSS/escaping or unsafe raw slots; keyboard/focus/screen-reader failures; unusable responsive layout; misleading or nonfunctional interaction; motion without reduced-motion safety; contrast or error-state defects that block task completion
Desirable	token/recipe consistency; visual parity; removal of duplicate spacing/color literals; component naming/adoption
Deferred	zero raw hex values; zero inline geometry styles; full legacy strangler cleanup; broad participant redesign; internal/operator DS convergence

Style purity blocks a release only when tied to a concrete security, accessibility, interaction or material maintainability risk.

Decision register

D-MIG-01

ID	D-MIG-01
Decision	Adopt Alembic or justify a custom ordered runner; decide supported SQLite parity, expand/contract, locking, rollback and forward-repair contract.
Owner / proposer	Luke / Alan
Affected PRs	P02-01, P02-02, P02-03
Latest point	Before P02-02
Recommendation / treatment	Prefer an established framework unless the ADR proves current SQL compatibility makes it unsafe or disproportionate.
Project item	D-MIG-01

D-AUTH-01

ID	D-AUTH-01
Decision	Which low-risk fields may a link-only claimant see, when is recipient/issuer confirmation required, and what consent/version binds each access elevation?
Owner / proposer	Luke / Alan
Affected PRs	P03-02, P03-03
Latest point	Before P03-02
Recommendation / treatment	Link possession creates pending workspace claim only; paid-case access always requires authenticated binding and explicit consent.
Project item	D-AUTH-01

D-AUTH-02

ID	D-AUTH-02
Decision	Choose a short-lived signed/server-verified participant admission proof and key/claim lifecycle while retaining a separate internal allowlist path.
Owner / proposer	Luke / Alan
Affected PRs	P03-04
Latest point	Before P03-04
Recommendation / treatment	Bind proof to email, purpose, claim ID and expiry; it admits account creation only and never grants workspace/case membership.
Project item	D-AUTH-02

D-DATA-01

ID	D-DATA-01
Decision	Which provisional IP assertions may the GC see, which GC answers may the IP see, and which fields survive promotion/provider projection?
Owner / proposer	Luke / Alan
Affected PRs	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02
Latest point	Before P04-02
Recommendation / treatment	Default deny; expose only explicit purpose-limited projections. Class 1/2 and class 4 never automatically cross roles.
Project item	D-DATA-01

D-DATA-02

ID	D-DATA-02
Decision	How long to retain unselected candidate assertions and how are GC corrections, conflicting relationship answers, and private IP notes represented?
Owner / proposer	Luke / Alan
Affected PRs	P04-02, P04-03, P06-02
Latest point	Before P04-02
Recommendation / treatment	Append revisions with supersedes/derived-from; never silently erase source; shortest justified retention for unselected candidates.
Project item	D-DATA-02

D-DATA-03

ID	D-DATA-03
Decision	Does any 0-100 candidate ranking or race-based preference survive, and what may provisional third-party answers gate?
Owner / proposer	Luke / Alan
Affected PRs	P05-02, P06-03
Latest point	Before P05-02
Recommendation / treatment	Remove candidate ranking and race preference; use source-labelled questions-to-clarify. Unverified IP answers cannot gate claim/onboarding/eligibility.
Project item	D-DATA-03

D-COM-01

ID	D-COM-01
Decision	Set the flat one-time price and tax treatment; it must remain fully creditable against the later JMF.
Owner / proposer	Luke / Luke
Affected PRs	P06-01, P07-01, P08-01
Latest point	Before activation of a Complete configuration
Recommendation / treatment	Keep no active production value until explicitly approved; test with synthetic configuration.
Project item	D-COM-01

D-COM-02

ID	D-COM-02
Decision	Ratify $8,900 JMF, $7,830 pay-in-full, $750 deposit and 20/30/20/20/10 post-deposit milestones as configuration v1 with effective date/currency.
Owner / proposer	Luke / Luke
Affected PRs	P07-01, P08-01, P08-03
Latest point	Before P07-01 activation data
Recommendation / treatment	Use the document anchors but require explicit version activation rather than code literals.
Project item	D-COM-02

D-COM-03

ID	D-COM-03
Decision	Approve the actual paid-service terms, privacy/consent disclosure and refundable-to-earned wording.
Owner / proposer	Luke / External/TBD
Affected PRs	P07-01, P07-02, P08-03
Latest point	Before P07-02 user acceptance
Recommendation / treatment	External legal/privacy/finance review before any real acceptance.
Project item	D-COM-03

D-COM-04

ID	D-COM-04
Decision	Define evidence for enrollment funding confirmed, promotion authorization, and for onboarding/compatibility complete enough to move case active and earn/issue activation billing.
Owner / proposer	Luke / Alan
Affected PRs	P07-02, P07-03, P07-04, P08-03, P09-04
Latest point	Before P07-02
Recommendation / treatment	Funding confirmation is durable ledger evidence; promotion uses the sole typed authorization event; activation requires an explicit evidence checklist, actor and audit event.
Project item	D-COM-04

D-PAY-01

ID	D-PAY-01
Decision	Select processor, card/ACH capabilities, webhook/security/settlement semantics and operational ownership.
Owner / proposer	Luke / Alan
Affected PRs	P08-02, P08-03, P08-04
Latest point	Before P08-02 adapter
Recommendation / treatment	Choose only after domain/ledger contract; retain processor-neutral ports.
Project item	D-PAY-01

D-COM-05

ID	D-COM-05
Decision	Select financing partner and define when funding is sufficiently authorized/settled; otherwise hide the path.
Owner / proposer	Luke / External/TBD
Affected PRs	P07-02, P08-02
Latest point	Before financed option is exposed
Recommendation / treatment	Do not block card/ACH MVP; keep optional and disabled.
Project item	D-COM-05

D-COM-06

ID	D-COM-06
Decision	Decide retained revenue versus future-journey credit after heartbeat loss and contractual support window.
Owner / proposer	Luke / External/TBD
Affected PRs	P08-03
Latest point	Before automated pregnancy-stage refund logic
Recommendation / treatment	Manual/counsel-reviewed hook until policy is approved; never hard-code.
Project item	D-COM-06

D-COM-07

ID	D-COM-07
Decision	Ratify earned/uncharged/refunded treatment, GC withdrawal/replacement and pay-in-full/financed reconciliation.
Owner / proposer	Luke / External/TBD
Affected PRs	P08-03
Latest point	Before real billing
Recommendation / treatment	Use document schedule as test anchor; require legal/finance approval for live policy.
Project item	D-COM-07

D-COM-08

ID	D-COM-08
Decision	Define reminder cadence, grace period, pauseable services and safety-critical/document access that can never be blocked.
Owner / proposer	Luke / Alan
Affected PRs	P08-04
Latest point	Before P08-04
Recommendation / treatment	Default to continued safety and entitled-document access; pause only clearly non-critical future services.
Project item	D-COM-08

D-DOC-01

ID	D-DOC-01
Decision	Set retention/legal-hold/deletion and choose presigned versus app-proxied downloads by document class.
Owner / proposer	Luke / Alan
Affected PRs	P09-01, P09-03
Latest point	Before P09-03
Recommendation / treatment	Purpose/role authorization independent of key; short-lived downloads; deletion tombstone/audit; legal/privacy review.
Project item	D-DOC-01

D-DOC-02

ID	D-DOC-02
Decision	Choose maximum size/types, scanner/quarantine mechanism, timeout/retry and fail-closed handling.
Owner / proposer	Luke / Alan
Affected PRs	P09-01, P09-02, P09-03
Latest point	Before P09-02
Recommendation / treatment	Narrow allowlist, checksum and content sniffing; unavailable scanner remains quarantined.
Project item	D-DOC-02

D-PROV-01

ID	D-PROV-01
Decision	Define permitted recipients/purposes, GC preview/consent, expiry, verification level and exact fact/document allowlists.
Owner / proposer	Luke / Alan
Affected PRs	P09-05
Latest point	Before P09-05
Recommendation / treatment	Class 1/2 excluded; class 4 only if GC explicitly selects for recipient/purpose; class 5 evidence linked.
Project item	D-PROV-01

D-AI-01

ID	D-AI-01
Decision	Approve scopes, allowed sources, model policy, proposal reviewers, class-5 evidence rail and degraded-mode copy.
Owner / proposer	Luke / Alan
Affected PRs	P10-01, P10-02, P10-03, P12-01
Latest point	Before P10-01
Recommendation / treatment	Generic case members cannot create reviewed/provider truth; intelligence remains advisory and scope explicit.
Project item	D-AI-01

D-OPS-01

ID	D-OPS-01
Decision	Select the revision/schema pair to recover dev, rollback/forward-repair criteria and evidence owner.
Owner / proposer	Luke / Alan
Affected PRs	P13-02
Latest point	Before Gate A0 recovery action
Recommendation / treatment	Use a tested immutable image and compatible schema; do not deploy from dirty or draft branches.
Project item	D-OPS-01

D-PHI-01

ID	D-PHI-01
Decision	Approve data classification, vendors/BAAs, encryption/logging/access/incident/backup controls and prohibited data at each gate.
Owner / proposer	Luke / External/TBD
Affected PRs	P09-01, P09-02, P09-05, P13-05, P13-06, P13-07
Latest point	Before B1 for PII scope; before B2 for PHI
Recommendation / treatment	B1 remains PII-only; B2 requires formal external security/privacy/compliance evidence.
Project item	D-PHI-01

D-PHI-02

ID	D-PHI-02
Decision	Name security/privacy/compliance approvers, evidence recency, exception authority and revalidation cadence.
Owner / proposer	Luke / External/TBD
Affected PRs	P13-06, P13-07
Latest point	Before B1
Recommendation / treatment	Luke is product go/no-go; independent qualified reviewers own their control attestations.
Project item	D-PHI-02

Risk register

ID	Class	Risk	Owner	Mitigation	Gate
R-01	Implementation risk	Runtime DDL or incompatible migration leaves dev unavailable.	Alan	WP01/WP02, advisory lock, one-off migrator, expand/contract, forward repair.	A0
R-02	Product/privacy risk	IP assertions are presented as GC truth or leak across roles.	Luke	Authorship matrix, D-DATA decisions, WP04/WP06 negative tests.	A1/A2
R-03	Security risk	Forwarded invitation or shared signup code grants sensitive access.	Alan	Separate claim/invite machines, claim-aware Cognito admission, consent/membership transaction.	A1/A2
R-04	Financial risk	Duplicate charge/case/invoice/refund or unreproducible balance.	Luke	Append-only ledgers, processor inbox, typed promotion event, idempotency/concurrency tests.	A2
R-05	Security/compliance risk	Unscanned/public/over-shared documents or provider packet expose sensitive data.	Luke	Quarantine/vault, KMS/IAM, purpose consent, access audit, RLS, formal gates.	A2/B2
R-06	AI safety risk	Private memory or provisional/AI fact becomes shared/provider truth.	Luke	Scoped envelopes, evidence rail, proposal authority, safety evals.	A2
R-07	Delivery risk	Useful work is deleted or Alan spends time polishing obsolete architecture.	Joint	Extraction ledger/checkpoint, destination tests, no cleanup before sign-off.	Pre-A0
R-08	Operational risk	Green CI is mistaken for live or PHI readiness.	Luke	Known-revision cloud gates and formal B1/B2 evidence/approvers.	A0-B2
R-09	UX/accessibility risk	Mobile or assistive users cannot complete the participant journey.	Alan	Release-blocking DS criteria and end-to-end mobile/keyboard tests.	A1/A2
R-10	Repository-boundary risk	Participant/internal DS, Learn/intelligence/app truth or app/infra ownership is blurred.	Luke	Pinned contracts, separate PRs/repositories, architecture-after-merge statements.	All

Release strategy and real-user/PHI boundaries

- Merge only after the consolidated plan is reviewed by Alan and accepted by Luke; proposed ownership is not a compulsory assignment before that sign-off.
- Use additive schemas, feature flags and dual-compatible reads/writes; contract only after telemetry/evidence proves the prior revision is unavailable.
- Recover and pass A0 first, then validate A1 core participant behavior before adding commercial/document complexity, then A2 before any real user.
- Keep all development synthetic through A2. B1 permits only the explicitly approved PII beta scope. PHI remains blocked until B2's formal human go/no-go.

- Every gate release identifies exact app/intelligence/infra revisions, migration heads, artifact pins and rollback/forward-repair response.
- Existing PRs/branches are closed or deleted only after extraction, replacement reachability, review reconciliation and explicit team confirmation.

A0/A1/A2 remain synthetic. B1 is a named, approved PII-only beta boundary. B2 is a formal PHI go/no-go that requires organizational, contractual, operational and technical evidence; successful tests or deployment alone are insufficient.

Document index and maintenance

Document	Purpose
README.md	Programme entrypoint, index, operating/maintenance/render instructions
existing-pr-extraction-ledger.md	PR/worktree preservation, extraction and closure/delete conditions
project-structure.md	GitHub Project fields, items and conventions
pr-manifest.md	Canonical proposed PR inventory and execution order
consolidated-plan.md	Full programme source for the PDF
consolidated-plan.pdf	Rendered review copy generated only from consolidated-plan.md
00-preservation-cloud-diagnosis-and-pr-reconciliation.md	Standalone execution plan for WP00 — Preservation, cloud diagnosis, and PR reconciliation
01-ci-and-postgres-evidence-gate.md	Standalone execution plan for WP01 — CI and PostgreSQL evidence gate
02-versioned-migrations-and-readiness.md	Standalone execution plan for WP02 — Versioned migrations and readiness
03-auth-invitations-and-resume.md	Standalone execution plan for WP03 — Authentication, invitations, and resume
04-node0-workspace-case-separation.md	Standalone execution plan for WP04 — Node 0 workspace and paid-case separation
05-node0-free-pathfinder-and-resume.md	Standalone execution plan for WP05 — Node 0 Free, Pathfinder, and resume
06-node0-complete-and-soft-onboarding.md	Standalone execution plan for WP06 — Node 0 Complete and GC soft onboarding
07-commercial-configuration-enrollment-and-promotion.md	Standalone execution plan for WP07 — Commercial configuration, enrollment, and promotion
08-payments-invoices-credits-and-refunds.md	Standalone execution plan for WP08 — Payments, invoices, credits, and refunds
09-documents-active-case-and-provider-handoffs.md	Standalone execution plan for WP09 — Documents, active case, and provider handoffs
10-intelligence-memory-proposals-and-ask-safety.md	Standalone execution plan for WP10 — Intelligence, memory, proposals, and Ask safety
11-learn-release-and-design-system-contract.md	Standalone execution plan for WP11 — Learn release and design-system contract
12-ask-learn-shell-and-mobile-cutover.md	Standalone execution plan for WP12 — Ask/Learn shell and mobile cutover
13-deployment-cloud-gates-and-phi-readiness.md	Standalone execution plan for WP13 — Deployment, cloud gates, and PHI readiness
programme_data.py	Canonical structured source used to keep IDs/contracts consistent
scripts/render_programme.py	Mechanical Markdown renderer
scripts/render_pdf.py	Markdown-to-PDF renderer
scripts/validate_programme.py	Cross-file/dependency/gate validator
scripts/sync_project.py	Idempotent GitHub Project inspector/synchronizer

Regenerate all Markdown:

```
/Users/lukerhodes/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 scripts/render_pro
gramme.py
```

Validate parity/dependencies:

```
/Users/lukerhodes/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 scripts/validate_p
rogramme.py
```

Render this exact Markdown to PDF:

```
/Users/lukerhodes/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 scripts/render_pdf
.py consolidated-plan.md consolidated-plan.pdf
```

Detailed work packages

Work package 00 — Preservation, cloud diagnosis, and PR reconciliation

Planning-only execution contract. Proposed implementers and owners remain recommendations until Alan reviews/signs off the consolidated plan and Luke accepts it. Project: MVP Product Flow Reshape.

Plan	Value
Work package	WP00
Plan file	`00-preservation-cloud-diagnosis-and-pr-reconciliation.md`
Proposed PRs	P00-01
Inspection snapshot	2026-07-10 20:12 America/New_York

1. Purpose and outcome

Problem. The live app is returning ALB 503 responses while overlapping draft PRs and protected dirty worktrees contain useful but contradictory implementations.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. A signed-off source-of-truth baseline, complete extraction ledger, Alan-review reconciliation, and explicit no-delete/no-close conditions before replacement work begins.

Explicit non-goals.

- No code porting
- No PR closure or retargeting
- No branch/worktree cleanup
- No deployment or AWS mutation

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-001	verified	live dev / 2026-07-10 20:12 America/New_York	https://dev.app.natalyx.health/ and /health	Both requests returned HTTP 503 from awselb/2.0; the deployed participant app is not presently usable.
E-002	verified limitation	AWS dev / 2026-07-10 20:12 America/New_York	aws sts get-caller-identity --profile natalyx-dev	Local AWS SSO was expired, so current ECS desired/running counts and task revision were not independently verified. Gate A0 must refresh access and capture them.
E-003	verified	GitHub / 2026-07-10 20:12 America/New_York	NatalyxPlatform open PR query	App PRs #112, #117, #119, #121, #122, #123 and #124 and design-system PRs #7/#8 are open; all app PRs are drafts.
E-004	verified	.github / main@6072ce9	GitHub Project 3 GraphQL inspection	MVP Product Flow Reshape existed with only default fields and zero items before this planning update.

3. Architecture and contracts

- Treat every existing branch and worktree as read-only evidence until its valuable files, tests, and reviewer findings have a named destination.
- Use the organisation Project as the programme control plane and this repository-neutral package as the reviewable contract.

- Do not infer live health from GitHub build success; Gate A0 requires a known revision and live evidence.

Contract area	Package rule
Ownership boundary	`.github`; repository-specific PRs remain separate.
Entities/state	No new runtime entity; evidence/coordination only.
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	No broad public contract; use the focused commands/events in each PR.
Storage	No new storage; preserve current state read-only for this package.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

4. Explicit PR sequence

1

Order	1
ID	P00-01
Title	Approve programme baseline and extraction ledger
Repository	.github
Dependencies	P00-01
Gate	Pre-A0
Size	Medium
Readiness	Needs plan approval

P00-01 — Approve programme baseline and extraction ledger

Field	Plan
Repository / base	`.github` (`main`)
Project item	P00-01
Gate / size / readiness	Pre-A0 / Medium / Needs plan approval
Dependencies	P00-01 plan sign-off only
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	All listed draft PRs and protected worktrees

Objective

Land the reviewed planning package and extraction baseline only after Alan has reviewed the consolidated document and Luke has accepted the programme contract.

Scope

- Record sign-off, unresolved decisions, source SHAs, and reconciliation procedure
- Freeze programme IDs used by the Project

Explicit non-goals

- No implementation code
- No assignment becomes mandatory before Alan sign-off

Expected files or modules

- `natalyx-repair-program/*.md`

- `natalyx-repair-program/consolidated-plan.pdf`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Repository-neutral planning and programme-control documents only.

Tests and validation

Objective tests:

- git diff --check
- document/project parity validator
- PDF visual QA

Local validation:

- `git diff --check`
- `/Users/lukerhodes/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 scripts/validate_programme.py`

Cloud validation: run only through Gate Pre-A0 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Merge only after document/sign-off checks; it changes no runtime. Re-sync the Project from the same canonical data.

Rollback or forward repair

Correct the canonical data and regenerate; do not rewrite implementation history.

Dependencies and parallel work

Hard dependencies: approved programme baseline. Parallel opportunities: None inside this package beyond review/test preparation.

Objective acceptance criteria

- □ Land the reviewed planning package and extraction baseline only after Alan has reviewed the consolidated document and Luke has accepted the programme contract.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Luke
Implementer	Luke (proposed; pending consolidated-plan sign-off)
Reviewer	Alan
Final approver	Luke
Required expertise	programme architecture, technical programme management
Higher-capability review	Risk-proportionate; not mandatory by programme default

5. Cross-repository coordination

Repository	Authority after merge
`.github`	programme/governance evidence and Project control plane

Merge order inside this package: `P00-01`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	SHA and timestamp reproducibility; Project/document identifier parity; No dirty-worktree mutation; HTTP and workflow evidence captured separately	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Type	ID / risk	Owner	Affected PRs / latest point	Project item / treatment
Implementation risk	Review findings arrive after decomposition	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	A protected worktree contains the only copy of useful code	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	A green PR is mistaken for live readiness	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#112	9605d88e31bd	Source-first 81-node/12-cluster surface map, participant-renderer boundaries, wave plan and cross-repo seams.	P00-01	Selective port, then supersede after extraction	Catalogued; content remains to be reconciled into replacement PR acceptance tests.
app#119	a8cd2f428535	Concise product-leverage framing and boundaries for the MVP flow.	P00-01	Retain as source or merge separately after plan decision	Catalogued; reconcile terminology into approved programme.

Package source inventory:

- App PRs #112, #117, #119, #121, #122, #123, #124
- Design-system PRs #7 and #8
- Provider-exchange, Node 0 onboarding/quiz, Learn-rendering, and intelligence dirty worktrees

9. Ownership recommendation

Luke owns product/architecture acceptance and the closure decision. Alan owns technical reconciliation recommendations. Both sign the checkpoint before Node 0, invitation/auth, or participant case replacement work starts.

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P00-01	Luke	Luke (proposed)	Alan	Luke	As risk warrants

Luke decides product/lifecycle/commercial/privacy/PHI/provider policy and accepts gates. Alan proposes implementation design, decomposition, migrations, code, tests, rollout and repair. This is a recommendation pending Alan's review/sign-off, not a compulsory assignment.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 01 — CI and PostgreSQL evidence gate

Planning-only execution contract. Proposed implementers and owners remain recommendations until Alan reviews/signs off the consolidated plan and Luke accepts it. Project: MVP Product Flow Reshape.

Plan	Value
Work package	WP01

Plan	Value
Plan file	`01-ci-and-postgres-evidence-gate.md`
Proposed PRs	P01-01
Inspection snapshot	2026-07-10 20:12 America/New_York

1. Purpose and outcome

Problem. Main builds containers and runs a deploy-contract smoke, but does not run the Python suite or prove current schema behavior against PostgreSQL.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. A first, deliberately CI-only PR that runs the current tests, exercises the existing schema on PostgreSQL, and publishes diagnosable evidence without introducing the future migration framework.

Explicit non-goals.

- No migration runner
- No product behavior changes
- No schema redesign
- No deployment

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-005	verified	app / origin/main@fdf2bb4	.github/workflows/build.yml:1-42; aws-app-deploy.yml:1-38	Main builds images and runs a local deploy-contract smoke but has no Python test workflow or PostgreSQL evidence job.
E-006	verified	app / PR #122@e853821	PR comment Review round 2 addressed and changed workflow/tests	Updated PR #122 reports 499 tests plus return_to, TTL, CAS, self-invite, lens and Learn fail-closed fixes; green checks were observed. These are salvage candidates, not blanket merge approval.
E-009	verified	app / origin/main@fdf2bb4	app/storage/connection.py:14-57,93-123	The app supports pooled PostgreSQL and SQLite with a SQL compatibility adapter; PostgreSQL is the cloud backend but the current suite is predominantly local.

3. Architecture and contracts

- PostgreSQL is production authority; SQLite remains a local compatibility target, not production evidence.
- Separate current-schema evidence from tests that require the future migration mechanism.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	No new runtime entity; evidence/coordination only.
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	No broad public contract; use the focused commands/events in each PR.
Storage	None; exercise the schema that main owns today.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.

Contract area	Package rule
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

4. Explicit PR sequence

1

Order	1
ID	P01-01
Title	Establish current-test and PostgreSQL CI evidence
Repository	app
Dependencies	P00-01
Gate	A0
Size	Medium
Readiness	Dependencies unmet

P01-01 — Establish current-test and PostgreSQL CI evidence

Field	Plan
Repository / base	`app` (`main`)
Project item	P01-01
Gate / size / readiness	A0 / Medium / Dependencies unmet
Dependencies	P00-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PR #122 unittest workflow and backend-neutral fixes after reconciliation

Objective

Run the current suite on pull requests/main and prove the existing schema boots and behaves on an ephemeral PostgreSQL service before any migration redesign.

Scope

- Python-version/dependency cache and current test suite
- Ephemeral PostgreSQL current-schema evidence
- Sanitized failure artifacts

Explicit non-goals

- No migration-version table
- No new-schema tests
- No application behavior

Expected files or modules

- `.github/workflows/python-tests.yml`
- `tests/python/test_postgres_current_schema.py`
- `scripts/ci_postgres_evidence.py`

Change map

Area	Planned change
Schema/model	None; exercise the schema that main owns today.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	Add blocking unit and PostgreSQL evidence jobs.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Current Python suite
- init_db twice on PostgreSQL
- representative create/read/authorization path
- intentional failure log-redaction check

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A0 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P00-01. Parallel opportunities: None inside this package beyond review/test preparation.

Objective acceptance criteria

- □ Run the current suite on pull requests/main and prove the existing schema boots and behaves on an ephemeral PostgreSQL service before any migration redesign.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Risk-proportionate; not mandatory by programme default

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P01-01`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Current unit suite; Fresh PostgreSQL database; Current init_db idempotency; Failure artifact and log clarity	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Type	ID / risk	Owner	Affected PRs / latest point	Project item / treatment
Implementation risk	CI exposes existing backend drift	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Tests accidentally depend on protected branch code	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Secrets or DB credentials leak in logs	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P01-01	Retain pending Alan reconciliation; selective port or rebase only after architecture approval	Reinspected at latest head. It is exactly e853821 (no newer delta); retention strengthened, but final extraction waits for Alan checkpoint.

Package source inventory:

- PR #122 Python-test workflow and PostgreSQL-compatible test fixes only after reconciliation

9. Ownership recommendation

Alan proposes the workflow and test matrix; Luke approves the evidence threshold. This lane can begin after programme sign-off while the Alan PR-reconciliation checkpoint is still open.

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P01-01	Alan	Alan (proposed)	Luke	Luke	As risk warrants

Luke decides product/lifecycle/commercial/privacy/PHI/provider policy and accepts gates. Alan proposes implementation design, decomposition, migrations, code, tests, rollout and repair. This is a recommendation pending Alan's review/sign-off, not a compulsory assignment.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 02 — Versioned migrations and readiness

Planning-only execution contract. Proposed implementers and owners remain recommendations until Alan reviews/signs off the consolidated plan and Luke accepts it. Project: MVP Product Flow Reshape.

Plan	Value
Work package	WP02
Plan file	`02-versioned-migrations-and-readiness.md`

Plan	Value
Proposed PRs	P02-01, P02-02, P02-03, P02-04
Inspection snapshot	2026-07-10 20:12 America/New_York

1. Purpose and outcome

Problem. Startup calls a monolithic CREATE/ALTER routine without an ordered migration ledger, advisory lock, expand/contract contract, or reliable partial-failure recovery.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. An approved migration ADR, versioned runner, production-safe locking/readiness behavior, baseline migration, and forward-repair procedure proven on PostgreSQL and compatible with supported SQLite development.

Explicit non-goals.

- No commercial schema yet
- No Node 0 semantic migration before WP04
- No destructive rollback promise
- No application feature work

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-007	verified	app / feature/byos-node0-consolidation@84a5a84	app/storage/schema.py:1-33,353-880	Schema ownership is a monolithic init_db/ensure_column routine with inline CREATE TABLE/ALTER TABLE and no ordered migration/version/checksum ledger.
E-008	verified	app / origin/main@fdf2bb4	app/main.py:295-313	Cloud startup validates auth/storage then calls storage.init_db in-process; readiness is not separated from migration execution.
E-009	verified	app / origin/main@fdf2bb4	app/storage/connection.py:14-57,93-123	The app supports pooled PostgreSQL and SQLite with a SQL compatibility adapter; PostgreSQL is the cloud backend but the current suite is predominantly local.

3. Architecture and contracts

- Alan evaluates Alembic versus a custom ordered runner; a custom runner is not the default without ADR evidence.
- Migrations are forward-only by default, transaction-scoped where PostgreSQL permits, advisory-locked, checksum-recorded, and expand/contract compatible with the previous app revision.

Contract area	Package rule
Ownership boundary	`app`, `intelligence`; repository-specific PRs remain separate.
Entities/state	No new runtime entity; evidence/coordination only.
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Add internal readiness detail behind safe health surface.
Storage	Add migration metadata only. Existing canonical schema becomes a checksummed baseline. Intelligence-owned tables enter a checksummed revision chain.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.

Contract area	Package rule
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Migration ADR contract

Options:

1. Alembic with reviewed operations/raw SQL
2. Custom ordered Python migrations with a demonstrated reason not to use an established framework
3. A standalone established runner such as Flyway/dbmate if its runtime/deployment burden fits

Provisional recommendation. Favor Alembic unless the ADR proves it cannot safely serve the current raw-SQL/SQLite compatibility architecture.

- PostgreSQL is production authority; SQLite is a supported local/test projection only if the ADR retains it.
- Fingerprint existing databases before baseline stamping; never blindly stamp unknown schema.
- App and intelligence use separate revision namespaces and PostgreSQL advisory-lock keys.
- Migration identities hold DDL; app/intelligence runtime identities do not after adoption.
- Run migrations in a deploy-controlled one-off task before app rollout, not opportunistically from web startup.
- Use expand in release N, bounded/backpressure-aware backfill, dual-compatible N+1, then contract only when N cannot run.
- CI covers fresh, current, one-revision-behind, N-1 app on expanded schema, concurrent runner, checksum drift and interrupted backfill.
- Irreversible changes require backup evidence, explicit approval and forward-repair plan. Schema rollback is not automatic.

4. Explicit PR sequence

1

Order	1
ID	P02-01
Title	Decide migration architecture and compatibility contract
Repository	app
Dependencies	P01-01
Gate	A0
Size	Small
Readiness	Blocked by decision

2

Order	2
ID	P02-02
Title	Introduce versioned migration runner and readiness checks
Repository	app
Dependencies	P02-01
Gate	A0
Size	Large
Readiness	Dependencies unmet

3

Order	3
ID	P02-03
Title	Baseline canonical schema and prove expand-contract repair
Repository	app
Dependencies	P02-02
Gate	A1
Size	Large
Readiness	Dependencies unmet

4

Order	4
ID	P02-04
Title	Move intelligence persistence to a revisioned migration namespace
Repository	intelligence
Dependencies	P02-01, P02-02
Gate	A1
Size	Large
Readiness	Dependencies unmet

P02-01 — Decide migration architecture and compatibility contract

Field	Plan
Repository / base	`app` (`main`)
Project item	P02-01
Gate / size / readiness	A0 / Small / Blocked by decision
Dependencies	P01-01
Blocking decisions/checkpoints	D-MIG-01
Source / salvage	app/storage/schema.py, app/storage/connection.py

Objective

Record an ADR choosing an established migration framework or justified ordered runner, including PostgreSQL authority, SQLite support, locks, expand/contract, backfills, rollback, and forward repair.

Scope

- Compare Alembic and custom runner against current SQL compatibility layer
- Specify old/new revision compatibility and irreversible-change policy

Explicit non-goals

- No migration code
- No schema change

Expected files or modules

- `docs/adr/ADR-00x-versioned-migrations.md`
- `docs/agents/storage-data-layer.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Decision record only; select framework before P02-02.

Tests and validation

Objective tests:

- ADR decision checklist
- example expand/contract and interrupted-backfill walkthrough

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A0 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P01-01. Parallel opportunities: P02-03

Objective acceptance criteria

- □ Record an ADR choosing an established migration framework or justified ordered runner, including PostgreSQL authority, SQLite support, locks, expand/contract, backfills, rollback, and forward repair.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	PostgreSQL migrations, zero-downtime deployment
Higher-capability review	Mandatory before merge

P02-02 — Introduce versioned migration runner and readiness checks

Field	Plan
Repository / base	`app` (`main`)
Project item	P02-02
Gate / size / readiness	A0 / Large / Dependencies unmet
Dependencies	P02-01
Blocking decisions/checkpoints	D-MIG-01
Source / salvage	Current init_db startup path

Objective

Implement the chosen ordered migration mechanism, checksum ledger, advisory lock, startup mode, and readiness signal without moving product tables yet.

Scope

- Version/checksum ledger and PostgreSQL advisory locking
- Separate migrate job/startup fail-closed policy
- Readiness reports pending/failed migration without sensitive detail

Explicit non-goals

- No Node 0/commercial tables
- No destructive down migration

Expected files or modules

- `migrations/`
- `app/storage/migrations.py`
- `app/readiness.py`
- `app/main.py`
- `tests/python/test_migrations.py`

Change map

Area	Planned change
Schema/model	Add migration metadata only.
API/routes	Add internal readiness detail behind safe health surface.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Bootstrap current database at a verified baseline.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Two concurrent migrators
- Checksum mismatch
- Partial failure and forward repair
- old app during expand window
- SQLite mode per ADR

Local validation:

- `python -m unittest discover -s tests/python`

- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A0 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P02-01. Parallel opportunities: None inside this package beyond review/test preparation.

Objective acceptance criteria

- □ Implement the chosen ordered migration mechanism, checksum ledger, advisory lock, startup mode, and readiness signal without moving product tables yet.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	PostgreSQL, migration framework, runtime operations
Higher-capability review	Mandatory before merge

P02-03 — Baseline canonical schema and prove expand-contract repair

Field	Plan
Repository / base	`app` (`main`)
Project item	P02-03
Gate / size / readiness	A1 / Large / Dependencies unmet
Dependencies	P02-02
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	app/storage/schema.py current definitions, PR #122 forward-migration tests after reconciliation

Objective

Move the existing canonical schema under version control, preserve compatible reads/writes, and supply rehearsal/backfill/repair tooling for future packages.

Scope

- Baseline existing tables and indexes
- Remove startup DDL only after parity is proved

- Document restore/forward-repair procedure

Explicit non-goals

- No semantic Node 0 split
- No commercial schema

Expected files or modules

- `migrations/versions/*_canonical_baseline.py`
- `app/storage/schema.py`
- `scripts/rehearse_migrations.py`
- `tests/migrations/`

Change map

Area	Planned change
Schema/model	Existing canonical schema becomes a checksummed baseline.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Expand/contract rehearsal from previous main and representative legacy fixtures.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Fresh install
- Upgrade from captured pre-baseline schema
- previous app compatibility
- large-table backfill rehearsal
- failure injection

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P02-02. Parallel opportunities: P02-01, P02-04

Objective acceptance criteria

- □ Move the existing canonical schema under version control, preserve compatible reads/writes, and supply rehearsal/backfill/repair tooling for future packages.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

P02-04 — Move intelligence persistence to a revisioned migration namespace

Field	Plan
Repository / base	`intelligence` (`main`)
Project item	P02-04
Gate / size / readiness	A1 / Large / Dependencies unmet
Dependencies	P02-01, P02-02
Blocking decisions/checkpoints	D-MIG-01
Source / salvage	src/natalyx_intelligence/shared/persistence.py runtime _ensure_schema

Objective

Replace intelligence runtime _ensure_schema DDL with the chosen framework, a separate revision namespace/advisory lock, deploy-time migration entrypoint, and cross-service compatibility evidence.

Scope

- Fingerprint and baseline intelligence-owned tables
- Separate revision/lock key from app
- Remove runtime DDL authority after adoption
- Shared-Postgres compatibility matrix

Explicit non-goals

- No context/product feature
- No app-owned table mutation

Expected files or modules

- `migrations/`
- `src/natalyx_intelligence/shared/persistence.py`
- `scripts/migrate.py`
- `tests/test_migrations.py`

Change map

Area	Planned change
Schema/model	Intelligence-owned tables enter a checksummed revision chain.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Deploy-controlled one-off task; runtime verifies head only.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Fresh/legacy PostgreSQL
- concurrent app/intelligence migrators
- old/new service compatibility
- interrupted migration
- runtime non-owner

Local validation:

- `uv run pytest`
- Run focused contract/migration/safety tests with redacted fixtures.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P02-01, P02-02. Parallel opportunities: P02-03

Objective acceptance criteria

- □ Replace intelligence runtime _ensure_schema DDL with the chosen framework, a separate revision namespace/advisory lock, deploy-time migration entrypoint, and cross-service compatibility evidence.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	PostgreSQL migrations, FastAPI service operations
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain
`intelligence`	advisory inference, scoped response/citation contract; no case truth

Merge order inside this package: `P02-01` → `P02-02` → `P02-03` → `P02-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Fresh and upgraded PostgreSQL; Concurrent migrators; Old/new app compatibility; Interrupted migration and forward repair; SQLite support decision	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Type	ID / risk	Owner	Affected PRs / latest point	Project item / treatment
Blocking/product/compliance decision	D-MIG-01 — Adopt Alembic or justify a custom ordered runner; decide supported SQLite parity, expand/contract, locking, rollback and forward-repair contract.	Luke	P02-01, P02-02, P02-03; Before P02-02	D-MIG-01 — Prefer an established framework unless the ADR proves current SQL compatibility makes it unsafe or disproportionate.
Implementation risk	Irreversible DDL	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Concurrent deploy race	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Long backfill lock	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	SQLite semantics hide PostgreSQL failure	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P02-03	Retain pending Alan reconciliation; selective port or rebase only after architecture approval	Reinspected at latest head. It is exactly e853821 (no newer delta); retention strengthened, but final extraction waits for Alan checkpoint.

Package source inventory:

- Current app/storage/schema.py definitions as baseline evidence; no direct copy of ad-hoc ensure_column sequencing

9. Ownership recommendation

Alan owns the technical proposal and implementation. Luke approves the architecture/compatibility contract. A higher-capability review is mandatory before merge.

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P02-01	Alan	Alan (proposed)	Luke	Luke	Mandatory
P02-02	Alan	Alan (proposed)	Luke	Luke	Mandatory
P02-03	Alan	Alan (proposed)	Luke	Luke	Mandatory
P02-04	Alan	Alan (proposed)	Luke	Luke	Mandatory

Luke decides product/lifecycle/commercial/privacy/PHI/provider policy and accepts gates. Alan proposes implementation design, decomposition, migrations, code, tests, rollout and repair. This is a recommendation pending Alan's review/sign-off, not a compulsory assignment.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 03 — Authentication, invitations, and resume

Planning-only execution contract. Proposed implementers and owners remain recommendations until Alan reviews/signs off the consolidated plan and Luke accepts it. Project: MVP Product Flow Reshape.

Plan	Value
Work package	WP03
Plan file	`03-auth-invitations-and-resume.md`
Proposed PRs	P03-01, P03-02, P03-03, P03-04
Inspection snapshot	2026-07-10 20:12 America/New_York

1. Purpose and outcome

Problem. Current invitation code mixes low-risk onboarding links with case membership, and existing flows need explicit recipient binding, return-to safety, consent, expiry, revocation, duplicate-claim, and mismatch semantics.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Separate workspace-claim and paid-case invitation lifecycles with safe authentication/resume behavior and no link-possession path to sensitive case access.

Explicit non-goals.

- No automatic paid-case membership
- No email deliverability product
- No commercial promotion
- No PHI enablement

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-006	verified	app / PR #122@e853821	PR comment Review round 2 addressed and changed workflow/tests	Updated PR #122 reports 499 tests plus return_to, TTL, CAS, self-invite, lens and Learn fail-closed fixes; green checks were observed. These are salvage candidates, not blanket merge approval.
E-011	verified	app / PR #124@84a5a84	app/storage/node0.py:400-462; app/routes/node0.py:605-652	Node 0 can issue an unaddressed GC invitation and redemption ultimately creates case membership, conflating low-risk onboarding possession with case access.
E-012	verified	app / PR #122@e853821	app/storage/invitations.py and PR review-round comment	The updated case invitation work adds 14-day ceiling, CAS transitions, mismatch confirmation, self-invite refusal, return_to preservation and paired lens tests; it has no Node 0 workspace target.

3. Architecture and contracts

- An email-less link may create a pending low-risk workspace claim only.
- Paid-case membership requires authenticated identity binding, recipient or issuer confirmation as appropriate, explicit consent, and auditable redemption.
- Return targets are internal allowlisted paths and survive login/signup/confirmation without becoming redirect authority.

Contract area	Package rule
Ownership boundary	`app`, `infra`; repository-specific PRs remain separate.
Entities/state	`case_invitation`, `entry_session`, `workspace_claim`
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Auth routes accept only allowlisted relative return targets. Issue, inspect, bind, consent, claim, revoke endpoints. Separate paid-case redeem/confirm flow. Client metadata carries opaque admission proof only.
Storage	Workspace claims and audit/idempotency fields. Case-invitation identity, consent, expiry, revocation, idempotency fields.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Applicable state machines

Participant entry and routing (`entry_session`)

Owner: natalyx-app. States: `unstarted`, `education`, `identity_required`, `authenticated`, `routed`, `handed_off`, `expired`.

unstarted

From	unstarted
To	education
Actor	visitor
Evidence	GET/explicit entry choice
Idempotency key	entry:{session}:start
Invalid transitions	No participant/case write
Audit event	entry_started
Retry	Return current session
Recovery	Create fresh anonymous session

education

From	education
To	identity_required
Actor	visitor/policy
Evidence	route requires saved workspace
Idempotency key	entry:{session}:identity
Invalid transitions	Cannot grant workspace/case
Audit event	identity_required
Retry	Preserve safe return intent
Recovery	Restart at public choice

identity_required

From	identity_required
To	authenticated
Actor	Cognito/auth provider
Evidence	verified subject/session
Idempotency key	auth:{subject}:{nonce}
Invalid transitions	Invalid/expired token
Audit event	identity_bound
Retry	Replay returns authenticated session
Recovery	Rechallenge; no partial membership

education|authenticated

From	education|authenticated
To	routed
Actor	participant + deterministic policy
Evidence	validated route answers + policy version
Idempotency key	route:{session}:{answer_hash}
Invalid transitions	Unknown/unsafe route goes review/education
Audit event	entry_routed
Retry	Same input returns same decision
Recovery	Allow corrected answers/new decision revision

routed

From	routed
To	handed_off
Actor	participant
Evidence	accepted destination and authorized target
Idempotency key	handoff:{session}:{route_version}
Invalid transitions	Target not authorized/available
Audit event	entry_handed_off
Retry	Resume same safe destination
Recovery	Return to route result with explanation

any nonterminal

From	any nonterminal
To	expired
Actor	system
Evidence	TTL elapsed/revoked session
Idempotency key	entry:{session}:expire
Invalid transitions	No transition from handed_off needed
Audit event	entry_expired
Retry	Idempotent
Recovery	New session; no case/access change

Low-risk workspace identity claim (`workspace_claim`)

Owner: natalyx-app. States: `issued`, `pending_identity`, `identity_bound`, `consented`, `claimed`, `mismatch_pending`, `revoked`, `expired`.

none

From	none
To	issued
Actor	workspace IP
Evidence	selected candidate + allowed claim scope
Idempotency key	claim:{workspace}:{candidate}:{generation}
Invalid transitions	Existing live claim superseded
Audit event	workspace_claim_issued
Retry	Return current token only at issuance
Recovery	Issue new generation; old revoked

issued

From	issued
To	pending_identity
Actor	link holder
Evidence	valid token landing
Idempotency key	claim:{id}:land
Invalid transitions	Expired/revoked token
Audit event	claim_opened
Retry	Safe repeat
Recovery	Authenticate or exit; no data grant

pending_identity

From	pending_identity
To	identity_bound|mismatch_pending
Actor	auth provider
Evidence	verified subject/email and binding policy
Idempotency key	claim:{id}:bind:{subject}
Invalid transitions	Already claimed/different subject
Audit event	claim_identity_bound|claim_mismatch
Retry	CAS no-op or mismatch state
Recovery	Issuer review/new claim

identity_bound

From	identity_bound
To	consented
Actor	GC claimant
Evidence	workspace consent version
Idempotency key	claim:{id}:consent:{version}
Invalid transitions	Stale/missing consent
Audit event	claim_consented
Retry	Idempotent
Recovery	Display current consent

consented

From	consented
To	claimed
Actor	claim service
Evidence	authorized low-risk workspace membership transaction
Idempotency key	claim:{id}:complete
Invalid transitions	Case membership requested or conflicting claimant
Audit event	workspace_claimed
Retry	Return same membership
Recovery	Transaction rollback/forward repair

issued|pending_identity|identity_bound|consented|mismatch_pending

From	issued|pending_identity|identity_bound|consented|mismatch_pending
To	revoked|expired
Actor	issuer/system
Evidence	revocation or TTL
Idempotency key	claim:{id}:terminal:{event}
Invalid transitions	No terminal overwrite
Audit event	claim_revoked|claim_expired
Retry	Idempotent
Recovery	Fresh claim only

Paid-case membership invitation (`case_invitation`)

Owner: natalyx-app. States: `issued`, `pending_identity`, `identity_bound`, `mismatch_pending`, `issuer_confirmed`, `consented`, `redeemed`, `declined`, `revoked`, `expired`, `superseded`.

none

From	none
To	issued
Actor	authorized IP case member
Evidence	enrollment_pending/active case + recipient binding
Idempotency key	case-invite:{case}:{role}:{generation}
Invalid transitions	Self/member/active-GC/unauthorized
Audit event	case_invite_issued
Retry	Return current; supersede prior live GC invite
Recovery	Issue corrected invite

issued

From	issued
To	pending_identity
Actor	recipient
Evidence	valid addressed token landing
Idempotency key	case-invite:{id}:land
Invalid transitions	Expired/revoked
Audit event	case_invite_opened
Retry	Safe repeat
Recovery	Authenticate with return_to

pending_identity

From	pending_identity
To	identity_bound|mismatch_pending
Actor	auth provider
Evidence	verified subject/email
Idempotency key	case-invite:{id}:bind:{subject}
Invalid transitions	Terminal/duplicate/different subject
Audit event	case_invite_bound|mismatch
Retry	CAS protects terminal
Recovery	Issuer confirmation or corrected invite

mismatch_pending

From	mismatch_pending
To	issuer_confirmed
Actor	authorized issuer
Evidence	explicit reviewed mismatch confirmation
Idempotency key	case-invite:{id}:confirm:{issuer}
Invalid transitions	Issuer unauthorized/recipient changed
Audit event	case_invite_mismatch_confirmed
Retry	CAS no-op
Recovery	Revoke and reissue

identity_bound|issuer_confirmed

From	identity_bound|issuer_confirmed
To	consented
Actor	invitee
Evidence	paid-case role/privacy consent version
Idempotency key	case-invite:{id}:consent:{version}
Invalid transitions	Stale/missing consent
Audit event	case_invite_consented
Retry	Idempotent
Recovery	Present current consent

consented

From	consented
To	redeemed
Actor	invitation service
Evidence	single transaction creates person/party/membership
Idempotency key	case-invite:{id}:redeem
Invalid transitions	Existing member/active GC/expired/revoked
Audit event	case_invite_redeemed
Retry	Return same membership
Recovery	Rollback entire transaction; forward repair from invite

issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented

From	issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented
To	declined|revoked|expired|superseded
Actor	invitee/issuer/system
Evidence	explicit decline/revoke/TTL/new generation
Idempotency key	case-invite:{id}:terminal:{event}
Invalid transitions	No terminal overwrite
Audit event	case_invite_terminal
Retry	Idempotent CAS
Recovery	Fresh invitation only

Invitation and identity safety

Retain from updated PR #122:

- hashed tokens
- purpose-specific TTL with a 14-day ceiling
- CAS terminal transitions
- atomic membership creation
- one-live-GC constraint
- self/existing-member refusal
- safe return_to
- portable integrity/concurrency tests

Required replacement contract:

- Exactly one target: workspace_id or case_id; include purpose, role, access_class, recipient_binding_mode and policy version.
- Email-less possession may only open pending low-risk workspace claim; it cannot reveal sensitive data or grant paid-case membership.
- Paid-case invitation is addressed, authenticated, consented and atomically redeemed; mismatch requires explicit policy/issuer handling and preferably fresh bound token.
- Cognito admission and app membership are separate decisions; account creation never equals workspace/case access.
- Verified identity comes from auth claims, not form-entered email; duplicate claims return the original result.
- Never put raw token/email/PHI in general participant timeline payloads; revoke/expire/supersede fail closed.
- Remove generic unconditional status mutation; expose authorized transition commands only.

4. Explicit PR sequence

1

Order	1
ID	P03-01
Title	Harden auth return-to and participant resume
Repository	app
Dependencies	P01-01
Gate	A1
Size	Medium
Readiness	Dependencies unmet

2

Order	2
ID	P03-02
Title	Add low-risk workspace claim lifecycle
Repository	app
Dependencies	P03-01, P04-01
Gate	A1
Size	Large
Readiness	Dependencies unmet

3

Order	3
ID	P03-03
Title	Bind paid-case invitations to identity and consent
Repository	app
Dependencies	P03-01, P04-01
Gate	A2
Size	Large
Readiness	Dependencies unmet

4

Order	4
ID	P03-04
Title	Replace shared signup admission with claim-aware Cognito admission
Repository	infra
Dependencies	P03-02, P03-03
Gate	A1
Size	Large
Readiness	Blocked by decision

P03-01 — Harden auth return-to and participant resume

Field	Plan
Repository / base	`app` (`main`)
Project item	P03-01
Gate / size / readiness	A1 / Medium / Dependencies unmet
Dependencies	P01-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PR #122 return_to fixes

Objective

Make signup/login/confirmation/logout preserve a validated internal destination and restore the correct role-scoped workspace without query parameters becoming authority.

Scope

- Central safe_return_to parser
- Resume intent across login/signup/confirm/resend
- Session/cookie security and error states

Explicit non-goals

- No invitation issuance

- No case access grant

Expected files or modules

- `app/auth/routes.py`
- `app/auth/cloud.py`
- `app/auth/cognito.py`
- `tests/python/test_auth_routes.py`
- `tests/python/test_auth_cloud_integration.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Auth routes accept only allowlisted relative return targets.
UI/product surface	Clear expired/failed session recovery.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Open redirect corpus
- logged-out deep link
- signup confirmation
- logout/login resume
- secure/samesite cookie matrix

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P01-01. Parallel opportunities: P03-04

Objective acceptance criteria

- □ Make signup/login/confirmation/logout preserve a validated internal destination and restore the correct role-scoped workspace without query parameters becoming authority.
- □ Every listed objective test passes locally and at the required gate where applicable.

- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	Cognito, web session security, FastAPI
Higher-capability review	Risk-proportionate; not mandatory by programme default

P03-02 — Add low-risk workspace claim lifecycle

Field	Plan
Repository / base	`app` (`main`)
Project item	P03-02
Gate / size / readiness	A1 / Large / Dependencies unmet
Dependencies	P03-01, P04-01
Blocking decisions/checkpoints	D-AUTH-01, CHECK-ALAN-122-124
Source / salvage	PRs #123/#124, dirty onboarding-link worktree

Objective

Create a pending Node 0 workspace claim that can start from an unaddressed link but requires identity binding, consent, expiry, revocation, and audit before low-risk access.

Scope

- Hashed one-time token and claim state machine
- Identity bind/consent/revoke/duplicate/mismatch behavior
- No sensitive case scope

Explicit non-goals

- No paid-case membership
- No possession-only private access

Expected files or modules

- `app/storage/workspace_claims.py`
- `app/routes/node0_claims.py`
- `app/templates/node0/claim*.html`
- `tests/python/test_workspace_claims.py`

Change map

Area	Planned change
Schema/model	Workspace claims and audit/idempotency fields.
API/routes	Issue, inspect, bind, consent, claim, revoke endpoints.
UI/product surface	Low-risk claim and mismatch screens.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Forwarded link
- duplicate/concurrent claim
- expiry/revocation
- identity mismatch
- issuer visibility

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P03-01, P04-01. Parallel opportunities: P03-03

Objective acceptance criteria

- □ Create a pending Node 0 workspace claim that can start from an unaddressed link but requires identity binding, consent, expiry, revocation, and audit before low-risk access.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

P03-03 — Bind paid-case invitations to identity and consent

Field	Plan
Repository / base	`app` (`main`)
Project item	P03-03
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P03-01, P04-01
Blocking decisions/checkpoints	D-AUTH-01, CHECK-ALAN-122-124
Source / salvage	PR #122 CAS/TTL/self-invite fixes, current invitations.py

Objective

Implement a separate recipient-bound paid-case invitation that grants membership only after authenticated binding, required confirmation, consent, and case/role checks.

Scope

- Case invite CAS transitions
- Recipient/issuer mismatch confirmation
- Consent-version capture and membership transaction

Explicit non-goals

- No workspace token reuse
- No automatic GC migration from soft onboarding

Expected files or modules

- `app/storage/case_invitations.py`
- `app/case_members.py`
- `app/routes/case.py`
- `tests/python/test_case_invitations.py`

Change map

Area	Planned change
Schema/model	Case-invitation identity, consent, expiry, revocation, idempotency fields.
API/routes	Separate paid-case redeem/confirm flow.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Membership grant is a single authorized transaction.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Replay
- self-invite
- active-member duplicate
- mismatch
- expired/revoked
- concurrent redemption rollback

Local validation:

- `python -m unittest discover -s tests/python`

- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P03-01, P04-01. Parallel opportunities: P03-02

Objective acceptance criteria

- □ Implement a separate recipient-bound paid-case invitation that grants membership only after authenticated binding, required confirmation, consent, and case/role checks.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

P03-04 — Replace shared signup admission with claim-aware Cognito admission

Field	Plan
Repository / base	`infra` (`main`)
Project item	P03-04
Gate / size / readiness	A1 / Large / Blocked by decision
Dependencies	P03-02, P03-03
Blocking decisions/checkpoints	D-AUTH-02
Source / salvage	infra platform-stack.ts current allowlist + shared invite code

Objective

Keep internal/dev allowlisting but add a short-lived, recipient-bound admission proof for approved workspace/case claimants so a shared invite code is never the participant trust boundary.

Scope

- One-time signed or server-verified admission proof
- Email/purpose/claim/expiry binding
- Replay/revocation/audit

- No membership grant in Cognito

Explicit non-goals

- No open self-signup
- No Cognito group as case authority
- No shared user-entered master code

Expected files or modules

- `lib/stacks/platform-stack.ts`
- `lib/constructs/auth-admission.ts`
- `test/auth-admission.test.ts`
- `docs/agents/platform-stack.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Client metadata carries opaque admission proof only.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	Harden pre-signup Lambda and key/claim verification contract; keep missing config fail closed.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Wrong email/purpose
- expired/replayed proof
- revoked claim
- internal allowlist path
- missing key/config
- membership still denied before app redemption

Local validation:

- `npm test`
- `npm run build`
- `Synthesize the affected CDK stack with billable/live changes disabled.`

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Synthesize and review first; deploy only through the guarded dev workflow with billable/desired-count defaults unchanged, exact revision recorded, and explicit confirmation outside this planning task.

Rollback or forward repair

Return to the prior reviewed stack/task definition when compatible; otherwise keep resources fail-closed and forward-repair under the runbook. Never delete data-bearing resources as an automatic rollback.

Dependencies and parallel work

Hard dependencies: P03-02, P03-03. Parallel opportunities: P03-01

Objective acceptance criteria

- □ Keep internal/dev allowlisting but add a short-lived, recipient-bound admission proof for approved workspace/case claimants so a shared invite code is never the participant trust boundary.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	Cognito, AWS Lambda/KMS, identity security
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain
`infra`	AWS resources, IAM/KMS/network/deployment/backup/audit contracts

Merge order inside this package: `P03-01` → `P03-02` → `P03-03` → `P03-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Logged-out resume; Token replay/expiry/revocation; Duplicate and concurrent claims; Identity mismatch; Workspace-to-case access escalation	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Type	ID / risk	Owner	Affected PRs / latest point	Project item / treatment
Blocking/product/compliance decision	D-AUTH-01 — Which low-risk fields may a link-only claimant see, when is recipient/issuer confirmation required, and what consent/version binds each access elevation?	Luke	P03-02, P03-03; Before P03-02	D-AUTH-01 — Link possession creates pending workspace claim only; paid-case access always requires authenticated binding and explicit consent.
Blocking/product/compliance decision	D-AUTH-02 — Choose a short-lived signed/server-verified participant admission proof and key/claim lifecycle while retaining a separate internal allowlist path.	Luke	P03-04; Before P03-04	D-AUTH-02 — Bind proof to email, purpose, claim ID and expiry; it admits account creation only and never grants workspace/case membership.
Implementation risk	Shared links forwarded	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Account mismatch	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Confused-deputy redirects	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Workspace data exposed during upgrade	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P03-01, P03-03	Retain pending Alan reconciliation; selective port or rebase only after architecture approval	Reinspected at latest head. It is exactly e853821 (no newer delta); retention strengthened, but final extraction waits for Alan checkpoint.
app#123	34078f810b63	GC onboarding form/copy, hosted auth return, invite handoff and role-specific surface work.	P03-01, P03-02	Selective UX/test port and rewrite	Unextracted; likely partially copied into #124 but not proven complete.
feature/node0-surrogate-onboarding-link	529d377	substantial uncommitted onboarding/invitation work	P03-02/P03-03/P06-02	Selective port from protected worktree	Preserve until file/test extraction is signed off.

Package source inventory:

- PR #122 return_to, CAS, TTL, self-invite and mismatch fixes
- PRs #123/#124 hosted GC onboarding UX
- Dirty onboarding-link worktree

9. Ownership recommendation

Alan proposes implementation and threat-model details; Luke approves role/privacy policy. Replacement work waits for the Alan PR-reconciliation checkpoint.

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P03-01	Alan	Alan (proposed)	Luke	Luke	As risk warrants
P03-02	Alan	Alan (proposed)	Luke	Luke	Mandatory
P03-03	Alan	Alan (proposed)	Luke	Luke	Mandatory
P03-04	Alan	Alan (proposed)	Luke	Luke	Mandatory

Luke decides product/lifecycle/commercial/privacy/PHI/provider policy and accepts gates. Alan proposes implementation design, decomposition, migrations, code, tests, rollout and repair. This is a recommendation pending Alan's review/sign-off, not a compulsory assignment.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 04 — Node 0 workspace and paid-case separation

Planning-only execution contract. Proposed implementers and owners remain recommendations until Alan reviews/signs off the consolidated plan and Luke accepts it. Project: MVP Product Flow Reshape.

Plan	Value
Work package	WP04

Plan	Value
Plan file	`04-node0-workspace-case-separation.md`
Proposed PRs	P04-01, P04-02, P04-03
Inspection snapshot	2026-07-10 20:12 America/New_York

1. Purpose and outcome

Problem. PR #124 creates a synthetic canonical case as soon as Node 0 is opened, collapsing free exploration, workspace membership, commercial commitment, and paid-case truth.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. A distinct Node 0 workspace aggregate with attributable revisions and an idempotent, purpose-limited projection into one paid case only after the authorized commercial event.

Explicit non-goals.

- No payment processor
- No full Node 0 UI
- No clinical or legal truth promotion
- No deletion of legacy rows during compatibility window

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-010	verified	app / PR #124@84a5a84	app/routes/node0.py:63-90	Opening Node 0 calls create_byos_case_for_ip with is_synthetic=True; there is no separate free workspace or authorized promotion.
E-011	verified	app / PR #124@84a5a84	app/storage/node0.py:400-462; app/routes/node0.py:605-652	Node 0 can issue an unaddressed GC invitation and redemption ultimately creates case membership, conflating low-risk onboarding possession with case access.
E-013	verified	app / PR #124@84a5a84	app/node0.py:252-482,622-878,975-1001	Nineteen IP-entered candidate answers about relationship and third-party circumstances/health derive a 0-100 score, readiness/status, ordering, guide route and evidence packet without per-field provenance or verification.
E-014	verified	app / PR #124@84a5a84	app/storage/node0.py:42-200,509-600; tests/python/test_node0.py:444-490	IP/co-IP own candidate reads; GC onboarding writes separate shared/private facts and does not revise the IP assertion. The GC cannot see or correct the IP candidate record.
E-015	verified	app / PR #122@e853821	app/storage/facts.py:19-69,107-127	Facts have broad provenance and visibility values but no subject-versus-author role, verification state, derived-from, or supersedes link.

3. Architecture and contracts

- Workspace, entitlement, commercial, payment, and case lifecycles are separate entities and state machines.
- Promotion copies only allowlisted projections with provenance, visibility, and verification intact.
- Compatibility adapters read old case-shaped Node 0 records while new writes target the workspace model.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	`node0_workspace`
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	No broad public contract; use the focused commands/events in each PR.
Storage	Add node0_workspaces, workspace_memberships, candidate assertion linkage and constraints. Append-only assertion/revision and projection-audit records. Idempotent backfill with source identifiers and exception report.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Applicable state machines

Node 0 workspace (`node0_workspace`)

Owner: natalyx-app. States: `draft`, `active`, `match_ready`, `promotion_pending`, `promoted`, `archived`.

none

From	none
To	draft
Actor	authenticated IP
Evidence	first explicit Node 0 start
Idempotency key	workspace:{owner}:create
Invalid transitions	Existing active workspace returns existing
Audit event	workspace_created
Retry	Unique owner/idempotency key
Recovery	Forward repair orphan membership

draft

From	draft
To	active
Actor	IP
Evidence	minimum workspace consent/profile
Idempotency key	workspace:{id}:activate:{revision}
Invalid transitions	Missing consent/owner
Audit event	workspace_activated
Retry	Return current active revision
Recovery	Correct inputs and retry

active

From	active
To	match_ready
Actor	IP + selected GC evidence policy
Evidence	selected candidate + required provisional/GC-authored fields
Idempotency key	workspace:{id}:ready:{revision}
Invalid transitions	Critical unknown/decline/revoked claim
Audit event	workspace_match_ready
Retry	No duplicate event
Recovery	Reopen active with reason revision

match_ready

From	match_ready
To	promotion_pending
Actor	promotion event consumer
Evidence	valid byos.workspace_promotion_authorized.v1
Idempotency key	promotion:{commitment}:v1
Invalid transitions	No authorization; already promoted; stale config
Audit event	workspace_promotion_requested
Retry	Replay resumes/finds same promotion
Recovery	Forward-repair incomplete transaction

promotion_pending

From	promotion_pending
To	promoted
Actor	promotion service
Evidence	paid case + manifest committed
Idempotency key	promotion:{commitment}:v1
Invalid transitions	Projection invariant/privacy failure
Audit event	workspace_promoted
Retry	Return same case ID
Recovery	Repair from immutable manifest; never duplicate

draft|active|match_ready

From	draft|active|match_ready
To	archived
Actor	IP/operator per policy
Evidence	explicit archive + retention policy
Idempotency key	workspace:{id}:archive:{request}
Invalid transitions	Promotion pending/promoted
Audit event	workspace_archived
Retry	Idempotent
Recovery	Restore only through new audited revision if policy permits

Data authorship/provenance rules

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.

- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

4. Explicit PR sequence

1

Order	1
ID	P04-01
Title	Separate Node 0 workspace storage from paid cases
Repository	app
Dependencies	P02-03
Gate	A1
Size	Exceptional
Readiness	Dependencies unmet

2

Order	2
ID	P04-02
Title	Add attributable workspace revision and projection contracts
Repository	app
Dependencies	P04-01
Gate	A1
Size	Large
Readiness	Dependencies unmet

3

Order	3
ID	P04-03
Title	Adapt legacy Node 0 case-shaped data without semantic promotion
Repository	app
Dependencies	P04-02
Gate	A1
Size	Large
Readiness	Dependencies unmet

P04-01 — Separate Node 0 workspace storage from paid cases

Field	Plan
Repository / base	`app` (`main`)
Project item	P04-01
Gate / size / readiness	A1 / Exceptional / Dependencies unmet
Dependencies	P02-03
Blocking decisions/checkpoints	CHECK-ALAN-122-124
Source / salvage	PR #124 _ensure_node0_case as behavior to replace, candidate worktrees

Objective

Introduce workspace, workspace membership, candidate, and selection entities so opening Node 0 never creates a canonical paid case.

Scope

- Workspace aggregate and memberships
- Candidate rows linked to workspace
- Case tables remain paid/manageable aggregate

Explicit non-goals

- No UI rewrite
- No promotion
- No legacy deletion

Expected files or modules

- `migrations/versions/*_node0_workspaces.py`
- `app/storage/workspaces.py`
- `app/storage/workspace_memberships.py`
- `app/storage/node0.py`

Change map

Area	Planned change
Schema/model	Add node0_workspaces, workspace_memberships, candidate assertion linkage and constraints.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Expand only; old records remain readable.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Opening Node 0 creates no case
- workspace authorization
- candidate isolation
- concurrent workspace creation
- old app compatibility

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P02-03. Parallel opportunities: P04-03

Objective acceptance criteria

- □ Introduce workspace, workspace membership, candidate, and selection entities so opening Node 0 never creates a canonical paid case.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	domain modelling, PostgreSQL, authorization
Higher-capability review	Mandatory before merge

P04-02 — Add attributable workspace revision and projection contracts

Field	Plan
Repository / base	`app` (`main`)
Project item	P04-02
Gate / size / readiness	A1 / Large / Dependencies unmet
Dependencies	P04-01
Blocking decisions/checkpoints	D-DATA-01, D-DATA-02, CHECK-ALAN-122-124
Source / salvage	Current facts/memory models, GC authorship audit

Objective

Represent IP assertions, GC confirmations, private notes, and reviewer/provider confirmations as attributable append-only revisions with purpose-limited projections.

Scope

- Authorship classes and supersedes links
- Verification/visibility/purpose metadata
- Projection allowlists for IP, GC, case, intelligence, provider

Explicit non-goals

- No provider transmission
- No silent in-place overwrite

Expected files or modules

- `migrations/versions/*_authored_assertions.py`
- `app/storage/assertions.py`
- `app/domain/projections.py`
- `tests/python/test_authorship_projections.py`

Change map

Area	Planned change
Schema/model	Append-only assertion/revision and projection-audit records.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Apply through the named versioned migration; no ad-hoc startup DDL. Backfill is none unless explicitly listed in scope.
Security/privacy	Default deny between roles; explicit purpose projection.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- IP provisional remains attributed
- GC revision does not erase source
- private cross-role denial
- AI cannot upgrade verification
- provider projection allowlist

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P04-01. Parallel opportunities: None inside this package beyond review/test preparation.

Objective acceptance criteria

- □ Represent IP assertions, GC confirmations, private notes, and reviewer/provider confirmations as attributable append-only revisions with purpose-limited projections.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	data provenance, privacy, authorization
Higher-capability review	Mandatory before merge

P04-03 — Adapt legacy Node 0 case-shaped data without semantic promotion

Field	Plan
Repository / base	`app` (`main`)
Project item	P04-03
Gate / size / readiness	A1 / Large / Dependencies unmet
Dependencies	P04-02
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PRs #117/#121/#123/#124 and dirty candidate worktrees

Objective

Provide a compatibility reader and controlled backfill that maps existing Node 0 rows to workspaces while preserving provenance and never treating legacy scores as verified truth.

Scope

- Dry-run audit and deterministic mapping
- Dual-read compatibility window
- Forward repair and per-row exception ledger

Explicit non-goals

- No destructive legacy cleanup
- No case creation

Expected files or modules

- `migrations/versions/*_node0_legacy_backfill.py`
- `app/storage/node0_compat.py`
- `scripts/audit_node0_backfill.py`
- `tests/fixtures/node0_legacy/`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Idempotent backfill with source identifiers and exception report.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Legacy fixtures

- retry after interruption
- duplicate prevention
- mixed old/new writes
- privacy/provenance preservation

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Counts/checksums and unmapped-row alerts.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P04-02. Parallel opportunities: P04-01

Objective acceptance criteria

- □ Provide a compatibility reader and controlled backfill that maps existing Node 0 rows to workspaces while preserving provenance and never treating legacy scores as verified truth.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P04-01` → `P04-02` → `P04-03`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	One workspace/one promotion; Retry/concurrency; No implicit fact upgrade; Private candidate isolation; Old/new revision coexistence	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Type	ID / risk	Owner	Affected PRs / latest point	Project item / treatment
Blocking/product/compliance decision	D-DATA-01 — Which provisional IP assertions may the GC see, which GC answers may the IP see, and which fields survive promotion/provider projection?	Luke	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02	D-DATA-01 — Default deny; expose only explicit purpose-limited projections. Class 1/2 and class 4 never automatically cross roles.
Blocking/product/compliance decision	D-DATA-02 — How long to retain unselected candidate assertions and how are GC corrections, conflicting relationship answers, and private IP notes represented?	Luke	P04-02, P04-03, P06-02; Before P04-02	D-DATA-02 — Append revisions with supersedes/derived-from; never silently erase source; shortest justified retention for unselected candidates.
Implementation risk	Semantic data corruption during backfill	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Duplicate cases	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Private candidate leakage	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Older app revision writes legacy shape	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#117	5ac09f0cdd92	Deterministic candidate questionnaire, scoring/status calculation, conversation-guide resource, routes and extensive Node 0 tests.	P04-01, P04-02	Selective port and rewrite	Unextracted; algorithm/test review required after Alan checkpoint and Luke field/score decision.
app#119	a8cd2f428535	Concise product-leverage framing and boundaries for the MVP flow.	P04-01	Retain as source or merge separately after plan decision	Catalogued; reconcile terminology into approved programme.
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P04-02	Retain pending Alan reconciliation; selective port or rebase only after architecture approval	Reinspected at latest head. It is exactly e853821 (no newer delta); retention strengthened, but final extraction waits for Alan checkpoint.
app#124	84a5a847fea7	Current consolidation of Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitations, DS assets and large focused test suites.	P04-01, P04-02, P04-03	Retain unchanged as evidence; selective port/rewrite after checkpoint	Unextracted; Alan review explicitly incomplete and unresolved.
detached candidate rewrite	9605d88	substantial uncommitted Node 0 quiz rewrite	P04-02/P05-02	Selective port from protected worktree	Preserve; compare with predecessor and #117/#124 before any discard.
detached predecessor candidate model	dc6a2f0	substantial uncommitted predecessor model	P04-02/P05-02 if unique	Selective port from protected worktree	Preserve until rewrite diff/extraction ledger is complete.

Package source inventory:

- PR #124 candidate/outreach behavior and tests by selective port
- PR #122 case projection primitives after reconciliation
- Older candidate-model and rewrite worktrees

9. Ownership recommendation

Alan owns domain/schema implementation; Luke approves entity boundaries and promotion contract. Joint high-capability review is mandatory.

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P04-01	Alan	Alan (proposed)	Luke	Luke	Mandatory
P04-02	Alan	Alan (proposed)	Luke	Luke	Mandatory
P04-03	Alan	Alan (proposed)	Luke	Luke	Mandatory

Luke decides product/lifecycle/commercial/privacy/PHI/provider policy and accepts gates. Alan proposes implementation design, decomposition, migrations, code, tests, rollout and repair. This is a recommendation pending Alan's review/sign-off, not a compulsory assignment.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.

- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 05 — Node 0 Free, Pathfinder, and resume

Planning-only execution contract. Proposed implementers and owners remain recommendations until Alan reviews/signs off the consolidated plan and Luke accepts it. Project: MVP Product Flow Reshape.

Plan	Value
Work package	WP05
Plan file	`05-node0-free-pathfinder-and-resume.md`
Proposed PRs	P05-01, P05-02, P05-03
Inspection snapshot	2026-07-10 20:12 America/New_York

1. Purpose and outcome

Problem. Pathfinder contains service/LLM-era routing complexity and Node 0 Free must satisfy the complete-the-match boundary without payment while preserving explicit IP authorship.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. A deterministic entry router and resumable Node 0 Free flow for education, candidate tracking, preliminary safety checks, outreach guidance, and a willing candidate without a paid case.

Explicit non-goals.

- No weighted ranking
- No GC authenticated soft onboarding
- No paid case
- No final Learn content

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-010	verified	app / PR #124@84a5a84	app/routes/node0.py:63-90	Opening Node 0 calls create_byos_case_for_ip with is_synthetic=True; there is no separate free workspace or authorized promotion.
E-013	verified	app / PR #124@84a5a84	app/node0.py:252-482,622-878, 975-1001	Nineteen IP-entered candidate answers about relationship and third-party circumstances/health derive a 0-100 score, readiness/status, ordering, guide route and evidence packet without per-field provenance or verification.

3. Architecture and contracts

- Routing is deterministic policy, not LLM authority.
- Candidate responses are provisional IP assertions, never GC-authored or medically verified facts.
- Safety-critical information and complete-the-match capability remain free.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	`entry_session`, `node0_workspace`

Contract area	Package rule
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Deterministic route decision response with policy version. Workspace-scoped candidate and guide commands. UI calls workspace-scoped commands only.
Storage	Use workspace candidates/assertions from WP04.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Applicable state machines

Participant entry and routing (`entry_session`)

Owner: natalyx-app. States: `unstarted`, `education`, `identity_required`, `authenticated`, `routed`, `handed_off`, `expired`.

unstarted

From	unstarted
To	education
Actor	visitor
Evidence	GET/explicit entry choice
Idempotency key	entry:{session}:start
Invalid transitions	No participant/case write
Audit event	entry_started
Retry	Return current session
Recovery	Create fresh anonymous session

education

From	education
To	identity_required
Actor	visitor/policy
Evidence	route requires saved workspace
Idempotency key	entry:{session}:identity
Invalid transitions	Cannot grant workspace/case
Audit event	identity_required
Retry	Preserve safe return intent
Recovery	Restart at public choice

identity_required

From	identity_required
To	authenticated
Actor	Cognito/auth provider
Evidence	verified subject/session
Idempotency key	auth:{subject}:{nonce}
Invalid transitions	Invalid/expired token
Audit event	identity_bound
Retry	Replay returns authenticated session
Recovery	Rechallenge; no partial membership

education|authenticated

From	education|authenticated
To	routed
Actor	participant + deterministic policy
Evidence	validated route answers + policy version
Idempotency key	route:{session}:{answer_hash}
Invalid transitions	Unknown/unsafe route goes review/education
Audit event	entry_routed
Retry	Same input returns same decision
Recovery	Allow corrected answers/new decision revision

routed

From	routed
To	handed_off
Actor	participant
Evidence	accepted destination and authorized target
Idempotency key	handoff:{session}:{route_version}
Invalid transitions	Target not authorized/available
Audit event	entry_handed_off
Retry	Resume same safe destination
Recovery	Return to route result with explanation

any nonterminal

From	any nonterminal
To	expired
Actor	system
Evidence	TTL elapsed/revoked session
Idempotency key	entry:{session}:expire
Invalid transitions	No transition from handed_off needed
Audit event	entry_expired
Retry	Idempotent
Recovery	New session; no case/access change

Node 0 workspace (`node0_workspace`)

Owner: natalyx-app. States: `draft`, `active`, `match_ready`, `promotion_pending`, `promoted`, `archived`.

none

From	none
To	draft
Actor	authenticated IP
Evidence	first explicit Node 0 start
Idempotency key	workspace:{owner}:create
Invalid transitions	Existing active workspace returns existing
Audit event	workspace_created
Retry	Unique owner/idempotency key
Recovery	Forward repair orphan membership

draft

From	draft
To	active
Actor	IP
Evidence	minimum workspace consent/profile
Idempotency key	workspace:{id}:activate:{revision}
Invalid transitions	Missing consent/owner
Audit event	workspace_activated
Retry	Return current active revision
Recovery	Correct inputs and retry

active

From	active
To	match_ready
Actor	IP + selected GC evidence policy
Evidence	selected candidate + required provisional/GC-authored fields
Idempotency key	workspace:{id}:ready:{revision}
Invalid transitions	Critical unknown/decline/revoked claim
Audit event	workspace_match_ready
Retry	No duplicate event
Recovery	Reopen active with reason revision

match_ready

From	match_ready
To	promotion_pending
Actor	promotion event consumer
Evidence	valid byos.workspace_promotion_authorized.v1
Idempotency key	promotion:{commitment}:v1
Invalid transitions	No authorization; already promoted; stale config
Audit event	workspace_promotion_requested
Retry	Replay resumes/finds same promotion
Recovery	Forward-repair incomplete transaction

promotion_pending

From	promotion_pending
To	promoted
Actor	promotion service
Evidence	paid case + manifest committed
Idempotency key	promotion:{commitment}:v1
Invalid transitions	Projection invariant/privacy failure
Audit event	workspace_promoted
Retry	Return same case ID
Recovery	Repair from immutable manifest; never duplicate

draft|active|match_ready

From	draft|active|match_ready
To	archived
Actor	IP/operator per policy
Evidence	explicit archive + retention policy
Idempotency key	workspace:{id}:archive:{request}
Invalid transitions	Promotion pending/promoted
Audit event	workspace_archived
Retry	Idempotent
Recovery	Restore only through new audited revision if policy permits

Data authorship/provenance rules

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.
- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

4. Explicit PR sequence

1

Order	1
ID	P05-01
Title	Replace Pathfinder routing with deterministic entry decisions
Repository	app
Dependencies	P03-01, P04-01
Gate	A1
Size	Medium
Readiness	Dependencies unmet

2

Order	2
ID	P05-02
Title	Rebuild Node 0 Free candidate and outreach domain
Repository	app
Dependencies	P04-02, P05-01
Gate	A1
Size	Exceptional
Readiness	Dependencies unmet

3

Order	3
ID	P05-03
Title	Ship resumable Node 0 Free participant surface
Repository	app
Dependencies	P03-01, P05-02, P11-03
Gate	A1
Size	Large
Readiness	Dependencies unmet

P05-01 — Replace Pathfinder routing with deterministic entry decisions

Field	Plan
Repository / base	`app` (`main`)
Project item	P05-01
Gate / size / readiness	A1 / Medium / Dependencies unmet
Dependencies	P03-01, P04-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Current pathfinder_destinations.py, PR #119 product-alignment doc

Objective

Route users predictably to Node 0, full IP setup, GC claim, Learn/Ask education, or provider/partner paths using explicit inputs and policy versioning.

Scope

- Small versioned decision table
- Clear reason/next-step output
- No LLM dependency for route selection

Explicit non-goals

- No final educational content
- No eligibility clearance

Expected files or modules

- `app/pathfinder_destinations.py`
- `app/pathfinder_policy.py`
- `app/routes/participant.py`
- `tests/python/test_pathfinder_policy.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Deterministic route decision response with policy version.
UI/product surface	Clear entry choices and safe review states.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Decision table coverage
- ambiguous/malformed inputs
- GC invite deep link
- service unavailable
- mobile entry

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P03-01, P04-01. Parallel opportunities: P05-03

Objective acceptance criteria

- □ Route users predictably to Node 0, full IP setup, GC claim, Learn/Ask education, or provider/partner paths using explicit inputs and policy versioning.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Risk-proportionate; not mandatory by programme default

P05-02 — Rebuild Node 0 Free candidate and outreach domain

Field	Plan
Repository / base	`app` (`main`)
Project item	P05-02
Gate / size / readiness	A1 / Exceptional / Dependencies unmet
Dependencies	P04-02, P05-01
Blocking decisions/checkpoints	CHECK-ALAN-122-124
Source / salvage	PR #117 direct algorithm/test salvage, PR #124, quiz rewrite worktree

Objective

Port the useful deterministic candidate, preliminary safety, and outreach behavior onto workspace/provenance contracts while preserving the complete-the-match free boundary.

Scope

- Unlimited candidates
- Preliminary hard-criteria check with limitations
- Basic outreach sequencing and shareable intro packet

Explicit non-goals

- No weighted ranking
- No medical clearance
- No paid feature

Expected files or modules

- `app/domain/node0_candidates.py`
- `app/storage/node0.py`
- `app/node0_conversation_guide.py`
- `app/resources/node0_conversation_guide_v1.json`
- `tests/python/test_node0_domain.py`

Change map

Area	Planned change
Schema/model	Use workspace candidates/assertions from WP04.
API/routes	Workspace-scoped candidate and guide commands.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Deterministic metrics
- critical safety blocking
- author attribution
- candidate isolation
- free feature contract

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P04-02, P05-01. Parallel opportunities: None inside this package beyond review/test preparation.

Objective acceptance criteria

- □ Port the useful deterministic candidate, preliminary safety, and outreach behavior onto workspace/provenance contracts while preserving the complete-the-match free boundary.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

P05-03 — Ship resumable Node 0 Free participant surface

Field	Plan
Repository / base	`app` (`main`)
Project item	P05-03
Gate / size / readiness	A1 / Large / Dependencies unmet
Dependencies	P03-01, P05-02, P11-03
Blocking decisions/checkpoints	CHECK-ALAN-122-124
Source / salvage	PRs #121/#124 selective UI port, DS PR #8

Objective

Provide a coherent mobile-capable Node 0 Free workspace shell for candidate tracking, outreach, guide use, and resume without paid-case language.

Scope

- Workspace home and candidate flow
- Honest preliminary labels/empty/error states
- Logout/login resume and mobile navigation

Explicit non-goals

- No Complete-only features
- No paid case/journey hero

Expected files or modules

- `app/routes/node0.py`
- `app/node0_surface.py`
- `app/templates/node0/*.html`
- `app/static/node0.*`
- `tests/python/test_node0_ui.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	UI calls workspace-scoped commands only.
UI/product surface	Participant DS recipes with responsive interaction and no fake content.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Clickable Free path
- resume
- authorization
- no-JS/keyboard
- 320/375/768px
- dependency failure

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P03-01, P05-02, P11-03. Parallel opportunities: P05-01

Objective acceptance criteria

- □ Provide a coherent mobile-capable Node 0 Free workspace shell for candidate tracking, outreach, guide use, and resume without paid-case language.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Risk-proportionate; not mandatory by programme default

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P05-01` → `P05-02` → `P05-03`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	All deterministic route outcomes; Free candidate/outreach boundary; Resume across sessions; Critical-flag fail-safe; Mobile interaction	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Type	ID / risk	Owner	Affected PRs / latest point	Project item / treatment
Blocking/product/compliance decision	D-DATA-03 — Does any 0-100 candidate ranking or race-based preference survive, and what may provisional third-party answers gate?	Luke	P05-02, P06-03; Before P05-02	D-DATA-03 — Remove candidate ranking and race preference; use source-labelled questions-to-clarify. Unverified IP answers cannot gate claim/onboarding/eligibility.
Implementation risk	Paid features leak into Free	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Provisional score is overclaimed	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Route drift	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	LLM dependency reintroduced	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#117	5ac09f0cdd92	Deterministic candidate questionnaire, scoring/status calculation, conversation-guide resource, routes and extensive Node 0 tests.	P05-02, P05-03	Selective port and rewrite	Unextracted; algorithm/test review required after Alan checkpoint and Luke field/score decision.
app#119	a8cd2f428535	Concise product-leverage framing and boundaries for the MVP flow.	P05-01	Retain as source or merge separately after plan decision	Catalogued; reconcile terminology into approved programme.
app#121	529d377e2337	Primary Node 0 participant model/UI, PDF/guide presentation and focused surface tests.	P05-03	Selective UI/test port, then supersede	Unextracted; #124 may contain later copies but equivalence must be proved by file/commit comparison.
app#124	84a5a847fea7	Current consolidation of Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitations, DS assets and large focused test suites.	P05-02, P05-03	Retain unchanged as evidence; selective port/rewrite after checkpoint	Unextracted; Alan review explicitly incomplete and unresolved.
detached candidate rewrite	9605d88	substantial uncommitted Node 0 quiz rewrite	P04-02/P05-02	Selective port from protected worktree	Preserve; compare with predecessor and #117/#124 before any discard.
detached predecessor candidate model	dc6a2f0	substantial uncommitted predecessor model	P04-02/P05-02 if unique	Selective port from protected worktree	Preserve until rewrite diff/extraction ledger is complete.

Package source inventory:

- PR #117 quiz/guide algorithms and tests
- PR #121 primary surface
- PR #124 consolidated UI
- Quiz rewrite worktree

9. Ownership recommendation

Luke owns entry/product boundary; Alan implements and validates. Replacement work waits for the Alan checkpoint and WP04 contracts.

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P05-01	Alan	Alan (proposed)	Luke	Luke	As risk warrants
P05-02	Alan	Alan (proposed)	Luke	Luke	Mandatory
P05-03	Alan	Alan (proposed)	Luke	Luke	As risk warrants

Luke decides product/lifecycle/commercial/privacy/PHI/provider policy and accepts gates. Alan proposes implementation design, decomposition, migrations, code, tests, rollout and repair. This is a recommendation pending Alan's review/sign-off, not a compulsory assignment.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 06 — Node 0 Complete and GC soft onboarding

Planning-only execution contract. Proposed implementers and owners remain recommendations until Alan reviews/signs off the consolidated plan and Luke accepts it. Project: MVP Product Flow Reshape.

Plan	Value
Work package	WP06
Plan file	`06-node0-complete-and-soft-onboarding.md`
Proposed PRs	P06-01, P06-02, P06-03
Inspection snapshot	2026-07-10 20:12 America/New_York

1. Purpose and outcome

Problem. Optional paid depth, GC soft onboarding, and readiness reporting need clear entitlement gates without creating a managed case or treating IP assertions as GC answers.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Versioned Node 0 Complete entitlement, authenticated low-risk GC onboarding, attributable GC confirmation/replacement, and a readiness report that remains preliminary.

Explicit non-goals.

- No managed case activation
- No provider access
- No medical/legal clearance
- No hard-coded price

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-010	verified	app / PR #124@84a5a84	app/routes/node0.py:63-90	Opening Node 0 calls create_byos_case_for_ip with is_synthetic=True; there is no separate free workspace or authorized promotion.
E-011	verified	app / PR #124@84a5a84	app/storage/node0.py:400-462; app/routes/node0.py:605-652	Node 0 can issue an unaddressed GC invitation and redemption ultimately creates case membership, conflating low-risk onboarding possession with case access.
E-013	verified	app / PR #124@84a5a84	app/node0.py:252-482,622-878,975-1001	Nineteen IP-entered candidate answers about relationship and third-party circumstances/health derive a 0-100 score, readiness/status, ordering, guide route and evidence packet without per-field provenance or verification.
E-014	verified	app / PR #124@84a5a84	app/storage/node0.py:42-200,509-600; tests/python/test_node0.py:444-490	IP/co-IP own candidate reads; GC onboarding writes separate shared/private facts and does not revise the IP assertion. The GC cannot see or correct the IP candidate record.

3. Architecture and contracts

- Node 0 Complete is a flat configurable entitlement fully creditable against the JMF.
- GC confirmation appends an attributable revision; it never silently overwrites the IP source.

- GC-private answers stay private; only explicit purpose-limited projections reach the IP.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	`node0_workspace`, `workspace_claim`, `workspace_entitlement`
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Read-only readiness projection with policy version.
Storage	Append-only entitlement events plus current projection. Use assertion/revision contract; onboarding progress events.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Applicable state machines

Node 0 workspace (`node0_workspace`)

Owner: natalyx-app. States: `draft`, `active`, `match_ready`, `promotion_pending`, `promoted`, `archived`.

none

From	none
To	draft
Actor	authenticated IP
Evidence	first explicit Node 0 start
Idempotency key	workspace:{owner}:create
Invalid transitions	Existing active workspace returns existing
Audit event	workspace_created
Retry	Unique owner/idempotency key
Recovery	Forward repair orphan membership

draft

From	draft
To	active
Actor	IP
Evidence	minimum workspace consent/profile
Idempotency key	workspace:{id}:activate:{revision}
Invalid transitions	Missing consent/owner
Audit event	workspace_activated
Retry	Return current active revision
Recovery	Correct inputs and retry

active

From	active
To	match_ready
Actor	IP + selected GC evidence policy
Evidence	selected candidate + required provisional/GC-authored fields
Idempotency key	workspace:{id}:ready:{revision}
Invalid transitions	Critical unknown/decline/revoked claim
Audit event	workspace_match_ready
Retry	No duplicate event
Recovery	Reopen active with reason revision

match_ready

From	match_ready
To	promotion_pending
Actor	promotion event consumer
Evidence	valid byos.workspace_promotion_authorized.v1
Idempotency key	promotion:{commitment}:v1
Invalid transitions	No authorization; already promoted; stale config
Audit event	workspace_promotion_requested
Retry	Replay resumes/finds same promotion
Recovery	Forward-repair incomplete transaction

promotion_pending

From	promotion_pending
To	promoted
Actor	promotion service
Evidence	paid case + manifest committed
Idempotency key	promotion:{commitment}:v1
Invalid transitions	Projection invariant/privacy failure
Audit event	workspace_promoted
Retry	Return same case ID
Recovery	Repair from immutable manifest; never duplicate

draft|active|match_ready

From	draft|active|match_ready
To	archived
Actor	IP/operator per policy
Evidence	explicit archive + retention policy
Idempotency key	workspace:{id}:archive:{request}
Invalid transitions	Promotion pending/promoted
Audit event	workspace_archived
Retry	Idempotent
Recovery	Restore only through new audited revision if policy permits

Entitlement (`workspace_entitlement`)

Owner: natalyx-app commercial domain. States: `free`, `complete_pending`, `complete_active`, `suspended`, `revoked`.

none

From	none
To	free
Actor	system
Evidence	active Node 0 workspace
Idempotency key	entitlement:{workspace}:free
Invalid transitions	No workspace
Audit event	entitlement_granted
Retry	Return current
Recovery	Repair projection from event

free

From	free
To	complete_pending
Actor	IP/commercial service
Evidence	accepted Complete offer/config
Idempotency key	entitlement:{workspace}:{commitment}
Invalid transitions	Unknown price/config/terms
Audit event	complete_requested
Retry	Return pending
Recovery	Cancel/expire commitment

complete_pending

From	complete_pending
To	complete_active
Actor	payment domain
Evidence	settled Complete ledger entry
Idempotency key	entitlement:{payment}:activate
Invalid transitions	Amount/currency/config mismatch
Audit event	complete_activated
Retry	Replay no-op
Recovery	Reconcile and retry from inbox

complete_active

From	complete_active
To	suspended
Actor	authorized commercial policy
Evidence	refund/reversal or approved pause reason
Idempotency key	entitlement:{id}:suspend:{event}
Invalid transitions	Cannot remove free/safety capability
Audit event	entitlement_suspended
Retry	No-op if same event
Recovery	Reinstate or revoke through new event

complete_active|suspended

From	complete_active|suspended
To	revoked
Actor	authorized operator/policy
Evidence	fraud/refund/terms revocation evidence
Idempotency key	entitlement:{id}:revoke:{event}
Invalid transitions	No silent case cancellation
Audit event	entitlement_revoked
Retry	Idempotent
Recovery	Manual reviewed regrant creates new entitlement event

Low-risk workspace identity claim (`workspace_claim`)

Owner: natalyx-app. States: `issued`, `pending_identity`, `identity_bound`, `consented`, `claimed`, `mismatch_pending`, `revoked`, `expired`.

none

From	none
To	issued
Actor	workspace IP
Evidence	selected candidate + allowed claim scope
Idempotency key	claim:{workspace}:{candidate}:{generation}
Invalid transitions	Existing live claim superseded
Audit event	workspace_claim_issued
Retry	Return current token only at issuance
Recovery	Issue new generation; old revoked

issued

From	issued
To	pending_identity
Actor	link holder
Evidence	valid token landing
Idempotency key	claim:{id}:land
Invalid transitions	Expired/revoked token
Audit event	claim_opened
Retry	Safe repeat
Recovery	Authenticate or exit; no data grant

pending_identity

From	pending_identity
To	identity_bound|mismatch_pending
Actor	auth provider
Evidence	verified subject/email and binding policy
Idempotency key	claim:{id}:bind:{subject}
Invalid transitions	Already claimed/different subject
Audit event	claim_identity_bound|claim_mismatch
Retry	CAS no-op or mismatch state
Recovery	Issuer review/new claim

identity_bound

From	identity_bound
To	consented
Actor	GC claimant
Evidence	workspace consent version
Idempotency key	claim:{id}:consent:{version}
Invalid transitions	Stale/missing consent
Audit event	claim_consented
Retry	Idempotent
Recovery	Display current consent

consented

From	consented
To	claimed
Actor	claim service
Evidence	authorized low-risk workspace membership transaction
Idempotency key	claim:{id}:complete
Invalid transitions	Case membership requested or conflicting claimant
Audit event	workspace_claimed
Retry	Return same membership
Recovery	Transaction rollback/forward repair

issued|pending_identity|identity_bound|consented|mismatch_pending

From	issued|pending_identity|identity_bound|consented|mismatch_pending
To	revoked|expired
Actor	issuer/system
Evidence	revocation or TTL
Idempotency key	claim:{id}:terminal:{event}
Invalid transitions	No terminal overwrite
Audit event	claim_revoked|claim_expired
Retry	Idempotent
Recovery	Fresh claim only

Invitation and identity safety

Retain from updated PR #122:

- hashed tokens
- purpose-specific TTL with a 14-day ceiling
- CAS terminal transitions
- atomic membership creation
- one-live-GC constraint
- self/existing-member refusal
- safe return_to
- portable integrity/concurrency tests

Required replacement contract:

- Exactly one target: workspace_id or case_id; include purpose, role, access_class, recipient_binding_mode and policy version.
- Email-less possession may only open pending low-risk workspace claim; it cannot reveal sensitive data or grant paid-case membership.

- Paid-case invitation is addressed, authenticated, consented and atomically redeemed; mismatch requires explicit policy/issuer handling and preferably fresh bound token.
- Cognito admission and app membership are separate decisions; account creation never equals workspace/case access.
- Verified identity comes from auth claims, not form-entered email; duplicate claims return the original result.
- Never put raw token/email/PHI in general participant timeline payloads; revoke/expire/supersede fail closed.
- Remove generic unconditional status mutation; expose authorized transition commands only.

Data authorship/provenance rules

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.
- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

4. Explicit PR sequence

1

Order	1
ID	P06-01
Title	Add Node 0 Complete entitlement and versioned feature gates
Repository	app
Dependencies	P04-01, P07-01, P08-01
Gate	A2
Size	Large
Readiness	Dependencies unmet

2

Order	2
ID	P06-02
Title	Add GC soft onboarding and attributable confirmation
Repository	app
Dependencies	P03-02, P04-02, P06-01
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet

3

Order	3
ID	P06-03
Title	Add readiness report and role-scoped progress surfaces
Repository	app
Dependencies	P06-02, P10-02, P11-04
Gate	A2
Size	Large
Readiness	Dependencies unmet

P06-01 — Add Node 0 Complete entitlement and versioned feature gates

Field	Plan
Repository / base	`app` (`main`)
Project item	P06-01
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P04-01, P07-01, P08-01
Blocking decisions/checkpoints	D-COM-01
Source / salvage	Business-model Node 0 feature table

Objective

Model optional Complete access independently from workspace and case, pin the commercial configuration, and expose server-side gates for the approved paid-depth features.

Scope

- Grant/suspend/revoke entitlement events
- Commercial-version and payment evidence reference
- Credit accrual link

Explicit non-goals

- No hard-coded price
- No paid-case creation

Expected files or modules

- `migrations/versions/*_entitlements.py`
- `app/domain/entitlements.py`
- `app/storage/entitlements.py`
- `tests/python/test_entitlements.py`

Change map

Area	Planned change
Schema/model	Append-only entitlement events plus current projection.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Apply through the named versioned migration; no ad-hoc startup DDL. Backfill is none unless explicitly listed in scope.
Security/privacy	Every Complete feature checks server entitlement.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Grant replay
- suspend/revoke
- config version
- free-feature unaffected
- no case created

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P04-01, P07-01, P08-01. Parallel opportunities: P06-03

Objective acceptance criteria

- □ Model optional Complete access independently from workspace and case, pin the commercial configuration, and expose server-side gates for the approved paid-depth features.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

P06-02 — Add GC soft onboarding and attributable confirmation

Field	Plan
Repository / base	`app` (`main`)
Project item	P06-02
Gate / size / readiness	A2 / Exceptional / Dependencies unmet
Dependencies	P03-02, P04-02, P06-01
Blocking decisions/checkpoints	D-DATA-01, D-DATA-02, CHECK-ALAN-122-124
Source / salvage	PRs #123/#124 UX only, GC authorship audit

Objective

Let an authenticated GC claim low-risk workspace access, provide GC-owned answers, confirm/replace selected provisional facts, and preserve private boundaries.

Scope

- GC-authored profile/intake revisions
- Field-by-field confirmation/replacement
- Purpose-limited IP progress projection

Explicit non-goals

- No paid-case membership
- No provider share
- No medical eligibility determination

Expected files or modules

- `app/routes/gc_workspace.py`
- `app/storage/gc_onboarding.py`
- `app/templates/node0/gc_onboarding*.html`
- `tests/python/test_gc_soft_onboarding.py`

Change map

Area	Planned change
Schema/model	Use assertion/revision contract; onboarding progress events.
API/routes	No public API/route contract change.
UI/product surface	GC consent and privacy explanations.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Paired IP/GC visibility
- private contact/health fields
- revision lineage
- claim replay/mismatch
- mobile resume

Local validation:

- `python -m unittest discover -s tests/python`

- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P03-02, P04-02, P06-01. Parallel opportunities: None inside this package beyond review/test preparation.

Objective acceptance criteria

- □ Let an authenticated GC claim low-risk workspace access, provide GC-owned answers, confirm/replace selected provisional facts, and preserve private boundaries.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	privacy, accessible forms, authorization
Higher-capability review	Mandatory before merge

P06-03 — Add readiness report and role-scoped progress surfaces

Field	Plan
Repository / base	`app` (`main`)
Project item	P06-03
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P06-02, P10-02, P11-04
Blocking decisions/checkpoints	D-DATA-01
Source / salvage	PR #124 score/report concepts

Objective

Produce a preliminary, source-labelled Node 0 readiness report and separate IP/GC progress views without overclaiming clearance or exposing private answers.

Scope

- Source/verification labels
- IP and GC lenses
- Promotion-ready projection manifest

Explicit non-goals

- No provider-confirmed status
- No final case activation

Expected files or modules

- `app/domain/node0_readiness.py`
- `app/templates/node0/readiness*.html`
- `app/node0_surface.py`
- `tests/python/test_node0_readiness.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Read-only readiness projection with policy version.
UI/product surface	Role-specific report/progress cards.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Provisional labels
- critical missing data
- cross-role redaction
- stale revision
- print/mobile

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P06-02, P10-02, P11-04. Parallel opportunities: P06-01

Objective acceptance criteria

- □ Produce a preliminary, source-labelled Node 0 readiness report and separate IP/GC progress views without overclaiming clearance or exposing private answers.

- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P06-01` → `P06-02` → `P06-03`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Entitlement versioning; GC-authored revisions; IP/GC visibility pairs; Revoked/expired claims; No paid-case creation	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Type	ID / risk	Owner	Affected PRs / latest point	Project item / treatment
Blocking/product/compliance decision	D-DATA-01 — Which provisional IP assertions may the GC see, which GC answers may the IP see, and which fields survive promotion/provider projection?	Luke	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02	D-DATA-01 — Default deny; expose only explicit purpose-limited projections. Class 1/2 and class 4 never automatically cross roles.
Blocking/product/compliance decision	D-DATA-02 — How long to retain unselected candidate assertions and how are GC corrections, conflicting relationship answers, and private IP notes represented?	Luke	P04-02, P04-03, P06-02; Before P04-02	D-DATA-02 — Append revisions with supersedes/derived-from; never silently erase source; shortest justified retention for unselected candidates.
Blocking/product/compliance decision	D-DATA-03 — Does any 0-100 candidate ranking or race-based preference survive, and what may provisional third-party answers gate?	Luke	P05-02, P06-03; Before P05-02	D-DATA-03 — Remove candidate ranking and race preference; use source-labelled questions-to-clarify. Unverified IP answers cannot gate claim/onboarding/eligibility.
Blocking/product/compliance decision	D-COM-01 — Set the flat one-time price and tax treatment; it must remain fully creditable against the later JMF.	Luke	P06-01, P07-01, P08-01; Before activation of a Complete configuration	D-COM-01 — Keep no active production value until explicitly approved; test with synthetic configuration.
Implementation risk	Entitlement mistaken for case access	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	GC answers exposed to IP	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	IP assertions silently converted	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Price hard-coded	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#119	a8cd2f428535	Concise product-leverage framing and boundaries for the MVP flow.	P06-01	Retain as source or merge separately after plan decision	Catalogued; reconcile terminology into approved programme.
app#123	34078f810b63	GC onboarding form/copy, hosted auth return, invite handoff and role-specific surface work.	P06-02, P06-03	Selective UX/test port and rewrite	Unextracted; likely partially copied into #124 but not proven complete.
app#124	84a5a847fea7	Current consolidation of Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitations, DS assets and large focused test suites.	P06-02, P06-03	Retain unchanged as evidence; selective port/rewrite after checkpoint	Unextracted; Alan review explicitly incomplete and unresolved.
feature/node0-surrogate-onboarding-link	529d377	substantial uncommitted onboarding/invitation work	P03-02/P03-03/P06-02	Selective port from protected worktree	Preserve until file/test extraction is signed off.

Package source inventory:

- PRs #123/#124 onboarding screens
- Dirty onboarding-link branch
- PR #122 paired-lens tests

9. Ownership recommendation

Luke decides product visibility and entitlement boundaries; Alan implements. External privacy review is recommended for field-level projections.

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P06-01	Alan	Alan (proposed)	Luke	Luke	Mandatory
P06-02	Alan	Alan (proposed)	Luke	Luke	Mandatory

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P06-03	Alan	Alan (proposed)	Luke	Luke	Mandatory

Luke decides product/lifecycle/commercial/privacy/PHI/provider policy and accepts gates. Alan proposes implementation design, decomposition, migrations, code, tests, rollout and repair. This is a recommendation pending Alan's review/sign-off, not a compulsory assignment.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 07 — Commercial configuration, enrollment, and promotion

Planning-only execution contract. Proposed implementers and owners remain recommendations until Alan reviews/signs off the consolidated plan and Luke accepts it. Project: MVP Product Flow Reshape.

Plan	Value
Work package	WP07
Plan file	`07-commercial-configuration-enrollment-and-promotion.md`
Proposed PRs	P07-01, P07-02, P07-03, P07-04
Inspection snapshot	2026-07-10 20:12 America/New_York

1. Purpose and outcome

Problem. The business model requires versioned price/terms, explicit commitment, and one precise processor-neutral event before workspace-to-case promotion; none is currently authoritative.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Versioned commercial configuration and acceptances, a separate commitment/enrollment lifecycle, one processor-neutral byos.workspace_promotion_authorized.v1 event, and exactly-once paid-case creation by its consumer.

Explicit non-goals.

- No processor-specific webhook
- No automated unresolved pricing policy
- No activation before onboarding evidence
- No browser-redirect authority

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-025	design authority	business model / Natalyx_Payment_and_Business_Model_BYOS_MVP.docx v1.0	Sections 3-12 and implementation tables	Node 0 Free must complete the match; Complete is optional/creditable; $8,900 JMF and $750 enrollment anchor the configurable model; promotion follows explicit commitment and funding, not willingness or redirect.

3. Architecture and contracts

- Every commitment, invoice, payment, credit, refund, and promotion records the activated commercial-configuration version.
- Only the app-owned commitment/commercial domain emits byos.workspace_promotion_authorized.v1 after durable funding/terms/credit validation; a processor callback or redirect never emits it.
- The consumer creates/fetches one enrollment_pending case and restricted IP membership; GC case access follows the paid-case invitation contract.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	`case`, `case_invitation`, `journey_commitment`, `node0_workspace`
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Read active config and record acceptance commands. Create, accept, cancel, inspect enrollment. Internal event producer only. Internal event handler only.
Storage	Commercial configuration and acceptance ledgers. Commitment events/current projection. Promotion ledger and source-workspace unique constraint.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Applicable state machines

Node 0 workspace (`node0_workspace`)

Owner: natalyx-app. States: `draft`, `active`, `match_ready`, `promotion_pending`, `promoted`, `archived`.

none

From	none
To	draft
Actor	authenticated IP
Evidence	first explicit Node 0 start
Idempotency key	workspace:{owner}:create
Invalid transitions	Existing active workspace returns existing
Audit event	workspace_created
Retry	Unique owner/idempotency key
Recovery	Forward repair orphan membership

draft

From	draft
To	active
Actor	IP
Evidence	minimum workspace consent/profile
Idempotency key	workspace:{id}:activate:{revision}
Invalid transitions	Missing consent/owner
Audit event	workspace_activated
Retry	Return current active revision
Recovery	Correct inputs and retry

active

From	active
To	match_ready
Actor	IP + selected GC evidence policy
Evidence	selected candidate + required provisional/GC-authored fields
Idempotency key	workspace:{id}:ready:{revision}
Invalid transitions	Critical unknown/decline/revoked claim
Audit event	workspace_match_ready
Retry	No duplicate event
Recovery	Reopen active with reason revision

match_ready

From	match_ready
To	promotion_pending
Actor	promotion event consumer
Evidence	valid byos.workspace_promotion_authorized.v1
Idempotency key	promotion:{commitment}:v1
Invalid transitions	No authorization; already promoted; stale config
Audit event	workspace_promotion_requested
Retry	Replay resumes/finds same promotion
Recovery	Forward-repair incomplete transaction

promotion_pending

From	promotion_pending
To	promoted
Actor	promotion service
Evidence	paid case + manifest committed
Idempotency key	promotion:{commitment}:v1
Invalid transitions	Projection invariant/privacy failure
Audit event	workspace_promoted
Retry	Return same case ID
Recovery	Repair from immutable manifest; never duplicate

draft|active|match_ready

From	draft|active|match_ready
To	archived
Actor	IP/operator per policy
Evidence	explicit archive + retention policy
Idempotency key	workspace:{id}:archive:{request}
Invalid transitions	Promotion pending/promoted
Audit event	workspace_archived
Retry	Idempotent
Recovery	Restore only through new audited revision if policy permits

Commercial commitment (`journey_commitment`)

Owner: natalyx-app commercial domain. States: `draft`, `accepted`, `funding_pending`, `funded`, `cancelled`, `expired`.

none

From	none
To	draft
Actor	IP
Evidence	active workspace + displayed config/terms
Idempotency key	commitment:{workspace}:{config}
Invalid transitions	No active config/workspace
Audit event	commitment_drafted
Retry	Return existing draft
Recovery	Create new draft after expiry/version change

draft

From	draft
To	accepted
Actor	IP
Evidence	terms/privacy consent + payment path
Idempotency key	commitment:{id}:accept:{terms_hash}
Invalid transitions	Stale/missing acceptance
Audit event	commitment_accepted
Retry	Return accepted
Recovery	New acceptance revision if version changes

accepted

From	accepted
To	funding_pending
Actor	commercial service
Evidence	enrollment invoice/payment intent created
Idempotency key	commitment:{id}:funding
Invalid transitions	No invoice or invalid amount
Audit event	funding_requested
Retry	Resume same invoice
Recovery	Cancel/reissue under audited adjustment

funding_pending

From	funding_pending
To	funded
Actor	payment domain
Evidence	net settled enrollment funding meets versioned requirement
Idempotency key	enrollment-funding:{id}:v1
Invalid transitions	Redirect/pending/failed/reversed evidence
Audit event	commercial.enrollment_funding_confirmed.v1
Retry	Exactly-once event/outbox replay
Recovery	Reconcile inbox/ledger; no browser repair

draft|accepted|funding_pending

From	draft|accepted|funding_pending
To	cancelled
Actor	IP/authorized operator
Evidence	explicit cancellation and refund policy
Idempotency key	commitment:{id}:cancel:{request}
Invalid transitions	Already funded/promoted uses case cancellation rail
Audit event	commitment_cancelled
Retry	Idempotent
Recovery	New commitment required

draft|accepted

From	draft|accepted
To	expired
Actor	system
Evidence	offer/terms TTL or config deactivation
Idempotency key	commitment:{id}:expire
Invalid transitions	No expiry after funding
Audit event	commitment_expired
Retry	Idempotent
Recovery	Draft new version

Paid case lifecycle (`case`)

Owner: natalyx-app. States: `enrollment_pending`, `active`, `paused`, `closed`, `cancelled`.

none

From	none
To	enrollment_pending
Actor	promotion service
Evidence	byos.workspace_promotion_authorized.v1 + projection manifest
Idempotency key	promotion:{commitment}:v1
Invalid transitions	No authorization/duplicate workspace
Audit event	paid_case_created
Retry	Return same case
Recovery	Forward-repair same promotion

enrollment_pending

From	enrollment_pending
To	active
Actor	authorized coordinator/domain service
Evidence	approved onboarding/compatibility/intake/education evidence
Idempotency key	case:{id}:activate:{evidence_hash}
Invalid transitions	Missing/revoked evidence or GC consent
Audit event	case_activated
Retry	No duplicate activation/invoice
Recovery	Return pending with unmet checklist

active

From	active
To	paused
Actor	authorized commercial/operations policy
Evidence	audited non-critical pause reason
Idempotency key	case:{id}:pause:{event}
Invalid transitions	Safety-critical access cannot be paused
Audit event	case_paused
Retry	Idempotent
Recovery	Resume after evidence; preserve entitled docs

paused

From	paused
To	active
Actor	authorized operator/domain service
Evidence	pause reason resolved
Idempotency key	case:{id}:resume:{event}
Invalid transitions	Unresolved safety/authorization block
Audit event	case_resumed
Retry	Idempotent
Recovery	Escalate manual review

active|paused

From	active|paused
To	closed
Actor	authorized coordinator
Evidence	birth/close or completed termination handoff evidence
Idempotency key	case:{id}:close:{evidence_hash}
Invalid transitions	Open critical obligations
Audit event	case_closed
Retry	No-op if same evidence
Recovery	Reopen only by exceptional audited event

enrollment_pending|active|paused

From	enrollment_pending|active|paused
To	cancelled
Actor	authorized operator under policy
Evidence	withdrawal/non-clearance/cancellation + financial disposition
Idempotency key	case:{id}:cancel:{request}
Invalid transitions	Cannot erase records/refund ledger
Audit event	case_cancelled
Retry	Idempotent
Recovery	Replacement is a new case/workspace policy event

Paid-case membership invitation (`case_invitation`)

Owner: natalyx-app. States: `issued`, `pending_identity`, `identity_bound`, `mismatch_pending`, `issuer_confirmed`, `consented`, `redeemed`, `declined`, `revoked`, `expired`, `superseded`.

none

From	none
To	issued
Actor	authorized IP case member
Evidence	enrollment_pending/active case + recipient binding
Idempotency key	case-invite:{case}:{role}:{generation}
Invalid transitions	Self/member/active-GC/unauthorized
Audit event	case_invite_issued
Retry	Return current; supersede prior live GC invite
Recovery	Issue corrected invite

issued

From	issued
To	pending_identity
Actor	recipient
Evidence	valid addressed token landing
Idempotency key	case-invite:{id}:land
Invalid transitions	Expired/revoked
Audit event	case_invite_opened
Retry	Safe repeat
Recovery	Authenticate with return_to

pending_identity

From	pending_identity
To	identity_bound|mismatch_pending
Actor	auth provider
Evidence	verified subject/email
Idempotency key	case-invite:{id}:bind:{subject}
Invalid transitions	Terminal/duplicate/different subject
Audit event	case_invite_bound|mismatch
Retry	CAS protects terminal
Recovery	Issuer confirmation or corrected invite

mismatch_pending

From	mismatch_pending
To	issuer_confirmed
Actor	authorized issuer
Evidence	explicit reviewed mismatch confirmation
Idempotency key	case-invite:{id}:confirm:{issuer}
Invalid transitions	Issuer unauthorized/recipient changed
Audit event	case_invite_mismatch_confirmed
Retry	CAS no-op
Recovery	Revoke and reissue

identity_bound|issuer_confirmed

From	identity_bound|issuer_confirmed
To	consented
Actor	invitee
Evidence	paid-case role/privacy consent version
Idempotency key	case-invite:{id}:consent:{version}
Invalid transitions	Stale/missing consent
Audit event	case_invite_consented
Retry	Idempotent
Recovery	Present current consent

consented

From	consented
To	redeemed
Actor	invitation service
Evidence	single transaction creates person/party/membership
Idempotency key	case-invite:{id}:redeem
Invalid transitions	Existing member/active GC/expired/revoked
Audit event	case_invite_redeemed
Retry	Return same membership
Recovery	Rollback entire transaction; forward repair from invite

issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented

From	issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented
To	declined|revoked|expired|superseded
Actor	invitee/issuer/system
Evidence	explicit decline/revoke/TTL/new generation
Idempotency key	case-invite:{id}:terminal:{event}
Invalid transitions	No terminal overwrite
Audit event	case_invite_terminal
Retry	Idempotent CAS
Recovery	Fresh invitation only

Data authorship/provenance rules

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.

- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

Sole promotion-authorizing event

`event_type: byos.workspace_promotion_authorized.v1`

Producer: natalyx-app commitment/commercial domain after deterministic validation. Consumer: natalyx-app promotion service.

Payload:

```
event_id
idempotency_key
workspace_id
commitment_id
owner_person_id
owner_subject
commercial_config_version
terms_acceptance_ids[]
payment_plan_id
funding_evidence.kind
funding_evidence.commercial_event_id
funding_evidence.settled_amount_minor
funding_evidence.currency
credit_allocation_ids[]
initial_case_state=enrollment_pending
occurred_at
correlation_id
causation_id
schema_version
```

Authorization conditions:

- Workspace is active/match-ready, unarchived and not already promoted.
- Commitment references the same workspace, has current terms/privacy acceptances and is in funding_pending/funded per the approved state contract.
- Payment path and activated commercial configuration are pinned.
- Required enrollment funding has a verified settled ledger entry or separately approved financing/manual-funding evidence; a processor webhook is first normalized and cannot itself authorize promotion.
- Credit allocation is non-negative, reproducible and cannot exceed the applicable obligation.
- No cancellation, compliance hold, reversal that invalidates funding, or existing promotion is present.

Unique constraints:

- one promotion per workspace
- one case per promotion event
- one result per idempotency key
- one primary owner membership per created case

No browser success/return route and no raw processor callback can emit or substitute for this event.

4. Explicit PR sequence

1

Order	1
ID	P07-01
Title	Add versioned commercial configuration and terms acceptance
Repository	app
Dependencies	P02-03
Gate	A2
Size	Large
Readiness	Dependencies unmet

2

Order	2
ID	P07-02
Title	Implement commitment and enrollment-pending lifecycle
Repository	app
Dependencies	P07-01, P08-01
Gate	A2
Size	Large
Readiness	Dependencies unmet

3

Order	3
ID	P07-03
Title	Emit processor-neutral workspace promotion authorization
Repository	app
Dependencies	P04-02, P07-02, P08-02
Gate	A2
Size	Large
Readiness	Dependencies unmet

4

Order	4
ID	P07-04
Title	Promote workspace exactly once from authorization event
Repository	app
Dependencies	P03-03, P04-02, P07-03
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet

P07-01 — Add versioned commercial configuration and terms acceptance

Field	Plan
Repository / base	`app` (`main`)
Project item	P07-01
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P02-03
Blocking decisions/checkpoints	D-COM-01, D-COM-02, D-COM-03
Source / salvage	Payment/business-model DOCX

Objective

Store activated price books, milestone schedules, Node 0 credit rules, and immutable terms/privacy acceptance versions without scattering literals.

Scope

- Immutable activated config versions
- JMF/deposit/discount/milestones/credit policy
- Terms and privacy acceptance records

Explicit non-goals

- No unresolved Node 0 price value
- No payment provider

Expected files or modules

- `migrations/versions/*_commercial_configuration.py`
- `app/domain/commercial_config.py`
- `app/storage/commercial_config.py`
- `tests/python/test_commercial_config.py`

Change map

Area	Planned change
Schema/model	Commercial configuration and acceptance ledgers.
API/routes	Read active config and record acceptance commands.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Apply through the named versioned migration; no ad-hoc startup DDL. Backfill is none unless explicitly listed in scope.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Activation immutability
- currency/amount validation
- acceptance version
- no retroactive mutation

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P02-03. Parallel opportunities: P07-03, P07-04

Objective acceptance criteria

- □ Store activated price books, milestone schedules, Node 0 credit rules, and immutable terms/privacy acceptance versions without scattering literals.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	commercial systems, PostgreSQL
Higher-capability review	Mandatory before merge

P07-02 — Implement commitment and enrollment-pending lifecycle

Field	Plan
Repository / base	`app` (`main`)
Project item	P07-02
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P07-01, P08-01
Blocking decisions/checkpoints	D-COM-04
Source / salvage	Business-model conversion trigger

Objective

Create a deliberate managed-journey commitment with accepted terms, selected payment path, pinned config, and an enrollment invoice request independent from case state.

Scope

- Draft/accepted/cancelled/expired commitment transitions
- Payment-path choice
- Enrollment invoice command

Explicit non-goals

- No browser-triggered promotion
- No activation

Expected files or modules

- `app/domain/commitments.py`
- `app/storage/commitments.py`
- `app/routes/enrollment.py`
- `tests/python/test_commitments.py`

Change map

Area	Planned change
Schema/model	Commitment events/current projection.
API/routes	Create, accept, cancel, inspect enrollment.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Acceptance prerequisites
- duplicate accept
- stale config
- cancel/expire
- resume

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P07-01, P08-01. Parallel opportunities: P07-04

Objective acceptance criteria

- □ Create a deliberate managed-journey commitment with accepted terms, selected payment path, pinned config, and an enrollment invoice request independent from case state.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

P07-03 — Emit processor-neutral workspace promotion authorization

Field	Plan
Repository / base	`app` (`main`)
Project item	P07-03
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P04-02, P07-02, P08-02
Blocking decisions/checkpoints	D-COM-04, D-DATA-01
Source / salvage	Business-model promotion invariants

Objective

Validate commitment, terms, commercial version, settled funding and credit evidence, then emit exactly one byos.workspace_promotion_authorized.v1 event; processor callbacks and browser redirects cannot emit it.

Scope

- Deterministic authorization predicate
- Canonical payload and idempotency key
- Transactional event/outbox
- Funding/credit/config evidence references

Explicit non-goals

- No case creation
- No redirect/webhook-as-authority
- No GC membership

Expected files or modules

- `app/domain/promotion_authorization.py`
- `app/event_handlers/commercial.py`
- `app/storage/outbox.py`
- `tests/python/test_promotion_authorization.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Internal event producer only.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Only commercial-domain authority can emit the typed event.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Pending/failed/reversed funding
- stale terms/config
- duplicate/replayed inputs
- credit mismatch
- event payload/checksum

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P04-02, P07-02, P08-02. Parallel opportunities: P07-01

Objective acceptance criteria

- □ Validate commitment, terms, commercial version, settled funding and credit evidence, then emit exactly one byos.workspace_promotion_authorized.v1 event; processor callbacks and browser redirects cannot emit it.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	event-driven systems, payments, commercial policy
Higher-capability review	Mandatory before merge

P07-04 — Promote workspace exactly once from authorization event

Field	Plan
Repository / base	`app` (`main`)
Project item	P07-04
Gate / size / readiness	A2 / Exceptional / Dependencies unmet
Dependencies	P03-03, P04-02, P07-03
Blocking decisions/checkpoints	D-COM-04, D-DATA-01, CHECK-ALAN-122-124
Source / salvage	byos.workspace_promotion_authorized.v1 contract, PR #124 source linkage concepts

Objective

Consume byos.workspace_promotion_authorized.v1 to create/find one enrollment_pending paid case, link the source workspace, project allowlisted data, and grant only restricted IP membership.

Scope

- Unique promotion idempotency key
- Case/source link and config version
- Purpose-limited data manifest and audit event

Explicit non-goals

- No redirect authority
- No GC case membership transfer
- No activation

Expected files or modules

- `app/domain/promotion.py`
- `app/storage/promotions.py`
- `app/event_handlers/enrollment.py`
- `tests/python/test_workspace_promotion.py`

Change map

Area	Planned change
Schema/model	Promotion ledger and source-workspace unique constraint.
API/routes	Internal event handler only.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Projection defaults deny and preserves provenance.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Replay/concurrency
- partial transaction
- funding event validation
- private candidate exclusion
- GC invitation required

Local validation:

- `python -m unittest discover -s tests/python`

- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P03-03, P04-02, P07-03. Parallel opportunities: P07-01, P07-02

Objective acceptance criteria

- □ Consume byos.workspace_promotion_authorized.v1 to create/find one enrollment_pending paid case, link the source workspace, project allowlisted data, and grant only restricted IP membership.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	event-driven systems, privacy, PostgreSQL
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P07-01` → `P07-02` → `P07-03` → `P07-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Terms/config version pinning; Commitment retries; Exact-once promotion; Funding reversal handling; No GC access escalation	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Type	ID / risk	Owner	Affected PRs / latest point	Project item / treatment
Blocking/product/compliance decision	D-DATA-01 — Which provisional IP assertions may the GC see, which GC answers may the IP see, and which fields survive promotion/provider projection?	Luke	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02	D-DATA-01 — Default deny; expose only explicit purpose-limited projections. Class 1/2 and class 4 never automatically cross roles.
Blocking/product/compliance decision	D-COM-01 — Set the flat one-time price and tax treatment; it must remain fully creditable against the later JMF.	Luke	P06-01, P07-01, P08-01; Before activation of a Complete configuration	D-COM-01 — Keep no active production value until explicitly approved; test with synthetic configuration.
Blocking/product/compliance decision	D-COM-02 — Ratify $8,900 JMF, $7,830 pay-in-full, $750 deposit and 20/30/20/20/10 post-deposit milestones as configuration v1 with effective date/currency.	Luke	P07-01, P08-01, P08-03; Before P07-01 activation data	D-COM-02 — Use the document anchors but require explicit version activation rather than code literals.
Blocking/product/compliance decision	D-COM-03 — Approve the actual paid-service terms, privacy/consent disclosure and refundable-to-earned wording.	Luke	P07-01, P07-02, P08-03; Before P07-02 user acceptance	D-COM-03 — External legal/privacy/finance review before any real acceptance.
Blocking/product/compliance decision	D-COM-04 — Define evidence for enrollment funding confirmed, promotion authorization, and for onboarding/compatibility complete enough to move case active and earn/issue activation billing.	Luke	P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02	D-COM-04 — Funding confirmation is durable ledger evidence; promotion uses the sole typed authorization event; activation requires an explicit evidence checklist, actor and audit event.
Blocking/product/compliance decision	D-COM-05 — Select financing partner and define when funding is sufficiently authorized/settled; otherwise hide the path.	Luke	P07-02, P08-02; Before financed option is exposed	D-COM-05 — Do not block card/ACH MVP; keep optional and disabled.
Implementation risk	Configuration drift	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Duplicate promotion	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Redirect spoofing	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Unresolved policy encoded as default	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
No direct source	—	Build from approved contracts/evidence	Package PRs	New implementation	N/A

Package source inventory:

- Payment/business-model DOCX as authority
- PR #124 source workspace linkage concepts only

9. Ownership recommendation

Luke owns commercial semantics and final contract. Alan proposes/implements the state machine and idempotency. Finance/legal review unresolved policy.

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P07-01	Alan	Alan (proposed)	Luke	Luke	Mandatory
P07-02	Alan	Alan (proposed)	Luke	Luke	Mandatory
P07-03	Alan	Alan (proposed)	Luke	Luke	Mandatory
P07-04	Alan	Alan (proposed)	Luke	Luke	Mandatory

Luke decides product/lifecycle/commercial/privacy/PHI/provider policy and accepts gates. Alan proposes implementation design, decomposition, migrations, code, tests, rollout and repair. This is a recommendation pending Alan's review/sign-off, not a compulsory assignment.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 08 — Payments, invoices, credits, and refunds

Planning-only execution contract. Proposed implementers and owners remain recommendations until Alan reviews/signs off the consolidated plan and Luke accepts it. Project: MVP Product Flow Reshape.

Plan	Value
Work package	WP08
Plan file	`08-payments-invoices-credits-and-refunds.md`
Proposed PRs	P08-01, P08-02, P08-03, P08-04
Inspection snapshot	2026-07-10 20:12 America/New_York

1. Purpose and outcome

Problem. No append-only commercial ledger or processor-neutral payment boundary currently supports the $8,900 JMF, $750 enrollment, Node 0 credit, milestones, deterministic refunds, or failure recovery.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Immutable commercial ledgers, idempotent provider inbox, milestone engine, reproducible credits/refunds, reconciliation, and safe non-critical pause behavior.

Explicit non-goals.

- No custody of pass-through funds
- No financing assumption
- No pregnancy-loss automation before policy
- No accounting-general-ledger replacement

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-025	design authority	business model / Natalyx_Payment_and_Business_Model_BYOS_MVP.docx v1.0	Sections 3-12 and implementation tables	Node 0 Free must complete the match; Complete is optional/creditable; $8,900 JMF and $750 enrollment anchor the configurable model; promotion follows explicit commitment and funding, not willingness or redirect.

3. Architecture and contracts

- Payments, credits, refunds, adjustments, invoices, and commercial events are append-only; mutable balances are derived projections.
- Processor events enter an idempotent inbox and become internal commands/events only after signature and amount/currency checks.
- Billing follows durable case evidence and never acts as first notification of clinical events.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	`journey_commitment`, `payment_obligation + append-only entries`, `workspace_entitlement`
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Processor webhook and internal payment commands. Internal evidence-to-invoice commands and explainable refund quote.
Storage	Append-only financial ledgers and unique idempotency constraints. Processor-event inbox/outbox and mapping records.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Applicable state machines

Entitlement (`workspace_entitlement`)

Owner: natalyx-app commercial domain. States: `free`, `complete_pending`, `complete_active`, `suspended`, `revoked`.

none

From	none
To	free
Actor	system
Evidence	active Node 0 workspace
Idempotency key	entitlement:{workspace}:free
Invalid transitions	No workspace
Audit event	entitlement_granted
Retry	Return current
Recovery	Repair projection from event

free

From	free
To	complete_pending
Actor	IP/commercial service
Evidence	accepted Complete offer/config
Idempotency key	entitlement:{workspace}:{commitment}
Invalid transitions	Unknown price/config/terms
Audit event	complete_requested
Retry	Return pending
Recovery	Cancel/expire commitment

complete_pending

From	complete_pending
To	complete_active
Actor	payment domain
Evidence	settled Complete ledger entry
Idempotency key	entitlement:{payment}:activate
Invalid transitions	Amount/currency/config mismatch
Audit event	complete_activated
Retry	Replay no-op
Recovery	Reconcile and retry from inbox

complete_active

From	complete_active
To	suspended
Actor	authorized commercial policy
Evidence	refund/reversal or approved pause reason
Idempotency key	entitlement:{id}:suspend:{event}
Invalid transitions	Cannot remove free/safety capability
Audit event	entitlement_suspended
Retry	No-op if same event
Recovery	Reinstate or revoke through new event

complete_active|suspended

From	complete_active|suspended
To	revoked
Actor	authorized operator/policy
Evidence	fraud/refund/terms revocation evidence
Idempotency key	entitlement:{id}:revoke:{event}
Invalid transitions	No silent case cancellation
Audit event	entitlement_revoked
Retry	Idempotent
Recovery	Manual reviewed regrant creates new entitlement event

Commercial commitment (`journey_commitment`)

Owner: natalyx-app commercial domain. States: `draft`, `accepted`, `funding_pending`, `funded`, `cancelled`, `expired`.

none

From	none
To	draft
Actor	IP
Evidence	active workspace + displayed config/terms
Idempotency key	commitment:{workspace}:{config}
Invalid transitions	No active config/workspace
Audit event	commitment_drafted
Retry	Return existing draft
Recovery	Create new draft after expiry/version change

draft

From	draft
To	accepted
Actor	IP
Evidence	terms/privacy consent + payment path
Idempotency key	commitment:{id}:accept:{terms_hash}
Invalid transitions	Stale/missing acceptance
Audit event	commitment_accepted
Retry	Return accepted
Recovery	New acceptance revision if version changes

accepted

From	accepted
To	funding_pending
Actor	commercial service
Evidence	enrollment invoice/payment intent created
Idempotency key	commitment:{id}:funding
Invalid transitions	No invoice or invalid amount
Audit event	funding_requested
Retry	Resume same invoice
Recovery	Cancel/reissue under audited adjustment

funding_pending

From	funding_pending
To	funded
Actor	payment domain
Evidence	net settled enrollment funding meets versioned requirement
Idempotency key	enrollment-funding:{id}:v1
Invalid transitions	Redirect/pending/failed/reversed evidence
Audit event	commercial.enrollment_funding_confirmed.v1
Retry	Exactly-once event/outbox replay
Recovery	Reconcile inbox/ledger; no browser repair

draft|accepted|funding_pending

From	draft|accepted|funding_pending
To	cancelled
Actor	IP/authorized operator
Evidence	explicit cancellation and refund policy
Idempotency key	commitment:{id}:cancel:{request}
Invalid transitions	Already funded/promoted uses case cancellation rail
Audit event	commitment_cancelled
Retry	Idempotent
Recovery	New commitment required

draft|accepted

From	draft|accepted
To	expired
Actor	system
Evidence	offer/terms TTL or config deactivation
Idempotency key	commitment:{id}:expire
Invalid transitions	No expiry after funding
Audit event	commitment_expired
Retry	Idempotent
Recovery	Draft new version

Payment (`payment_obligation + append-only entries`)

Owner: natalyx-app payment domain. States: `created`, `processing`, `settled`, `failed`, `partially_refunded`, `refunded`, `reversed`.

none

From	none
To	created
Actor	billing policy
Evidence	valid invoice/obligation
Idempotency key	payment:{invoice}:create
Invalid transitions	Duplicate or invalid invoice
Audit event	payment_created
Retry	Return current intent
Recovery	Void/reissue through adjustment

created|failed

From	created|failed
To	processing
Actor	processor adapter
Evidence	signed provider acknowledgement
Idempotency key	processor:{provider_event}
Invalid transitions	Amount/currency/customer mismatch
Audit event	payment_processing
Retry	Inbox replay no-op
Recovery	Retry same intent or audited replacement

processing|created

From	processing|created
To	settled
Actor	payment domain
Evidence	verified settlement event
Idempotency key	processor:{provider_event}
Invalid transitions	Browser redirect or unverified event
Audit event	payment_settled
Retry	Exactly-once append
Recovery	Reconcile provider and append repair event

created|processing

From	created|processing
To	failed
Actor	payment domain
Evidence	verified failure/timeout policy
Idempotency key	processor:{provider_event}
Invalid transitions	Cannot overwrite settled
Audit event	payment_failed
Retry	Replay no-op
Recovery	Retry/new attempt linked to obligation

settled|partially_refunded

From	settled|partially_refunded
To	partially_refunded|refunded
Actor	authorized refund service
Evidence	approved refund calculation + processor confirmation
Idempotency key	refund:{refund_request}
Invalid transitions	Exceeds settled less prior refunds
Audit event	refund_settled
Retry	Replay no-op
Recovery	Reconcile/adjust; never mutate original

settled

From	settled
To	reversed
Actor	payment domain
Evidence	verified chargeback/reversal
Idempotency key	processor:{provider_event}
Invalid transitions	Unsupported/manual assertion
Audit event	payment_reversed
Retry	Replay no-op
Recovery	Open discrepancy and apply policy; case is not deleted

Append-only commercial contract

Authoritative ledgers:

- `commercial_events`
- `processor_events`
- `invoice_events`
- `payment_ledger_entries`
- `credit_ledger_entries`
- `refund_ledger_entries`
- `adjustment_ledger_entries`
- `performance_obligation_events`
- `promotion_events`

Derived projections:

- payment plan state

- invoice status
- settled cash
- available credit
- recognized/deferred amount
- refundable amount
- delinquency/pause status

Rules:

- Every record carries currency in integer minor units, commercial configuration version, source/idempotency/correlation identifiers and immutable timestamps.
- Mutable balances/statuses are projections only; reversals, chargebacks, credits, refunds and adjustments append entries.
- Processor events are signature-verified, stored once and normalized before domain use.
- Node 0 Complete credit cannot create a negative JMF obligation and stays distinct from cash/refund entries.
- Pass-through clinic/legal/psych/background/escrow/insurance/GC funds never enter Natalyx revenue ledgers.
- Clinical-linked invoices require durable evidence plus dignity-of-timing delay and can never be the first clinical notification.
- Refundable cash = max(0, settled JMF cash - earned JMF - prior cash refunds - chargebacks/reversals), with policy hooks for unresolved cases.

4. Explicit PR sequence

1

Order	1
ID	P08-01
Title	Add append-only commercial and money ledgers
Repository	app
Dependencies	P02-03, P07-01
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet

2

Order	2
ID	P08-02
Title	Add processor-neutral payment command and webhook inbox
Repository	app
Dependencies	P08-01, P07-02
Gate	A2
Size	Exceptional
Readiness	Blocked by decision

3

Order	3
ID	P08-03
Title	Implement milestone invoicing, credits, and deterministic refunds
Repository	app
Dependencies	P08-02, P07-04, P09-04
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet

4

Order	4
ID	P08-04
Title	Add payment reconciliation and safe service-pause operations
Repository	app
Dependencies	P08-03
Gate	A2
Size	Large
Readiness	Dependencies unmet

P08-01 — Add append-only commercial and money ledgers

Field	Plan
Repository / base	`app` (`main`)
Project item	P08-01
Gate / size / readiness	A2 / Exceptional / Dependencies unmet
Dependencies	P02-03, P07-01
Blocking decisions/checkpoints	D-COM-01, D-COM-02, D-COM-03
Source / salvage	Payment/business-model DOCX

Objective

Create immutable commercial-event, invoice, payment, credit, refund, and adjustment entries with deterministic projections and config-version references.

Scope

- Ledger invariants and currency precision
- Derived balances/status projections
- Audit/source/idempotency references

Explicit non-goals

- No processor
- No mutable balance authority
- No pass-through custody

Expected files or modules

- `migrations/versions/*_commercial_ledgers.py`
- `app/storage/ledgers.py`
- `app/domain/money.py`
- `tests/python/test_ledgers.py`

Change map

Area	Planned change
Schema/model	Append-only financial ledgers and unique idempotency constraints.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Apply through the named versioned migration; no ad-hoc startup DDL. Backfill is none unless explicitly listed in scope.
Security/privacy	Least-privilege write paths and redacted logs.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Double-entry-like conservation checks
- duplicate key
- credit floor
- refund totals
- concurrent append

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P02-03, P07-01. Parallel opportunities: P08-03, P08-04

Objective acceptance criteria

- □ Create immutable commercial-event, invoice, payment, credit, refund, and adjustment entries with deterministic projections and config-version references.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	payments, financial ledgers, PostgreSQL
Higher-capability review	Mandatory before merge

P08-02 — Add processor-neutral payment command and webhook inbox

Field	Plan
Repository / base	`app` (`main`)
Project item	P08-02
Gate / size / readiness	A2 / Exceptional / Blocked by decision
Dependencies	P08-01, P07-02
Blocking decisions/checkpoints	D-PAY-01, D-COM-05
Source / salvage	Business-model processor-neutral requirement

Objective

Define provider ports and a signed, idempotent webhook inbox that emits internal settlement/failure/refund events, including commercial.enrollment_funding_confirmed.v1 when domain conditions hold.

Scope

- Provider-neutral intents and event mapping
- Signature/replay/order handling
- Settlement predicate and outbox

Explicit non-goals

- No browser redirect settlement
- No financing implementation unless separately approved

Expected files or modules

- `app/payments/ports.py`
- `app/payments/webhook_inbox.py`
- `app/routes/payment_webhooks.py`
- `tests/python/test_payment_webhooks.py`

Change map

Area	Planned change
Schema/model	Processor-event inbox/outbox and mapping records.
API/routes	Processor webhook and internal payment commands.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	Secrets/endpoints deferred to guarded environment configuration.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Signature failure
- replay/out-of-order
- amount/currency mismatch
- timeout/retry
- settlement event exactly once

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P08-01, P07-02. Parallel opportunities: P08-04

Objective acceptance criteria

- □ Define provider ports and a signed, idempotent webhook inbox that emits internal settlement/failure/refund events, including commercial.enrollment_funding_confirmed.v1 when domain conditions hold.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	payment security, webhooks, event systems
Higher-capability review	Mandatory before merge

P08-03 — Implement milestone invoicing, credits, and deterministic refunds

Field	Plan
Repository / base	`app` (`main`)
Project item	P08-03
Gate / size / readiness	A2 / Exceptional / Dependencies unmet
Dependencies	P08-02, P07-04, P09-04
Blocking decisions/checkpoints	D-COM-04, D-COM-06, D-COM-07
Source / salvage	Payment/business-model milestone/refund tables

Objective

Turn explicit evidence-backed case transitions into one invoice each, apply Node 0 credit, and calculate reproducible earned/uncharged/refunded amounts from immutable state.

Scope

- Enrollment/activation/clearance/legal-escrow/pregnancy/birth milestones
- Credit allocation
- Non-clearance and cancellation tables
- Dignity delay for clinical-linked invoices

Explicit non-goals

- No unresolved pregnancy-loss automatic disposition
- No promise of reimbursement

Expected files or modules

- `app/domain/billing_policy.py`
- `app/domain/refunds.py`
- `app/event_handlers/case_billing.py`
- `tests/python/test_billing_policy.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Internal evidence-to-invoice commands and explainable refund quote.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Only authorized domain events trigger billing.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- All milestone/refund rows
- duplicate/stale transition
- Node 0 credit
- pay-in-full/deposit math
- clinical notification separation

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P08-02, P07-04, P09-04. Parallel opportunities: P08-01

Objective acceptance criteria

- □ Turn explicit evidence-backed case transitions into one invoice each, apply Node 0 credit, and calculate reproducible earned/uncharged/refunded amounts from immutable state.

- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	payments, commercial policy, case state
Higher-capability review	Mandatory before merge

P08-04 — Add payment reconciliation and safe service-pause operations

Field	Plan
Repository / base	`app` (`main`)
Project item	P08-04
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P08-03
Blocking decisions/checkpoints	D-COM-08
Source / salvage	Business-model failure handling

Objective

Expose operator-safe reconciliation, discrepancy, receipt, retry, and non-critical pause controls without blocking safety information or altering clinical truth.

Scope

- Ledger-vs-processor reconciliation
- Itemized receipt data
- Grace/reminder/soft-pause policy
- Audited manual adjustments

Explicit non-goals

- No organization-wide ambient access
- No critical-service lockout

Expected files or modules

- `app/payments/reconciliation.py`
- `app/routes/operator_billing.py`
- `app/templates/internal/billing/`
- `tests/python/test_reconciliation.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	Internal role-gated discrepancy and adjustment surface.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Role denial
- mismatch workflow
- manual adjustment audit
- critical content remains accessible
- provider outage

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Metrics/alerts for settlement, refund, mismatch and reserve inputs.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P08-03. Parallel opportunities: P08-01, P08-02

Objective acceptance criteria

- □ Expose operator-safe reconciliation, discrepancy, receipt, retry, and non-critical pause controls without blocking safety information or altering clinical truth.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	payment operations, authorization, observability
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P08-01` → `P08-02` → `P08-03` → `P08-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Webhook replay/order; Ledger invariants; Milestone idempotency; Refund table; Node 0 credit floor; Reconciliation mismatch	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Type	ID / risk	Owner	Affected PRs / latest point	Project item / treatment
Blocking/product/compliance decision	D-COM-01 — Set the flat one-time price and tax treatment; it must remain fully creditable against the later JMF.	Luke	P06-01, P07-01, P08-01; Before activation of a Complete configuration	D-COM-01 — Keep no active production value until explicitly approved; test with synthetic configuration.
Blocking/product/compliance decision	D-COM-02 — Ratify $8,900 JMF, $7,830 pay-in-full, $750 deposit and 20/30/20/20/10 post-deposit milestones as configuration v1 with effective date/currency.	Luke	P07-01, P08-01, P08-03; Before P07-01 activation data	D-COM-02 — Use the document anchors but require explicit version activation rather than code literals.
Blocking/product/compliance decision	D-COM-03 — Approve the actual paid-service terms, privacy/consent disclosure and refundable-to-earned wording.	Luke	P07-01, P07-02, P08-03; Before P07-02 user acceptance	D-COM-03 — External legal/privacy/finance review before any real acceptance.
Blocking/product/compliance decision	D-COM-04 — Define evidence for enrollment funding confirmed, promotion authorization, and for onboarding/compatibility complete enough to move case active and earn/issue activation billing.	Luke	P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02	D-COM-04 — Funding confirmation is durable ledger evidence; promotion uses the sole typed authorization event; activation requires an explicit evidence checklist, actor and audit event.
Blocking/product/compliance decision	D-PAY-01 — Select processor, card/ACH capabilities, webhook/security/settlement semantics and operational ownership.	Luke	P08-02, P08-03, P08-04; Before P08-02 adapter	D-PAY-01 — Choose only after domain/ledger contract; retain processor-neutral ports.
Blocking/product/compliance decision	D-COM-05 — Select financing partner and define when funding is sufficiently authorized/settled; otherwise hide the path.	Luke	P07-02, P08-02; Before financed option is exposed	D-COM-05 — Do not block card/ACH MVP; keep optional and disabled.
Blocking/product/compliance decision	D-COM-06 — Decide retained revenue versus future-journey credit after heartbeat loss and contractual support window.	Luke	P08-03; Before automated pregnancy-stage refund logic	D-COM-06 — Manual/counsel-reviewed hook until policy is approved; never hard-code.
Blocking/product/compliance decision	D-COM-07 — Ratify earned/uncharged/refunded treatment, GC withdrawal/replacement and pay-in-full/financed reconciliation.	Luke	P08-03; Before real billing	D-COM-07 — Use document schedule as test anchor; require legal/finance approval for live policy.
Blocking/product/compliance decision	D-COM-08 — Define reminder cadence, grace period, pauseable services and safety-critical/document access that can never be blocked.	Luke	P08-04; Before P08-04	D-COM-08 — Default to continued safety and entitled-document access; pause only clearly non-critical future services.
Implementation risk	Double charge/refund	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Processor lock-in	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Clinical timing harm	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Pass-through commingling	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Policy gaps	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
No direct source	—	Build from approved contracts/evidence	Package PRs	New implementation	N/A

Package source inventory:

- Business-model schedules and smoke paths; no current code is authoritative

9. Ownership recommendation

Luke owns price/refund/pause policy; Alan implements. Payment, finance, security, and legal review are required before live money movement.

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P08-01	Alan	Alan (proposed)	Luke	Luke	Mandatory
P08-02	Alan	Alan (proposed)	Luke	Luke	Mandatory
P08-03	Alan	Alan (proposed)	Luke	Luke	Mandatory
P08-04	Alan	Alan (proposed)	Luke	Luke	Mandatory

Luke decides product/lifecycle/commercial/privacy/PHI/provider policy and accepts gates. Alan proposes implementation design, decomposition, migrations, code, tests, rollout and repair. This is a recommendation pending Alan's review/sign-off, not a compulsory assignment.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 09 — Documents, active case, and provider handoffs

Planning-only execution contract. Proposed implementers and owners remain recommendations until Alan reviews/signs off the consolidated plan and Luke accepts it. Project: MVP Product Flow Reshape.

Plan	Value
Work package	WP09
Plan file	`09-documents-active-case-and-provider-handoffs.md`
Proposed PRs	P09-01, P09-02, P09-03, P09-04, P09-05
Inspection snapshot	2026-07-10 20:12 America/New_York

1. Purpose and outcome

Problem. Current documents are metadata-only; the dirty provider branch is unreviewed and can over-project facts. There is no durable private S3/KMS/quarantine/recovery pipeline.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. A secure document lifecycle, resilient storage/scan/download pipeline, usable active-case surfaces, and consent-scoped immutable provider snapshots with operational review.

Explicit non-goals.

- No ambient provider case access
- No clinical approval by Natalyx
- No direct public objects
- No automatic promotion of AI/IP assertions

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-006	verified	app / PR #122@e853821	PR comment Review round 2 addressed and changed workflow/tests	Updated PR #122 reports 499 tests plus return_to, TTL, CAS, self-invite, lens and Learn fail-closed fixes; green checks were observed. These are salvage candidates, not blanket merge approval.
E-012	verified	app / PR #122@e853821	app/storage/invitations.py and PR review-round comment	The updated case invitation work adds 14-day ceiling, CAS transitions, mismatch confirmation, self-invite refusal, return_to preservation and paired lens tests; it has no Node 0 workspace target.
E-018	verified	app / PR #124@84a5a84	app/storage/documents.py:1-114; app/storage/schema.py:785-801	Documents are explicitly metadata-only; storage_uri/checksum fields exist but no durable upload, quarantine, scan, authorization-by-purpose, retention or recovery pipeline.
E-019	verified protected work	app / dirty feature/provider-exchange-layer-v0@84a5a84	app/provider_handoffs.py:109-133 and uncommitted handoff modules	The protected provider work is not a PR. Its broad GC-owner fact selection can include private/provisional data, so only selective port after provenance/purpose review is safe.

3. Architecture and contracts

- Private bucket, block-public-access, KMS, opaque keys, application authorization, size/type/checksum gates, quarantine/scan, audit, retention, and orphan cleanup are mandatory.
- Provider handoffs are purpose-bound snapshots with checksum/revision/expiry and consent; provider-visible items are allowlisted, not inferred from broad visibility.
- ECS replacement must not lose upload/task state; recovery resumes from durable metadata and object state.

Contract area	Package rule
Ownership boundary	`app`, `infra`; repository-specific PRs remain separate.
Entities/state	`case`, `case_invitation`, `document + document_version + upload`
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Intent/finalize/status/download/replace/delete commands. Membership-scoped reads and evidence commands.
Storage	Versioned document/upload/scan/object metadata. Handoff/packet/snapshot/transmission/evidence ledgers under migrations.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Applicable state machines

Paid case lifecycle (`case`)

Owner: natalyx-app. States: `enrollment_pending`, `active`, `paused`, `closed`, `cancelled`.

none

From	none
To	enrollment_pending
Actor	promotion service
Evidence	byos.workspace_promotion_authorized.v1 + projection manifest
Idempotency key	promotion:{commitment}:v1
Invalid transitions	No authorization/duplicate workspace
Audit event	paid_case_created
Retry	Return same case
Recovery	Forward-repair same promotion

enrollment_pending

From	enrollment_pending
To	active
Actor	authorized coordinator/domain service
Evidence	approved onboarding/compatibility/intake/education evidence
Idempotency key	case:{id}:activate:{evidence_hash}
Invalid transitions	Missing/revoked evidence or GC consent
Audit event	case_activated
Retry	No duplicate activation/invoice
Recovery	Return pending with unmet checklist

active

From	active
To	paused
Actor	authorized commercial/operations policy
Evidence	audited non-critical pause reason
Idempotency key	case:{id}:pause:{event}
Invalid transitions	Safety-critical access cannot be paused
Audit event	case_paused
Retry	Idempotent
Recovery	Resume after evidence; preserve entitled docs

paused

From	paused
To	active
Actor	authorized operator/domain service
Evidence	pause reason resolved
Idempotency key	case:{id}:resume:{event}
Invalid transitions	Unresolved safety/authorization block
Audit event	case_resumed
Retry	Idempotent
Recovery	Escalate manual review

active|paused

From	active|paused
To	closed
Actor	authorized coordinator
Evidence	birth/close or completed termination handoff evidence
Idempotency key	case:{id}:close:{evidence_hash}
Invalid transitions	Open critical obligations
Audit event	case_closed
Retry	No-op if same evidence
Recovery	Reopen only by exceptional audited event

enrollment_pending|active|paused

From	enrollment_pending|active|paused
To	cancelled
Actor	authorized operator under policy
Evidence	withdrawal/non-clearance/cancellation + financial disposition
Idempotency key	case:{id}:cancel:{request}
Invalid transitions	Cannot erase records/refund ledger
Audit event	case_cancelled
Retry	Idempotent
Recovery	Replacement is a new case/workspace policy event

Paid-case membership invitation (`case_invitation`)

Owner: natalyx-app. States: `issued`, `pending_identity`, `identity_bound`, `mismatch_pending`, `issuer_confirmed`, `consented`, `redeemed`, `declined`, `revoked`, `expired`, `superseded`.

none

From	none
To	issued
Actor	authorized IP case member
Evidence	enrollment_pending/active case + recipient binding
Idempotency key	case-invite:{case}:{role}:{generation}
Invalid transitions	Self/member/active-GC/unauthorized
Audit event	case_invite_issued
Retry	Return current; supersede prior live GC invite
Recovery	Issue corrected invite

issued

From	issued
To	pending_identity
Actor	recipient
Evidence	valid addressed token landing
Idempotency key	case-invite:{id}:land
Invalid transitions	Expired/revoked
Audit event	case_invite_opened
Retry	Safe repeat
Recovery	Authenticate with return_to

pending_identity

From	pending_identity
To	identity_bound|mismatch_pending
Actor	auth provider
Evidence	verified subject/email
Idempotency key	case-invite:{id}:bind:{subject}
Invalid transitions	Terminal/duplicate/different subject
Audit event	case_invite_bound|mismatch
Retry	CAS protects terminal
Recovery	Issuer confirmation or corrected invite

mismatch_pending

From	mismatch_pending
To	issuer_confirmed
Actor	authorized issuer
Evidence	explicit reviewed mismatch confirmation
Idempotency key	case-invite:{id}:confirm:{issuer}
Invalid transitions	Issuer unauthorized/recipient changed
Audit event	case_invite_mismatch_confirmed
Retry	CAS no-op
Recovery	Revoke and reissue

identity_bound|issuer_confirmed

From	identity_bound|issuer_confirmed
To	consented
Actor	invitee
Evidence	paid-case role/privacy consent version
Idempotency key	case-invite:{id}:consent:{version}
Invalid transitions	Stale/missing consent
Audit event	case_invite_consented
Retry	Idempotent
Recovery	Present current consent

consented

From	consented
To	redeemed
Actor	invitation service
Evidence	single transaction creates person/party/membership
Idempotency key	case-invite:{id}:redeem
Invalid transitions	Existing member/active GC/expired/revoked
Audit event	case_invite_redeemed
Retry	Return same membership
Recovery	Rollback entire transaction; forward repair from invite

issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented

From	issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented
To	declined|revoked|expired|superseded
Actor	invitee/issuer/system
Evidence	explicit decline/revoke/TTL/new generation
Idempotency key	case-invite:{id}:terminal:{event}
Invalid transitions	No terminal overwrite
Audit event	case_invite_terminal
Retry	Idempotent CAS
Recovery	Fresh invitation only

Document lifecycle (`document + document_version + upload`)

Owner: natalyx-app; storage in natalyx-infra. States: `upload_requested`, `uploading`, `quarantined`, `scanning`, `available`, `review`, `approved`, `replaced`, `deleted`, `failed`.

none

From	none
To	upload_requested
Actor	authorized case member
Evidence	document purpose/type/size intent
Idempotency key	upload:{case}:{client_request}
Invalid transitions	Unauthorized/type/size/retention block
Audit event	upload_requested
Retry	Return same intent
Recovery	Expire/orphan intent

upload_requested

From	upload_requested
To	uploading
Actor	client/storage service
Evidence	authorized upload starts
Idempotency key	upload:{id}:start
Invalid transitions	Expired intent/wrong key
Audit event	upload_started
Retry	Multipart resume
Recovery	Abort and mark failed/orphan cleanup

uploading

From	uploading
To	quarantined
Actor	finalize service
Evidence	object exists + size/checksum/content sniff
Idempotency key	upload:{id}:finalize:{checksum}
Invalid transitions	Mismatch/missing object
Audit event	upload_quarantined
Retry	Repeat verify
Recovery	Delete bad object; request new upload

quarantined

From	quarantined
To	scanning
Actor	scan worker
Evidence	durable queued task
Idempotency key	scan:{version}:{engine_version}
Invalid transitions	Already available/deleted
Audit event	scan_started
Retry	Leased retry
Recovery	Lease expiry requeues after ECS replacement

scanning

From	scanning
To	available|failed
Actor	scan service
Evidence	signed clean/malicious/error result
Idempotency key	scan:{version}:{result_id}
Invalid transitions	Unknown result/engine
Audit event	scan_completed|scan_failed
Retry	Inbox replay no-op
Recovery	Remain quarantined on timeout/error; security review

available

From	available
To	review|approved
Actor	authorized participant/reviewer
Evidence	purpose-specific review/evidence
Idempotency key	document:{version}:review:{event}
Invalid transitions	Wrong role/stale version
Audit event	document_reviewed
Retry	Idempotent
Recovery	New review revision

available|review|approved

From	available|review|approved
To	replaced
Actor	authorized owner
Evidence	new clean version linked
Idempotency key	document:{id}:replace:{new_version}
Invalid transitions	New version not available
Audit event	document_replaced
Retry	Return current version
Recovery	Keep prior per retention; roll current pointer forward

available|review|approved|replaced|failed

From	available|review|approved|replaced|failed
To	deleted
Actor	authorized policy job/operator
Evidence	retention/delete request and no hold
Idempotency key	document:{version}:delete:{request}
Invalid transitions	Legal hold/provider obligation
Audit event	document_deleted
Retry	Tombstone idempotent
Recovery	Object-delete retry/orphan alarm; metadata/audit retained

Data authorship/provenance rules

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.
- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

Durable document and S3 contract

Storage

- Private versioned quarantine bucket and clean vault with block-public-access, SSL-only and KMS encryption.
- App can create short-lived unique upload targets in quarantine but cannot read unscanned content; scanner moves/copies only clean versions to vault.
- Object key: env/{org_id}/{case_id}/{document_id}/{version_id}/{generated_object_id}; no names, emails, filenames or clinical labels.
- Authorization is always a Postgres membership/role/visibility/consent decision independent of the key.
- CloudTrail S3 data events and app document_access_event records provide purpose-aware access evidence.

Upload

- Configurable maximum size; provisional 25 MiB awaits D-DOC-02.
- Initial proposed allowlist PDF/JPEG/PNG; DOCX only after scanner/archive review; no macro-enabled Office formats.
- Unicode-normalize/sanitize display filename; byte-sniff MIME, expected length and required SHA-256 checksum.
- Never reuse a key. Finalize verifies object version, HeadObject size/checksum/type and upload-session ownership.
- S3 event -> durable queue -> scanner -> signed clean/rejected/error result; unavailable/timed-out scan remains quarantined.

Download

- App authenticates and authorizes exact version/purpose, records access event, then returns a very short-lived clean-vault GET with safe content disposition.
- App proxy is reserved for justified transformation/audit cases, not the default durable path.

Recovery

- Upload sessions/tasks live in Postgres/queue so ECS replacement does not lose them; leases expire and retry idempotently.
- Replacement creates a new immutable version and supersedes the prior current pointer.

- Deletion is an audited tombstone followed by retention/legal-hold controlled purge; object-delete failures alert and retry.
- Orphan reconciliation handles expired intents and objects without document versions; lifecycle aborts incomplete multipart uploads.
- Provider share uses a purpose-bound snapshot/checksum/expiry, not direct object-key authority.

4. Explicit PR sequence

1

Order	1
ID	P09-01
Title	Define document lifecycle, authorization, and retention contract
Repository	app
Dependencies	P00-01, P02-01
Gate	A2
Size	Medium
Readiness	Blocked by decision

2

Order	2
ID	P09-02
Title	Provision private encrypted document and quarantine storage
Repository	infra
Dependencies	P09-01, P13-02
Gate	A2
Size	Large
Readiness	Dependencies unmet

3

Order	3
ID	P09-03
Title	Implement resilient document upload, scan, and download
Repository	app
Dependencies	P09-02, P02-03
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet

4

Order	4
ID	P09-04
Title	Complete active-case participant document and progress surfaces
Repository	app
Dependencies	P03-03, P07-04, P09-03, P11-03
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet

5

Order	5
ID	P09-05
Title	Implement consent-scoped provider handoff snapshots and operations
Repository	app
Dependencies	P04-02, P09-03, P09-04, P10-02
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet

P09-01 — Define document lifecycle, authorization, and retention contract

Field	Plan
Repository / base	`app` (`main`)
Project item	P09-01
Gate / size / readiness	A2 / Medium / Blocked by decision
Dependencies	P00-01, P02-01
Blocking decisions/checkpoints	D-DOC-01, D-DOC-02, D-PHI-01
Source / salvage	Current metadata-only documents.py, provider branch

Objective

Record the document state machine, role/purpose authorization matrix, object-key independence, retention/deletion/replacement rules, and failure recovery before storage implementation.

Scope

- Upload/scan/available/review/replace/delete states
- Size/type/checksum and filename policy
- Presigned versus proxied download decision
- Orphan/retention jobs

Explicit non-goals

- No bucket or application code

Expected files or modules

- `docs/adr/ADR-00x-durable-documents.md`
- `docs/agents/documents.md`
- `docs/security/document-threat-model.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Decision and threat model only.

Tests and validation

Objective tests:

- Threat-model abuse cases

- state transition table review
- retention/legal-hold walkthrough

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P00-01, P02-01. Parallel opportunities: P09-03, P09-04, P09-05

Objective acceptance criteria

- □ Record the document state machine, role/purpose authorization matrix, object-key independence, retention/deletion/replacement rules, and failure recovery before storage implementation.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	application security, S3/KMS, privacy
Higher-capability review	Mandatory before merge

P09-02 — Provision private encrypted document and quarantine storage

Field	Plan
Repository / base	`infra` (`main`)
Project item	P09-02
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P09-01, P13-02
Blocking decisions/checkpoints	D-DOC-02, D-PHI-01
Source / salvage	Existing infra platform stack patterns

Objective

Provision dev/private document buckets, KMS policy, block-public-access, lifecycle/quarantine events, least-privilege task roles, and audit logging behind PHI/billable guardrails.

Scope

- Private primary/quarantine storage
- KMS and task-role policy
- Lifecycle/orphan signals
- No public ACL/policy

Explicit non-goals

- No production/PHI activation
- No application upload

Expected files or modules

- `lib/stacks/platform-stack.ts`
- `lib/constructs/document-storage.ts`
- `test/document-storage.test.ts`
- `docs/agents/platform-stack.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	CDK resources and SSM outputs in dev, desired counts unchanged by default.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- CDK assertions
- IAM negative cases
- public-access-block
- encryption/lifecycle/event contract

Local validation:

- `npm test`
- `npm run build`
- `Synthesize the affected CDK stack with billable/live changes disabled.`

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Synthesize and review first; deploy only through the guarded dev workflow with billable/desired-count defaults unchanged, exact revision recorded, and explicit confirmation outside this planning task.

Rollback or forward repair

Return to the prior reviewed stack/task definition when compatible; otherwise keep resources fail-closed and forward-repair under the runbook. Never delete data-bearing resources as an automatic rollback.

Dependencies and parallel work

Hard dependencies: P09-01, P13-02. Parallel opportunities: P09-04, P09-05

Objective acceptance criteria

- □ Provision dev/private document buckets, KMS policy, block-public-access, lifecycle/quarantine events, least-privilege task roles, and audit logging behind PHI/billable guardrails.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	AWS S3, KMS/IAM, CDK, security
Higher-capability review	Mandatory before merge

P09-03 — Implement resilient document upload, scan, and download

Field	Plan
Repository / base	`app` (`main`)
Project item	P09-03
Gate / size / readiness	A2 / Exceptional / Dependencies unmet
Dependencies	P09-02, P02-03
Blocking decisions/checkpoints	D-DOC-01, D-DOC-02
Source / salvage	Current metadata scaffold

Objective

Implement multipart-safe upload intent/finalize, checksum/type/size validation, quarantine/scan transitions, authorized download, replacement, deletion, and task-recovery behavior.

Scope

- Opaque object keys and sanitized display names
- Durable upload states
- Malware/quarantine integration
- Presigned/proxy path per ADR
- Audit events

Explicit non-goals

- No public object URL
- No provider share by key

Expected files or modules

- `app/documents/service.py`
- `app/documents/storage.py`
- `app/routes/documents.py`
- `app/jobs/document_recovery.py`

- `tests/python/test_documents_e2e.py`

Change map

Area	Planned change
Schema/model	Versioned document/upload/scan/object metadata.
API/routes	Intent/finalize/status/download/replace/delete commands.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Oversize/type spoof
- checksum mismatch
- scan fail/timeout
- ECS task replacement
- orphan cleanup
- authorization

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P09-02, P02-03. Parallel opportunities: P09-01

Objective acceptance criteria

- □ Implement multipart-safe upload intent/finalize, checksum/type/size validation, quarantine/scan transitions, authorized download, replacement, deletion, and task-recovery behavior.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	S3, secure uploads, background jobs, authorization
Higher-capability review	Mandatory before merge

P09-04 — Complete active-case participant document and progress surfaces

Field	Plan
Repository / base	`app` (`main`)
Project item	P09-04
Gate / size / readiness	A2 / Exceptional / Dependencies unmet
Dependencies	P03-03, P07-04, P09-03, P11-03
Blocking decisions/checkpoints	D-COM-04, CHECK-ALAN-122-124
Source / salvage	PR #122 selective/direct reuse after checkpoint, PR #124 case UI

Objective

Expose role-scoped active-case tasks, documents, progress, waiting and safe pause states after activation while keeping the server projection authoritative.

Scope

- enrollment_pending versus active surfaces
- IP/GC lenses
- Documents/tasks/timeline/notifications
- Mobile journey orientation

Explicit non-goals

- No client-owned state transition
- No operator/provider leakage

Expected files or modules

- `app/case_home.py`
- `app/case_journey.py`
- `app/routes/case.py`
- `app/templates/case/`
- `tests/python/test_case_home.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Membership-scoped reads and evidence commands.
UI/product surface	Participant control center and journey map backed by case projection.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- IP/GC paired lenses
- pending/active/paused
- no membership
- malformed/stale projection
- mobile/no-JS/reduced-motion

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P03-03, P07-04, P09-03, P11-03. Parallel opportunities: P09-01, P09-02

Objective acceptance criteria

- □ Expose role-scoped active-case tasks, documents, progress, waiting and safe pause states after activation while keeping the server projection authoritative.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	FastAPI/Jinja, accessibility, authorization, state projection
Higher-capability review	Mandatory before merge

P09-05 — Implement consent-scoped provider handoff snapshots and operations

Field	Plan
Repository / base	`app` (`main`)
Project item	P09-05
Gate / size / readiness	A2 / Exceptional / Dependencies unmet
Dependencies	P04-02, P09-03, P09-04, P10-02
Blocking decisions/checkpoints	D-PROV-01, D-PHI-01
Source / salvage	Protected provider-exchange worktree: selective port only after provenance fix

Objective

Create purpose-bound, reviewed, checksummed provider packets from explicitly allowed facts/documents and operate send/confirm/revoke/reroute without ambient provider access.

Scope

- Packet revisions/snapshots/checksums
- Consent, recipient, purpose, expiry
- Operator route review and transmission outcomes
- Inbound evidence becomes proposal

Explicit non-goals

- No raw private/provisional bulk projection
- No provider truth without review

Expected files or modules

- `app/provider_handoffs.py`
- `app/storage/handoffs.py`
- `app/routes/operator_handoffs.py`
- `app/templates/internal/handoffs/`
- `tests/python/test_provider_handoffs.py`

Change map

Area	Planned change
Schema/model	Handoff/packet/snapshot/transmission/evidence ledgers under migrations.
API/routes	No public API/route contract change.
UI/product surface	Case-scoped operator queue and GC review.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Purpose allowlist
- revoked consent
- stale route
- snapshot immutability
- delivery retry
- private/provisional exclusion

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P04-02, P09-03, P09-04, P10-02. Parallel opportunities: P09-01, P09-02

Objective acceptance criteria

- □ Create purpose-bound, reviewed, checksummed provider packets from explicitly allowed facts/documents and operate send/confirm/revoke/reroute without ambient provider access.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	provider workflows, privacy, document security, operations
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain
`infra`	AWS resources, IAM/KMS/network/deployment/backup/audit contracts

Merge order inside this package: `P09-01` → `P09-02` → `P09-03` → `P09-04` → `P09-05`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Unauthorized object access; Oversize/type/spoof; Scan failure/quarantine; Partial upload/task loss; Snapshot immutability/revocation; Mobile documents	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Type	ID / risk	Owner	Affected PRs / latest point	Project item / treatment
Blocking/product/compliance decision	D-DATA-01 — Which provisional IP assertions may the GC see, which GC answers may the IP see, and which fields survive promotion/provider projection?	Luke	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02	D-DATA-01 — Default deny; expose only explicit purpose-limited projections. Class 1/2 and class 4 never automatically cross roles.
Blocking/product/compliance decision	D-COM-04 — Define evidence for enrollment funding confirmed, promotion authorization, and for onboarding/compatibility complete enough to move case active and earn/issue activation billing.	Luke	P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02	D-COM-04 — Funding confirmation is durable ledger evidence; promotion uses the sole typed authorization event; activation requires an explicit evidence checklist, actor and audit event.
Blocking/product/compliance decision	D-DOC-01 — Set retention/legal-hold/deletion and choose presigned versus app-proxied downloads by document class.	Luke	P09-01, P09-03; Before P09-03	D-DOC-01 — Purpose/role authorization independent of key; short-lived downloads; deletion tombstone/audit; legal/privacy review.
Blocking/product/compliance decision	D-DOC-02 — Choose maximum size/types, scanner/quarantine mechanism, timeout/retry and fail-closed handling.	Luke	P09-01, P09-02, P09-03; Before P09-02	D-DOC-02 — Narrow allowlist, checksum and content sniffing; unavailable scanner remains quarantined.
Blocking/product/compliance decision	D-PROV-01 — Define permitted recipients/purposes, GC preview/consent, expiry, verification level and exact fact/document allowlists.	Luke	P09-05; Before P09-05	D-PROV-01 — Class 1/2 excluded; class 4 only if GC explicitly selects for recipient/purpose; class 5 evidence linked.
Blocking/product/compliance decision	D-PHI-01 — Approve data classification, vendors/BAAs, encryption/logging/access/incident/backup controls and prohibited data at each gate.	Luke	P09-01, P09-02, P09-05, P13-05, P13-06, P13-07; Before B1 for PII scope; before B2 for PHI	D-PHI-01 — B1 remains PII-only; B2 requires formal external security/privacy/compliance evidence.
Implementation risk	PHI exposure	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Malware	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Orphaned objects	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Stale provider packet	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Dirty branch leakage	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#112	9605d88e31bd	Source-first 81-node/12-cluster surface map, participant-renderer boundaries, wave plan and cross-repo seams.	P09-04	Selective port, then supersede after extraction	Catalogued; content remains to be reconciled into replacement PR acceptance tests.
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P09-04	Retain pending Alan reconciliation; selective port or rebase only after architecture approval	Reinspected at latest head. It is exactly e853821 (no newer delta); retention strengthened, but final extraction waits for Alan checkpoint.
app#124	84a5a847fea7	Current consolidation of Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitations, DS assets and large focused test suites.	P09-04	Retain unchanged as evidence; selective port/rewrite after checkpoint	Unextracted; Alan review explicitly incomplete and unresolved.
feature/provider-exchange-layer-v0	84a5a84	substantial uncommitted provider handoff/ops/document work	P09-01/P09-03/P09-05 after provenance/security review	Selective port from protected worktree	Do not stage, reset, stash, switch, commit, clean or overwrite.
main	current local	substantial uncommitted operator shell/design-system work	Not part of participant DS; review only if P09-05 operator UI needs approved internal recipes	Selective port from protected worktree	Preserve repo boundary and dirty worktree.

Package source inventory:

- Protected provider-exchange branch by selective port after review
- Current document metadata scaffold
- PR #122 case shell UX

9. Ownership recommendation

Alan owns implementation; Luke approves retention, provider purpose, and PHI boundary. External security/privacy and provider workflow review required.

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P09-01	Alan	Alan (proposed)	Luke	Luke	Mandatory
P09-02	Alan	Alan (proposed)	Luke	Luke	Mandatory
P09-03	Alan	Alan (proposed)	Luke	Luke	Mandatory
P09-04	Alan	Alan (proposed)	Luke	Luke	Mandatory
P09-05	Alan	Alan (proposed)	Luke	Luke	Mandatory

Luke decides product/lifecycle/commercial/privacy/PHI/provider policy and accepts gates. Alan proposes implementation design, decomposition, migrations, code, tests, rollout and repair. This is a recommendation pending Alan's review/sign-off, not a compulsory assignment.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.

- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 10 — Intelligence, memory, proposals, and Ask safety

Planning-only execution contract. Proposed implementers and owners remain recommendations until Alan reviews/signs off the consolidated plan and Luke accepts it. Project: MVP Product Flow Reshape.

Plan	Value
Work package	WP10
Plan file	`10-intelligence-memory-proposals-and-ask-safety.md`
Proposed PRs	P10-01, P10-02, P10-03, P10-04
Inspection snapshot	2026-07-10 20:12 America/New_York

1. Purpose and outcome

Problem. Private memory, shared case truth, Learn context, and AI proposals have partial contracts but current app code can place IP guide notes into shared memory and provider projections lack provenance enforcement.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Versioned scoped context envelopes, separate private/shared memory revisions, evidence-bound proposal approval, grounded Ask behavior, and cross-service safety evaluation/observability.

Explicit non-goals.

- No LLM writes to case truth
- No medical/legal advice
- No silent fallback success
- No final Learn corpus authoring

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-014	verified	app / PR #124@84a5a84	app/storage/node0.py:42-200,509-600; tests/python/test_node0.py:444-490	IP/co-IP own candidate reads; GC onboarding writes separate shared/private facts and does not revise the IP assertion. The GC cannot see or correct the IP candidate record.
E-015	verified	app / PR #122@e853821	app/storage/facts.py:19-69,107-127	Facts have broad provenance and visibility values but no subject-versus-author role, verification state, derived-from, or supersedes link.
E-016	verified	app / PR #122@e853821	app/storage/proposals.py:117-142	Generic proposal approval permits any case member and can pass caller-selected provenance to create_fact; a dedicated evidence/role rail is required before reviewed/provider-confirmed truth.
E-017	verified	app / PR #124@84a5a84	app/routes/node0.py:686-719	IP Conversation Guide notes are saved as shared memory and sent to intelligence, creating a future cross-role leakage risk.
E-019	verified protected work	app / dirty feature/provider-exchange-layer-v0@84a5a84	app/provider_handoffs.py:109-133 and uncommitted handoff modules	The protected provider work is not a PR. Its broad GC-owner fact selection can include private/provisional data, so only selective port after provenance/purpose review is safe.

ID	Class	Repository / ref	File, construct or location	Finding
E-020	verified	intelligence / origin/main@56f6c4b	src/natalyx_intelligence/case_context.py:8-147; docs/agents/case-context.md	The service returns advisory-only proposals and says durable writes are app-owned; this is a useful boundary but the surrounding context provenance remains incomplete.
E-021	verified	intelligence / origin/main@56f6c4b	src/natalyx_intelligence/ask_natalyx.py:34-42,365-378	Ask accepts known_intake_state without an authorship-class contract and serializes it into model context.

3. Architecture and contracts

- The app owns case truth, authorization, PHI policy, and durable approval; intelligence returns advisory proposals only.
- Every request declares general Learn, private-user, approved-shared-case, or provider-purpose scope; contexts are not merged.
- Failures degrade explicitly with source/boundary notices and no fabricated case answer.

Contract area	Package rule
Ownership boundary	`app`, `intelligence`; repository-specific PRs remain separate.
Entities/state	No new runtime entity; evidence/coordination only.
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Versioned context/advisory contract with strict validation and safe compatibility window. Responses carry source, scope, policy, safety and degraded-mode metadata.
Storage	Revision/supersedes/provenance/evidence fields under migrations.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Data authorship/provenance rules

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.
- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

4. Explicit PR sequence

1

Order	1
ID	P10-01
Title	Version the scoped intelligence context envelope
Repository	intelligence
Dependencies	P01-01, P04-02
Gate	A2
Size	Large
Readiness	Dependencies unmet

2

Order	2
ID	P10-02
Title	Enforce private/shared memory and evidence-bound proposal approval
Repository	app
Dependencies	P04-02, P10-01
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet

3

Order	3
ID	P10-03
Title	Harden Ask and Learn grounding and degraded-mode behavior
Repository	intelligence
Dependencies	P10-01, P11-02
Gate	A2
Size	Large
Readiness	Dependencies unmet

4

Order	4
ID	P10-04
Title	Add intelligence safety evaluations and cross-service observability
Repository	intelligence
Dependencies	P10-02, P10-03
Gate	A2
Size	Large
Readiness	Dependencies unmet

P10-01 — Version the scoped intelligence context envelope

Field	Plan
Repository / base	`intelligence` (`main`)
Project item	P10-01
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P01-01, P04-02
Blocking decisions/checkpoints	D-DATA-01, D-AI-01
Source / salvage	Current case_context advisory stub

Objective

Define and validate versioned request/response envelopes that keep general Learn, private-user memory, approved shared-case facts, and provider-purpose context distinct.

Scope

- Explicit scope and audience
- Source/provenance/verification metadata
- Policy/model/version and redacted trace metadata

Explicit non-goals

- No durable app write
- No merged contexts

Expected files or modules

- `src/natalyx_intelligence/contracts/context.py`
- `src/natalyx_intelligence/case_context.py`
- `docs/agents/case-context.md`
- `tests/test_context_contract.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Versioned context/advisory contract with strict validation and safe compatibility window.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Cross-scope rejection
- unknown version
- private/shared fixtures
- redaction
- service auth

Local validation:

- `uv run pytest`
- Run focused contract/migration/safety tests with redacted fixtures.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Release a versioned/pinned artifact or API contract first; consumers opt in explicitly and retain fail-closed/previous-version behavior during the compatibility window.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P01-01, P04-02. Parallel opportunities: P10-04

Objective acceptance criteria

- □ Define and validate versioned request/response envelopes that keep general Learn, private-user memory, approved shared-case facts, and provider-purpose context distinct.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	FastAPI/Pydantic, AI safety, privacy
Higher-capability review	Mandatory before merge

P10-02 — Enforce private/shared memory and evidence-bound proposal approval

Field	Plan
Repository / base	`app` (`main`)
Project item	P10-02
Gate / size / readiness	A2 / Exceptional / Dependencies unmet
Dependencies	P04-02, P10-01
Blocking decisions/checkpoints	D-DATA-01, D-AI-01
Source / salvage	Current facts/memory/proposals scaffolds, GC authorship audit

Objective

Store attributable private/shared memory revisions and require authorized evidence review before an intelligence proposal can affect case truth or provider material.

Scope

- Private and shared memory ledgers
- Proposal preview/approve/reject
- Evidence/source and reviewer authority
- Audit/outbox

Explicit non-goals

- No automatic AI approval
- No guide notes default shared

Expected files or modules

- `app/case_intelligence.py`
- `app/storage/memory.py`
- `app/storage/proposals.py`

- `app/storage/facts.py`
- `tests/python/test_case_intelligence.py`

Change map

Area	Planned change
Schema/model	Revision/supersedes/provenance/evidence fields under migrations.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Role and purpose authorization on every projection/write.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Private-to-shared denial
- wrong-role approval
- AI provenance remains proposal
- concurrent review
- provider projection

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P04-02, P10-01. Parallel opportunities: P10-03

Objective acceptance criteria

- □ Store attributable private/shared memory revisions and require authorized evidence review before an intelligence proposal can affect case truth or provider material.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	AI safety, authorization, data provenance
Higher-capability review	Mandatory before merge

P10-03 — Harden Ask and Learn grounding and degraded-mode behavior

Field	Plan
Repository / base	`intelligence` (`main`)
Project item	P10-03
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P10-01, P11-02
Blocking decisions/checkpoints	D-AI-01
Source / salvage	Current ask_natalyx.py and PR #124 Ask integration

Objective

Answer from the declared scope with approved sources/citations and explicit boundary/failure notices; never fabricate case context or silently fall across scope.

Scope

- Approved resource resolution
- Scope-specific prompt/context
- Timeout/unavailable/empty-source behavior
- Citation validation

Explicit non-goals

- No medical/legal advice
- No unapproved corpus fallback

Expected files or modules

- `src/natalyx_intelligence/ask_natalyx.py`
- `src/natalyx_intelligence/learn_resources.py`
- `tests/test_ask_natalyx.py`
- `docs/agents/ask-natalyx.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Responses carry source, scope, policy, safety and degraded-mode metadata.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Empty/unapproved corpus
- service/provider timeout
- case question without case scope
- citation mismatch
- safety escalation

Local validation:

- `uv run pytest`
- Run focused contract/migration/safety tests with redacted fixtures.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Release a versioned/pinned artifact or API contract first; consumers opt in explicitly and retain fail-closed/previous-version behavior during the compatibility window.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P10-01, P11-02. Parallel opportunities: P10-02

Objective acceptance criteria

- □ Answer from the declared scope with approved sources/citations and explicit boundary/failure notices; never fabricate case context or silently fall across scope.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	LLM orchestration, retrieval/citations, safety evaluation
Higher-capability review	Mandatory before merge

P10-04 — Add intelligence safety evaluations and cross-service observability

Field	Plan
Repository / base	`intelligence` (`main`)
Project item	P10-04
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P10-02, P10-03
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Current intelligence auth/telemetry tests

Objective

Add deterministic safety/grounding evaluations, correlation IDs, latency/failure metrics, and redacted traces across app-intelligence calls.

Scope

- Leakage/hallucination/fallback evals
- Correlation and contract version
- Safe logs and alarms

Explicit non-goals

- No prompt or participant payload in logs
- No PHI readiness claim

Expected files or modules

- `tests/evals/`
- `src/natalyx_intelligence/telemetry.py`
- `docs/runbooks/intelligence.md`
- `app/intelligence_auth.py (consumer follow-up PR if needed)`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- PII/PHI redaction fixtures
- correlation continuity
- failure classification
- model fallback
- eval threshold

Local validation:

- `uv run pytest`
- Run focused contract/migration/safety tests with redacted fixtures.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Metrics and redacted structured events; app-side consumer change must remain a separate coordinated PR if repository boundaries require it.

Rollout

Release a versioned/pinned artifact or API contract first; consumers opt in explicitly and retain fail-closed/previous-version behavior during the compatibility window.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P10-02, P10-03. Parallel opportunities: P10-01

Objective acceptance criteria

- □ Add deterministic safety/grounding evaluations, correlation IDs, latency/failure metrics, and redacted traces across app-intelligence calls.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	AI evaluation, observability, privacy
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain
`intelligence`	advisory inference, scoped response/citation contract; no case truth

Merge order inside this package: `P10-01` → `P10-02` → `P10-03` → `P10-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Scope isolation; Proposal approval/rejection; Citation/source validity; Unavailable service; Prompt/context leakage; Trace redaction	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Type	ID / risk	Owner	Affected PRs / latest point	Project item / treatment
Blocking/product/compliance decision	D-DATA-01 — Which provisional IP assertions may the GC see, which GC answers may the IP see, and which fields survive promotion/provider projection?	Luke	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02	D-DATA-01 — Default deny; expose only explicit purpose-limited projections. Class 1/2 and class 4 never automatically cross roles.
Blocking/product/compliance decision	D-AI-01 — Approve scopes, allowed sources, model policy, proposal reviewers, class-5 evidence rail and degraded-mode copy.	Luke	P10-01, P10-02, P10-03, P12-01; Before P10-01	D-AI-01 — Generic case members cannot create reviewed/provider truth; intelligence remains advisory and scope explicit.
Implementation risk	Private-to-shared leakage	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Hallucinated truth	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Sensitive logs	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Contract version drift	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#119	a8cd2f428535	Concise product-leverage framing and boundaries for the MVP flow.	P10-01	Retain as source or merge separately after plan decision	Catalogued; reconcile terminology into approved programme.
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P10-02	Retain pending Alan reconciliation; selective port or rebase only after architecture approval	Reinspected at latest head. It is exactly e853821 (no newer delta); retention strengthened, but final extraction waits for Alan checkpoint.
app#124	84a5a847fea7	Current consolidation of Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitations, DS assets and large focused test suites.	P10-02	Retain unchanged as evidence; selective port/rewrite after checkpoint	Unextracted; Alan review explicitly incomplete and unresolved.
main (behind origin/main)	9d824e8	dirty Learn registry/corpus plus uv.lock	P11-02/P10-03 as appropriate	Selective port from protected worktree	Do not pull/reset/stash; extract in isolated worktree later.

Package source inventory:

- Intelligence case_context advisory stub
- App proposal/fact/memory scaffolds
- Protected intelligence Learn corpus changes

9. Ownership recommendation

Alan implements service/app contracts; Luke approves truth/privacy policy. Higher-capability, security/privacy, and clinical/legal safety review required.

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P10-01	Alan	Alan (proposed)	Luke	Luke	Mandatory
P10-02	Alan	Alan (proposed)	Luke	Luke	Mandatory
P10-03	Alan	Alan (proposed)	Luke	Luke	Mandatory
P10-04	Alan	Alan (proposed)	Luke	Luke	Mandatory

Luke decides product/lifecycle/commercial/privacy/PHI/provider policy and accepts gates. Alan proposes implementation design, decomposition, migrations, code, tests, rollout and repair. This is a recommendation pending Alan's review/sign-off, not a compulsory assignment.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 11 — Learn release and design-system contract

Planning-only execution contract. Proposed implementers and owners remain recommendations until Alan reviews/signs off the consolidated plan and Luke accepts it. Project: MVP Product Flow Reshape.

Plan	Value
Work package	WP11
Plan file	`11-learn-release-and-design-system-contract.md`
Proposed PRs	P11-01, P11-02, P11-03, P11-04
Inspection snapshot	2026-07-10 20:12 America/New_York

1. Purpose and outcome

Problem. Learn artifact generation and catalog work is dirty/unreleased, while participant surfaces need governed recipes without making aesthetic purity a universal release gate.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Validated provenance-bearing Learn artifacts, approved-only serving, release-blocking DS accessibility/security/interaction contract, and app adoption of stable recipes/source cards.

Explicit non-goals.

- No invented final content
- No universal zero-hex rule
- No internal/operator DS merge
- No full legacy redesign

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-022	verified protected work	learn / dirty feature/learn-resource-rendering-contract@9c9a9c2	builder/validator/schema/templates/preview additions	A substantial artifact/rendering contract exists only as protected uncommitted work; no open Learn PR contains it.
E-023	verified protected work	intelligence / dirty main@9d824e8, behind origin/main	src/natalyx_intelligence/learn_resources.py; starting_natalyx_journey_v1.md	Catalog/resource edits are uncommitted and must not be overwritten or treated as released.
E-024	verified	design-system / main@43dab986; PR #7@cedfcc4; PR #8@4656426	components, tokens, Jinja attrs/parity guidance	Participant DS recipes and parity guidance exist; local brand assets are also dirty. Internal/operator DS is a separate repository and boundary.

3. Architecture and contracts

- Learn resources own authored artifacts/provenance; intelligence owns approved catalog delivery; app owns authorized presentation and context.
- Security, accessibility, responsive layout, and interaction defects can block release; token/style purity alone is backlog unless tied to concrete risk.
- Participant and internal/operator design systems remain separate.

Contract area	Package rule
Ownership boundary	`app`, `design-system`, `intelligence`, `learn`; repository-specific PRs remain separate.
Entities/state	No new runtime entity; evidence/coordination only.
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.

Contract area	Package rule
API/service	Versioned artifact manifest consumed by intelligence. Catalog response includes manifest/version/source/approval fields. Read-only catalog contract with fail-closed approval checks.
Storage	No new storage; preserve current state read-only for this package.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Design-system release rubric

Class	Items
Release-blocking	XSS/escaping or unsafe raw slots; keyboard/focus/screen-reader failures; unusable responsive layout; misleading or nonfunctional interaction; motion without reduced-motion safety; contrast or error-state defects that block task completion
Desirable	token/recipe consistency; visual parity; removal of duplicate spacing/color literals; component naming/adoption
Deferred	zero raw hex values; zero inline geometry styles; full legacy strangler cleanup; broad participant redesign; internal/operator DS convergence

Style purity blocks a release only when tied to a concrete security, accessibility, interaction or material maintainability risk.

4. Explicit PR sequence

1

Order	1
ID	P11-01
Title	Publish validated Learn artifact and provenance contract
Repository	learn
Dependencies	P00-01
Gate	A2
Size	Large
Readiness	Dependencies unmet

2

Order	2
ID	P11-02
Title	Serve approved Learn catalog versions from intelligence
Repository	intelligence
Dependencies	P11-01
Gate	A2
Size	Large
Readiness	Dependencies unmet

3

Order	3
ID	P11-03
Title	Release participant design-system accessibility and interaction contract
Repository	design-system
Dependencies	P00-01
Gate	A1
Size	Medium
Readiness	Dependencies unmet

4

Order	4
ID	P11-04
Title	Adopt approved Learn and source-card recipes in app
Repository	app
Dependencies	P11-02, P11-03
Gate	A2
Size	Large
Readiness	Dependencies unmet

P11-01 — Publish validated Learn artifact and provenance contract

Field	Plan
Repository / base	`learn` (`main`)
Project item	P11-01
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P00-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Protected feature/learn-resource-rendering-contract, feature/node0-beta-learn-resources

Objective

Turn the protected rendering-contract work into a small releaseable artifact schema/builder/validator with source provenance and no invented resources.

Scope

- Markdown/YAML source schema
- Article/PDF/context-packet artifacts
- Source/revision/audience/approval metadata
- Deterministic build

Explicit non-goals

- No final content beyond separately approved resources
- No app UI

Expected files or modules

- `schemas/`
- `src/natalyx_learn_resources/builder.py`
- `src/natalyx_learn_resources/validator.py`
- `tests/test_builder.py`
- `tests/test_validator.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Versioned artifact manifest consumed by intelligence.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Provenance and release procedure.

Tests and validation

Objective tests:

- Schema invalid cases
- deterministic output
- missing provenance
- unsafe links/HTML
- artifact checksum

Local validation:

- `uv run pytest`
- Run the builder/validator against approved fixture sources and compare deterministic checksums.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Release a versioned/pinned artifact or API contract first; consumers opt in explicitly and retain fail-closed/previous-version behavior during the compatibility window.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P00-01. Parallel opportunities: P11-03, P11-04

Objective acceptance criteria

- □ Turn the protected rendering-contract work into a small releaseable artifact schema/builder/validator with source provenance and no invented resources.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	content pipelines, schema validation, document generation
Higher-capability review	Risk-proportionate; not mandatory by programme default

P11-02 — Serve approved Learn catalog versions from intelligence

Field	Plan
Repository / base	`intelligence` (`main`)
Project item	P11-02
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P11-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Intelligence issue #14, protected dirty learn_resources.py

Objective

Load a pinned Learn artifact manifest, filter to approved audience/scope, and expose explicit empty/degraded behavior without hard-coded duplicate catalog truth.

Scope

- Manifest pin/checksum
- Approval/audience/cluster filters
- Source metadata
- Atomic catalog reload

Explicit non-goals

- No unapproved fallback catalog
- No app rendering

Expected files or modules

- `src/natalyx_intelligence/learn_resources.py`
- `src/natalyx_intelligence/catalog.py`
- `tests/pathfinder/test_corpus_loading.py`
- `docs/agents/learn-resources.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Catalog response includes manifest/version/source/approval fields.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Missing/bad manifest
- unapproved exclusion
- audience filter
- atomic reload
- stale version

Local validation:

- `uv run pytest`
- Run focused contract/migration/safety tests with redacted fixtures.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Release a versioned/pinned artifact or API contract first; consumers opt in explicitly and retain fail-closed/previous-version behavior during the compatibility window.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P11-01. Parallel opportunities: P11-03

Objective acceptance criteria

- □ Load a pinned Learn artifact manifest, filter to approved audience/scope, and expose explicit empty/degraded behavior without hard-coded duplicate catalog truth.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	content serving, FastAPI, provenance
Higher-capability review	Risk-proportionate; not mandatory by programme default

P11-03 — Release participant design-system accessibility and interaction contract

Field	Plan
Repository / base	`design-system` (`main`)
Project item	P11-03
Gate / size / readiness	A1 / Medium / Dependencies unmet
Dependencies	P00-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Design-system PRs #7 and #8, dirty local brand assets

Objective

Land the participant recipes/tokens/assets and parity guidance needed for secure escaping, accessible controls, responsive layouts, and stable interaction contracts.

Scope

- Jinja escaping/attrs contract
- 44px/keyboard/focus/reduced-motion requirements
- Responsive shell/recipe guidance
- Versioned vendoring

Explicit non-goals

- No universal raw-hex blocker
- No internal/operator DS
- No legacy app sweep

Expected files or modules

- `components/`
- `tokens/`
- `assets/`
- `docs/consumer-parity.md`
- `tests/`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	Participant design-system release only.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Template escaping
- raw-slot registry
- keyboard/focus
- responsive visual fixtures
- asset integrity

Local validation:

- `npm test`
- Run recipe escaping, accessibility and responsive fixture checks.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Release a versioned/pinned artifact or API contract first; consumers opt in explicitly and retain fail-closed/previous-version behavior during the compatibility window.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P00-01. Parallel opportunities: P11-01, P11-02

Objective acceptance criteria

- □ Land the participant recipes/tokens/assets and parity guidance needed for secure escaping, accessible controls, responsive layouts, and stable interaction contracts.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	design systems, accessibility, Jinja security
Higher-capability review	Risk-proportionate; not mandatory by programme default

P11-04 — Adopt approved Learn and source-card recipes in app

Field	Plan
Repository / base	`app` (`main`)
Project item	P11-04
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P11-02, P11-03
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PR #122 Learn slot, PR #124 dependency note

Objective

Consume pinned Learn/catalog and participant-DS versions for source cards, reading views, suggestions, and honest empty/error states.

Scope

- Version-pinned catalog client
- Source/citation cards and reading shell
- No-resource/degraded states
- Cluster/context handoff

Explicit non-goals

- No final content invention
- No case truth write

Expected files or modules

- `app/learn_resources.py`

- `app/templates/learn_workspace.html`
- `app/templates/components/participant/`
- `tests/python/test_learn_workspace.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Read-only catalog contract with fail-closed approval checks.
UI/product surface	Governed participant recipes.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Approved resource
- missing/unapproved resource
- service timeout
- source links
- mobile/keyboard

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P11-02, P11-03. Parallel opportunities: P11-01

Objective acceptance criteria

- □ Consume pinned Learn/catalog and participant-DS versions for source cards, reading views, suggestions, and honest empty/error states.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Risk-proportionate; not mandatory by programme default

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain
`design-system`	participant recipes/tokens/accessibility/interaction contract
`intelligence`	advisory inference, scoped response/citation contract; no case truth
`learn`	authored Learn artifacts, provenance and release manifest

Merge order inside this package: `P11-01` → `P11-03` → `P11-02` → `P11-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Artifact schema/provenance; Approved-only catalog; Missing-resource empty state; Recipe escaping; Keyboard/reduced-motion/mobile	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Type	ID / risk	Owner	Affected PRs / latest point	Project item / treatment
Implementation risk	Fake/unapproved content	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Cross-repo version drift	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Aesthetic review blocks recovery	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Internal DS leakage	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#112	9605d88e31bd	Source-first 81-node/12-cluster surface map, participant-renderer boundaries, wave plan and cross-repo seams.	P11-03, P11-04	Selective port, then supersede after extraction	Catalogued; content remains to be reconciled into replacement PR acceptance tests.
app#121	529d377e2337	Primary Node 0 participant model/UI, PDF/guide presentation and focused surface tests.	P11-03	Selective UI/test port, then supersede	Unextracted; #124 may contain later copies but equivalence must be proved by file/commit comparison.
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P11-03	Retain pending Alan reconciliation; selective port or rebase only after architecture approval	Reinspected at latest head. It is exactly e853821 (no newer delta); retention strengthened, but final extraction waits for Alan checkpoint.
app#124	84a5a847fea7	Current consolidation of Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitations, DS assets and large focused test suites.	P11-04	Retain unchanged as evidence; selective port/rewrite after checkpoint	Unextracted; Alan review explicitly incomplete and unresolved.
ds#7	cedfcc489b1f	Documents an escaping/attrs hardening follow-up relevant to participant recipe security.	P11-03	Retain for direct review/merge or fold into P11-03	Direct-reuse candidate; not yet merged.
ds#8	4656426e493e	Consumer parity and bounded app-owned layout guidance for Node 0/case/mobile adoption.	P11-03	Prefer direct review/merge if still correct	Direct-reuse candidate; local dirty asset delta separately protected.
feature/learn-resource-rendering-contract	9c9a9c2	substantial uncommitted builder/schema/templates/resource work	P11-01	Selective port from protected worktree	Do not update/switch/clean; create a future isolated extraction worktree.
feature/node0-beta-learn-resources	e1fb2d6	uncommitted Node 0 Learn implementation	P11-01 after provenance/no-fake-content review	Selective port from protected worktree	Preserve until compared with rendering-contract branch.
main (behind origin/main)	9d824e8	dirty Learn registry/corpus plus uv.lock	P11-02/P10-03 as appropriate	Selective port from protected worktree	Do not pull/reset/stash; extract in isolated worktree later.
feature/ds-consumer-parity-guidance	4656426	dirty brand assets beyond PR #8	P11-03 after asset review	Selective port from protected worktree	Remote PR and local asset delta are separate evidence; do not overwrite.

Package source inventory:

- Dirty Learn rendering-contract branch
- Learn issue #3
- Intelligence issue #14 and dirty catalog work
- DS PRs #7/#8

9. Ownership recommendation

Luke approves content/product and release-blocker rubric; Alan implements contracts/adoption. Accessibility and content/provenance review required.

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P11-01	Alan	Alan (proposed)	Luke	Luke	As risk warrants
P11-02	Alan	Alan (proposed)	Luke	Luke	As risk warrants
P11-03	Alan	Alan (proposed)	Luke	Luke	As risk warrants
P11-04	Alan	Alan (proposed)	Luke	Luke	As risk warrants

Luke decides product/lifecycle/commercial/privacy/PHI/provider policy and accepts gates. Alan proposes implementation design, decomposition, migrations, code, tests, rollout and repair. This is a recommendation pending Alan's review/sign-off, not a compulsory assignment.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 12 — Ask/Learn shell and mobile cutover

Planning-only execution contract. Proposed implementers and owners remain recommendations until Alan reviews/signs off the consolidated plan and Luke accepts it. Project: MVP Product Flow Reshape.

Plan	Value
Work package	WP12
Plan file	`12-ask-learn-shell-and-mobile-cutover.md`
Proposed PRs	P12-01, P12-02, P12-03
Inspection snapshot	2026-07-10 20:12 America/New_York

1. Purpose and outcome

Problem. Participant Ask/Learn entry points, case shell, legacy routes, and mobile interactions are not yet one coherent clickable journey with explicit context boundaries.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. A unified scope-labelled Ask/Learn shell, accessible mobile participant journey, and verified redirect-based legacy cutover with no broken or misleading controls.

Explicit non-goals.

- No final content authoring
- No staff/operator redesign
- No broad visual rebrand
- No removal before telemetry proves safe

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-006	verified	app / PR #122@e853821	PR comment Review round 2 addressed and changed workflow/tests	Updated PR #122 reports 499 tests plus return_to, TTL, CAS, self-invite, lens and Learn fail-closed fixes; green checks were observed. These are salvage candidates, not blanket merge approval.
E-012	verified	app / PR #122@e853821	app/storage/invitations.py and PR review-round comment	The updated case invitation work adds 14-day ceiling, CAS transitions, mismatch confirmation, self-invite refusal, return_to preservation and paired lens tests; it has no Node 0 workspace target.
E-024	verified	design-system / main@43dab986; PR #7@cedfcc4; PR #8@4656426	components, tokens, Jinja attrs/parity guidance	Participant DS recipes and parity guidance exist; local brand assets are also dirty. Internal/operator DS is a separate repository and boundary.

3. Architecture and contracts

- The shell labels general, private, and shared-case context and never implies a source exists when it does not.
- Server owns state and permissions; client code enhances interaction only.
- Legacy routes redirect through allowlisted, telemetry-observed compatibility paths before deletion.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	No new runtime entity; evidence/coordination only.
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Pass versioned scoped envelopes only. Stable redirects with query sanitization.
Storage	No new storage; preserve current state read-only for this package.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

4. Explicit PR sequence

1

Order	1
ID	P12-01
Title	Unify Ask and Learn shell with explicit context scopes
Repository	app
Dependencies	P10-03, P11-04, P09-04
Gate	A2
Size	Large
Readiness	Dependencies unmet

2

Order	2
ID	P12-02
Title	Complete mobile participant interaction and accessibility cutover
Repository	app
Dependencies	P05-03, P06-03, P09-04, P12-01
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet

3

Order	3
ID	P12-03
Title	Retire legacy participant routes behind verified redirects
Repository	app
Dependencies	P12-02, P13-04
Gate	B1
Size	Large
Readiness	Dependencies unmet

P12-01 — Unify Ask and Learn shell with explicit context scopes

Field	Plan
Repository / base	`app` (`main`)
Project item	P12-01
Gate / size / readiness	A2 / Large / Dependencies unmet
Dependencies	P10-03, P11-04, P09-04
Blocking decisions/checkpoints	D-AI-01
Source / salvage	Current Ask widget/Learn workspace, PR #124

Objective

Present general Learn, private personalization, and approved shared-case Ask as visibly distinct contexts with source cards and safe switching.

Scope

- Context scope badge/selector
- Suggested questions and citations
- Boundary and unavailable states
- Case-aware authorization

Explicit non-goals

- No context merge
- No case fact write

Expected files or modules

- `app/ask_natalyx.py`
- `app/templates/ask_natalyx*.html`
- `app/templates/learn_workspace.html`
- `app/static/js/learn_chat.js`
- `tests/python/test_ask_learn_shell.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Pass versioned scoped envelopes only.
UI/product surface	Unified but scope-explicit shell.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- General/private/shared views
- wrong-case denial
- logout/login
- empty/failure
- mobile/keyboard/screen-reader

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P10-03, P11-04, P09-04. Parallel opportunities: P12-03

Objective acceptance criteria

- □ Present general Learn, private personalization, and approved shared-case Ask as visibly distinct contexts with source cards and safe switching.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	privacy UX, FastAPI/Jinja, accessibility
Higher-capability review	Mandatory before merge

P12-02 — Complete mobile participant interaction and accessibility cutover

Field	Plan
Repository / base	`app` (`main`)
Project item	P12-02
Gate / size / readiness	A2 / Exceptional / Dependencies unmet
Dependencies	P05-03, P06-03, P09-04, P12-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PR #122 mobile journey, PR #124 Node 0, DS PR #8

Objective

Run a bounded participant-only mobile/accessibility pass across entry, auth, Node 0, GC claim, case, documents, Ask, and Learn using the released DS contract.

Scope

- Navigation/forms/cards/dialogs
- 320/375/768 widths
- Keyboard/focus/reduced motion
- Error/loading/empty states

Explicit non-goals

- No internal/operator surface
- No visual rebrand
- No zero-hex purity gate

Expected files or modules

- `app/templates/participant*/`
- `app/templates/node0/`
- `app/templates/case/`
- `app/static/participant*/`
- `tests/browser/participant_mobile.spec.*`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	Participant interaction defects and concrete DS adoption only.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- All required clickable journeys
- axe/manual keyboard
- reduced motion
- orientation/zoom
- slow dependency

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P05-03, P06-03, P09-04, P12-01. Parallel opportunities: None inside this package beyond review/test preparation.

Objective acceptance criteria

- □ Run a bounded participant-only mobile/accessibility pass across entry, auth, Node 0, GC claim, case, documents, Ask, and Learn using the released DS contract.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	responsive UI, accessibility, browser testing
Higher-capability review	Risk-proportionate; not mandatory by programme default

P12-03 — Retire legacy participant routes behind verified redirects

Field	Plan
Repository / base	`app` (`main`)
Project item	P12-03
Gate / size / readiness	B1 / Large / Dependencies unmet
Dependencies	P12-02, P13-04
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Current route inventory, PRs #112/#119 planning maps

Objective

Move legacy participant entry/case/Ask/Learn routes behind allowlisted redirects and telemetry, then remove only after a measured compatibility window.

Scope

- Route inventory and redirect map
- Bookmark/return_to compatibility
- Telemetry and rollback flag
- Later dead-code removal

Explicit non-goals

- No immediate destructive deletion
- No operator route change

Expected files or modules

- `app/routes/participant*.py`
- `app/main.py`
- `tests/python/route_inventory.json`
- `docs/runbooks/legacy-cutover.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Stable redirects with query sanitization.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Every legacy route
- query/open-redirect safety
- stale bookmarks
- rollback flag
- route parity

Local validation:

- `python -m unittest discover -s tests/python`

- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate B1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Legacy hit counters and unknown-route alerts.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P12-02, P13-04. Parallel opportunities: P12-01

Objective acceptance criteria

- □ Move legacy participant entry/case/Ask/Learn routes behind allowlisted redirects and telemetry, then remove only after a measured compatibility window.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Risk-proportionate; not mandatory by programme default

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P12-01` → `P12-02` → `P12-03`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Clickable IP/GC flows; Context labels; No-JS/reduced-motion/keyboard; Mobile widths; Redirect and stale bookmark; Logout/login/resume	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Type	ID / risk	Owner	Affected PRs / latest point	Project item / treatment
Blocking/product/compliance decision	D-AI-01 — Approve scopes, allowed sources, model policy, proposal reviewers, class-5 evidence rail and degraded-mode copy.	Luke	P10-01, P10-02, P10-03, P12-01; Before P10-01	D-AI-01 — Generic case members cannot create reviewed/provider truth; intelligence remains advisory and scope explicit.
Implementation risk	Context ambiguity	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Mobile-only dead ends	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Premature legacy removal	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Design drift	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#112	9605d88e31bd	Source-first 81-node/12-cluster surface map, participant-renderer boundaries, wave plan and cross-repo seams.	P12-03	Selective port, then supersede after extraction	Catalogued; content remains to be reconciled into replacement PR acceptance tests.
app#119	a8cd2f428535	Concise product-leverage framing and boundaries for the MVP flow.	P12-03	Retain as source or merge separately after plan decision	Catalogued; reconcile terminology into approved programme.
app#121	529d377e2337	Primary Node 0 participant model/UI, PDF/guide presentation and focused surface tests.	P12-02	Selective UI/test port, then supersede	Unextracted; #124 may contain later copies but equivalence must be proved by file/commit comparison.
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P12-02	Retain pending Alan reconciliation; selective port or rebase only after architecture approval	Reinspected at latest head. It is exactly e853821 (no newer delta); retention strengthened, but final extraction waits for Alan checkpoint.
app#124	84a5a847fea7	Current consolidation of Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitations, DS assets and large focused test suites.	P12-02	Retain unchanged as evidence; selective port/rewrite after checkpoint	Unextracted; Alan review explicitly incomplete and unresolved.
ds#8	4656426e493e	Consumer parity and bounded app-owned layout guidance for Node 0/case/mobile adoption.	P12-02	Prefer direct review/merge if still correct	Direct-reuse candidate; local dirty asset delta separately protected.

Package source inventory:

- PR #122 responsive journey and lens behavior
- PR #124 Node 0 surfaces
- DS PR #8 parity guidance

9. Ownership recommendation

Luke owns journey acceptance; Alan implements and verifies. Accessibility review recommended.

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P12-01	Alan	Alan (proposed)	Luke	Luke	Mandatory
P12-02	Alan	Alan (proposed)	Luke	Luke	As risk warrants
P12-03	Alan	Alan (proposed)	Luke	Luke	As risk warrants

Luke decides product/lifecycle/commercial/privacy/PHI/provider policy and accepts gates. Alan proposes implementation design, decomposition, migrations, code, tests, rollout and repair. This is a recommendation pending Alan's review/sign-off, not a compulsory assignment.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.

- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 13 — Deployment, cloud gates, and PHI readiness

Planning-only execution contract. Proposed implementers and owners remain recommendations until Alan reviews/signs off the consolidated plan and Luke accepts it. Project: MVP Product Flow Reshape.

Plan	Value
Work package	WP13
Plan file	`13-deployment-cloud-gates-and-phi-readiness.md`
Proposed PRs	P13-01, P13-02, P13-03, P13-04, P13-05, P13-06, P13-07
Inspection snapshot	2026-07-10 20:12 America/New_York

1. Purpose and outcome

Problem. The dev endpoint is presently 503, deployments do not prove application readiness, and test/deploy success cannot establish real-user or PHI readiness.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Known-revision diagnostics, guarded rollback/evidence capture, automated synthetic A1/A2 journeys, and explicit B1/B2 operational/privacy/PHI go/no-go packets.

Explicit non-goals.

- No deployment in this planning task
- No real user before B1
- No PHI before B2
- No compliance claim from green CI

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-001	verified	live dev / 2026-07-10 20:12 America/New_York	https://dev.app.natalyx.health/ and /health	Both requests returned HTTP 503 from awselb/2.0; the deployed participant app is not presently usable.
E-002	verified limitation	AWS dev / 2026-07-10 20:12 America/New_York	aws sts get-caller-identity --profile natalyx-dev	Local AWS SSO was expired, so current ECS desired/running counts and task revision were not independently verified. Gate A0 must refresh access and capture them.
E-007	verified	app / feature/byos-node0-consolidation@84a5a84	app/storage/schema.py:1-33,353-880	Schema ownership is a monolithic init_db/ensure_column routine with inline CREATE TABLE/ALTER TABLE and no ordered migration/version/checksum ledger.
E-008	verified	app / origin/main@fdf2bb4	app/main.py:295-313	Cloud startup validates auth/storage then calls storage.init_db in-process; readiness is not separated from migration execution.
E-026	verified	app / origin/main@fdf2bb4	app/main.py:145-148	/healthz returns only status ok when the process can serve; it does not report revision, database, migration, intelligence, storage, or readiness.

ID	Class	Repository / ref	File, construct or location	Finding
E-027	verified	app/infra / origin/main@fdf2bb4 / infra main@fec515f	app .github/workflows/aws-app-deploy.yml:60-190; infra CDK workflow	Deployment reads infra-owned SSM contracts and guards billable ECS updates, but successful image build/contract smoke does not prove a healthy service revision.

3. Architecture and contracts

- Health, readiness, revision, dependency, migration, and safe configuration signals are distinct.
- Every cloud gate has prerequisites, evidence, owner, verifier, and failure/rollback response.
- B2 requires formal security/privacy/compliance evidence and human go/no-go; it is not a software test result.

Contract area	Package rule
Ownership boundary	`.github`, `app`, `infra`; repository-specific PRs remain separate.
Entities/state	No new runtime entity; evidence/coordination only.
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Health/readiness endpoints with stable machine-readable reasons.
Storage	RLS policies and least-privilege grants under a reviewed migration.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Programme gate summary

Gate A0 — Infrastructure recovery and known-revision health

Field	Contract
Prerequisite PRs/gates	P01-01, P02-02, P13-01, P13-02
Prerequisite decisions	D-MIG-01, D-OPS-01
Responsible owner	Alan
Verifier	Luke

Acceptance matrix:

- □ CloudFormation/ECS/ALB state is captured from refreshed dev credentials and is stable.
- □ The running app and intelligence image SHAs/task definitions are known and map to reviewed commits.
- □ Liveness and readiness are both healthy; PostgreSQL, migrations and required private intelligence dependencies pass their bounded checks.
- □ A synthetic unauthenticated health request and authenticated minimal route succeed without PII/PHI.
- □ Rollback or forward-repair path is rehearsed and the prior compatible revision is identified.

Required evidence:

- AWS account/region and stack/service snapshots
- ALB status and target health
- image SHA/task definition
- readiness JSON/log correlation

- CI/PostgreSQL artifacts
- rollback rehearsal

Failure and rollback response. Do not proceed to participant validation. Freeze live updates, retain evidence, restore the known compatible image/schema or forward-repair under the approved runbook, and keep the Project gate blocked.

Gate A1 — Synthetic core participant journey

Field	Contract
Prerequisite PRs/gates	P02-04, P03-01, P03-02, P03-04, P04-01, P04-02, P04-03, P05-01, P05-02, P05-03, P11-03, P13-03
Prerequisite decisions	D-AUTH-01, D-AUTH-02, D-DATA-01, D-DATA-02, D-DATA-03
Responsible owner	Alan
Verifier	Luke

Acceptance matrix:

- □ A new synthetic IP signs up, is deterministically routed, enters Node 0 Free and no paid case is created.
- □ Candidate/outreach data remains IP-attributed and provisional; no score or answer becomes GC/provider truth.
- □ A synthetic GC begins a low-risk claim, binds identity/consent and sees only the allowed workspace projection.
- □ Cross-role negative tests prove class-2 IP assertions and class-4 GC-private answers do not leak.
- □ Both roles log out/in/resume and complete the mobile journey at tested widths.
- □ Known revision, audit events, correlation IDs and sanitized evidence are attached to the gate item.

Required evidence:

- A1 synthetic run artifact
- separate IP/GC subject IDs (synthetic only)
- authorization matrix results
- mobile screenshots/accessibility output
- audit event trace
- database invariant queries

Failure and rollback response. Stop before Complete, commercial or paid-case validation. Revert/disable the failing participant feature flag or deploy the known A0 revision; preserve synthetic data only for defect diagnosis, then purge under the runbook.

Gate A2 — Synthetic commercial, promotion, billing and document journey

Field	Contract
Prerequisite PRs/gates	P03-03, P06-01, P06-02, P06-03, P07-01, P07-02, P07-03, P07-04, P08-01, P08-02, P08-03, P08-04, P09-01, P09-02, P09-03, P09-04, P09-05, P10-01, P10-02, P10-03, P10-04, P11-01, P11-02, P11-04, P12-01, P12-02, P13-04
Prerequisite decisions	D-COM-01, D-COM-02, D-COM-03, D-COM-04, D-PAY-01, D-COM-05, D-COM-06, D-COM-07, D-COM-08, D-DOC-01, D-DOC-02, D-PROV-01, D-AI-01
Responsible owner	Alan
Verifier	Luke

Acceptance matrix:

- □ Synthetic Complete entitlement does not itself create a case and records its commercial configuration version/credit.
- □ Only byos.workspace_promotion_authorized.v1 authorizes exactly one enrollment_pending case; callbacks/redirects/replays create none/duplicates.
- □ Recipient-bound paid-case invitation and consent grant correct membership; activation waits for approved evidence.
- □ Enrollment/activation/milestone/credit/refund ledgers reconcile, including duplicate and out-of-order provider events.
- □ Synthetic files follow upload/quarantine/scan/available/download/replacement/revocation/recovery and never become public.

- □ Provider packet contains only purpose-allowed consented sources; intelligence scopes and proposals do not upgrade truth.

Required evidence:

- A2 synthetic run artifact
- processor simulator event ledger
- promotion idempotency query
- invoice/refund explanations
- S3/KMS/IAM/quarantine assertions
- provider snapshot checksum
- Ask/proposal safety evals

Failure and rollback response. Disable commercial/document/provider features, retain free/safety content, prevent live money or real documents, reconcile synthetic ledger inconsistencies, and restore the A1-known revision or forward-repair as approved.

Gate B1 — Real-user and PII beta readiness

Field	Contract
Prerequisite PRs/gates	A2, P12-03, P13-07
Prerequisite decisions	D-PHI-01, D-PHI-02
Responsible owner	Luke
Verifier	External/TBD

Acceptance matrix:

- □ A written B1 data-classification boundary permits named PII and expressly prohibits PHI/clinical documents.
- □ Named beta users, support/on-call, access review, incident response, backup/restore, retention/deletion and complaint routes are ready.
- □ Security/privacy review has no unresolved critical finding; vendors handling PII are approved.
- □ A2 evidence is current for the exact candidate revision and rollback/feature-disable controls are operational.
- □ Consent/privacy/terms copy is approved and data-subject correction/deletion workflows are testable.

Required evidence:

- Signed B1 checklist
- data inventory/classification
- access review
- incident tabletop
- backup/restore result
- vendor register
- current A2 bundle

Failure and rollback response. No real-user invitation. Continue synthetic-only, close control gaps, invalidate stale evidence and re-run the exact affected reviews/tests before a new go/no-go.

Gate B2 — PHI readiness and formal go/no-go

Field	Contract
Prerequisite PRs/gates	B1, P13-05, P13-06, P13-07
Prerequisite decisions	D-PHI-01, D-PHI-02
Responsible owner	Luke
Verifier	External security/privacy/compliance reviewers

Acceptance matrix:

- □ Formal PHI data flows, risk assessment, policies, BAAs/vendor controls, least privilege, audit/log retention, encryption/key management, incident/breach response and backup/restore are approved.
- □ Application, infrastructure, document, provider and intelligence threat models have current evidence and no unresolved critical/high risk without accepted exception.
- □ Operational staff training, access provisioning/deprovisioning, monitoring, support and emergency stop are complete.
- □ The exact release passed A2 and B1 revalidation; a signed human go/no-go records scope, date, evidence versions and approvers.
- □ The gate states explicitly that testing/deployment alone did not establish HIPAA/PHI readiness.

Required evidence:

- Signed go/no-go
- risk/security/privacy assessments
- BAA/vendor evidence
- access/log/incident/backup controls
- penetration/security testing
- training/operations evidence
- current technical gate bundles

Failure and rollback response. PHI remains prohibited. Keep B1 PII scope only if still independently valid; disable any PHI-capable upload/context path, remediate, and repeat formal review.

4. Explicit PR sequence

1

Order	1
ID	P13-01
Title	Expose known-revision health, readiness, and dependency diagnostics
Repository	app
Dependencies	P01-01
Gate	A0
Size	Large
Readiness	Dependencies unmet

2

Order	2
ID	P13-02
Title	Harden guarded deployment, rollback, and evidence capture
Repository	infra
Dependencies	P01-01, P13-01
Gate	A0
Size	Large
Readiness	Dependencies unmet

3

Order	3
ID	P13-03
Title	Automate synthetic A1 core participant journey verification
Repository	app
Dependencies	P03-02, P04-03, P05-03, P11-03, P13-02
Gate	A1
Size	Large
Readiness	Dependencies unmet

4

Order	4
ID	P13-04
Title	Automate synthetic A2 commercial and document journey verification
Repository	app
Dependencies	P06-03, P07-04, P08-04, P09-05, P10-04, P12-02, P13-03
Gate	A2
Size	Exceptional
Readiness	Dependencies unmet

5

Order	5
ID	P13-05
Title	Enforce case RLS and a non-owner PostgreSQL runtime
Repository	app
Dependencies	P02-03, P03-03, P04-03, P09-05, P10-02
Gate	B2
Size	Exceptional
Readiness	Dependencies unmet

6

Order	6
ID	P13-06
Title	Harden B1/B2 backups, audit, alarms, and cohort controls
Repository	infra
Dependencies	P09-02, P13-02, P13-05
Gate	B2
Size	Exceptional
Readiness	Dependencies unmet

7

Order	7
ID	P13-07
Title	Publish B1 and B2 privacy, PHI, and operational go-no-go runbooks
Repository	.github
Dependencies	P13-04, P13-05, P13-06
Gate	B2
Size	Large
Readiness	Blocked by decision

P13-01 — Expose known-revision health, readiness, and dependency diagnostics

Field	Plan
Repository / base	`app` (`main`)
Project item	P13-01
Gate / size / readiness	A0 / Large / Dependencies unmet
Dependencies	P01-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Current /healthz, direct 503 evidence

Objective

Separate liveness, readiness, build revision, migration, database, intelligence, storage, and safe-config status so Gate A0 can identify the actual failing dependency.

Scope

- Unauthenticated minimal liveness
- Safe readiness/dependency checks
- Revision/config identifiers without secrets
- Startup failure classification

Explicit non-goals

- No PHI/PII in diagnostics
- No dependency mutation

Expected files or modules

- `app/health.py`
- `app/main.py`
- `app/dev_diagnostics.py`
- `tests/python/test_health.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Health/readiness endpoints with stable machine-readable reasons.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- DB/migration/intelligence/storage failures
- timeout
- redaction
- known SHA
- ALB contract

Local validation:

- `python -m unittest discover -s tests/python`

- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A0 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Structured startup/readiness logs.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P01-01. Parallel opportunities: P13-03, P13-04, P13-05, P13-06, P13-07

Objective acceptance criteria

- □ Separate liveness, readiness, build revision, migration, database, intelligence, storage, and safe-config status so Gate A0 can identify the actual failing dependency.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	runtime operations, FastAPI, observability
Higher-capability review	Mandatory before merge

P13-02 — Harden guarded deployment, rollback, and evidence capture

Field	Plan
Repository / base	`infra` (`main`)
Project item	P13-02
Gate / size / readiness	A0 / Large / Dependencies unmet
Dependencies	P01-01, P13-01
Blocking decisions/checkpoints	D-OPS-01
Source / salvage	Existing guarded app/infra workflows

Objective

Make app/intelligence deployment record the image SHA/task definition, wait for health, capture non-secret failure evidence, and roll back or stop safely under explicit guards.

Scope

- Known-revision deploy and service stabilization
- Least-privilege one-off migration task identity
- Migration/app compatibility preflight

- Rollback/forward-repair decision
- Evidence artifact

Explicit non-goals

- No automatic billable activation
- No planning-task deployment

Expected files or modules

- `.github/workflows/cdk.yml`
- `lib/stacks/platform-stack.ts`
- `lib/constructs/migration-task.ts`
- `scripts/verify-service-health.*`
- `docs/runbooks/deploy-rollback.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	Guarded workflow/alarms/contracts; default desired counts remain safe.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- CDK/workflow contract
- unhealthy revision
- rollback compatibility
- billable flag false
- missing SSM output

Local validation:

- `npm test`
- `npm run build`
- `Synthesize the affected CDK stack with billable/live changes disabled.`

Cloud validation: run only through Gate A0 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Synthesize and review first; deploy only through the guarded dev workflow with billable/desired-count defaults unchanged, exact revision recorded, and explicit confirmation outside this planning task.

Rollback or forward repair

Return to the prior reviewed stack/task definition when compatible; otherwise keep resources fail-closed and forward-repair under the runbook. Never delete data-bearing resources as an automatic rollback.

Dependencies and parallel work

Hard dependencies: P01-01, P13-01. Parallel opportunities: P13-04, P13-05, P13-07

Objective acceptance criteria

- □ Make app/intelligence deployment record the image SHA/task definition, wait for health, capture non-secret failure evidence, and roll back or stop safely under explicit guards.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	AWS ECS/ALB, GitHub Actions, rollback operations
Higher-capability review	Mandatory before merge

P13-03 — Automate synthetic A1 core participant journey verification

Field	Plan
Repository / base	`app` (`main`)
Project item	P13-03
Gate / size / readiness	A1 / Large / Dependencies unmet
Dependencies	P03-02, P04-03, P05-03, P11-03, P13-02
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Existing synthetic sandbox/scenario utilities

Objective

Run the entry, auth, Node 0 Free, low-risk GC claim, role separation, resume, and mobile journey against a known dev revision and publish a redacted A1 evidence bundle.

Scope

- A1 core IP/GC workspace journey
- Known SHA and evidence bundle
- Role/authorization negative cases
- Cleanup of synthetic records

Explicit non-goals

- No payment/promotion/document journey
- No real user
- No PHI
- No gate auto-approval

Expected files or modules

- `tests/cloud/a1/`
- `scripts/run_synthetic_a1.py`
- `.github/workflows/synthetic-cloud-gates.yml`

- `docs/runbooks/synthetic-gates.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	Manual/guarded A1 synthetic runner with immutable artifact summary.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Happy path
- authorization denial
- retry/duplicate
- logout/login
- mobile
- dependency outage
- cleanup failure

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P03-02, P04-03, P05-03, P11-03, P13-02. Parallel opportunities: P13-01, P13-05, P13-06, P13-07

Objective acceptance criteria

- □ Run the entry, auth, Node 0 Free, low-risk GC claim, role separation, resume, and mobile journey against a known dev revision and publish a redacted A1 evidence bundle.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	browser automation, AWS dev operations, end-to-end testing
Higher-capability review	Mandatory before merge

P13-04 — Automate synthetic A2 commercial and document journey verification

Field	Plan
Repository / base	`app` (`main`)
Project item	P13-04
Gate / size / readiness	A2 / Exceptional / Dependencies unmet
Dependencies	P06-03, P07-04, P08-04, P09-05, P10-04, P12-02, P13-03
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Payment document smoke paths, A1 harness

Objective

Extend the known-revision cloud harness with commitment, funding, exactly-once promotion, case invite, activation, billing, refund, document, handoff, Ask/Learn, and recovery evidence.

Scope

- A2 commercial/promotion/billing/document journey
- Duplicate/retry/refund/failure cases
- Known SHA and evidence bundle
- Synthetic cleanup

Explicit non-goals

- No real money
- No real user
- No PHI
- No gate auto-approval

Expected files or modules

- `tests/cloud/a2/`
- `scripts/run_synthetic_a2.py`
- `.github/workflows/synthetic-cloud-gates.yml`
- `docs/runbooks/synthetic-gates.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	Guarded A2 synthetic runner; uses processor simulator and synthetic documents only.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Funding event exactly once
- promotion replay
- case invite
- milestone/refund
- upload/scan failure
- Ask scope
- rollback evidence

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P06-03, P07-04, P08-04, P09-05, P10-04, P12-02, P13-03. Parallel opportunities: P13-01, P13-02, P13-05, P13-06

Objective acceptance criteria

- □ Extend the known-revision cloud harness with commitment, funding, exactly-once promotion, case invite, activation, billing, refund, document, handoff, Ask/Learn, and recovery evidence.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	browser automation, payments testing, AWS dev operations, end-to-end testing
Higher-capability review	Mandatory before merge

P13-05 — Enforce case RLS and a non-owner PostgreSQL runtime

Field	Plan
Repository / base	`app` (`main`)
Project item	P13-05
Gate / size / readiness	B2 / Exceptional / Dependencies unmet
Dependencies	P02-03, P03-03, P04-03, P09-05, P10-02
Blocking decisions/checkpoints	D-PHI-01
Source / salvage	case-shell.md current app-layer-only/RLS gap

Objective

Move app runtime to a least-privilege non-DDL database role and enforce tenant/case membership policies with RLS/FORCE RLS before any PHI gate can open.

Scope

- Runtime/migrator role separation
- RLS/FORCE RLS on case/participant/document/commercial tables
- Transaction-local subject/org/case context
- Policy-negative and pool-reset tests

Explicit non-goals

- No PHI gate activation
- No trust in app guards alone
- No runtime DDL

Expected files or modules

- `migrations/versions/*_case_rls.py`
- `app/storage/connection.py`
- `app/storage/session_context.py`
- `tests/security/test_case_rls.py`
- `docs/security/rls.md`

Change map

Area	Planned change
Schema/model	RLS policies and least-privilege grants under a reviewed migration.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Apply through the named versioned migration; no ad-hoc startup DDL. Backfill is none unless explicitly listed in scope.
Security/privacy	Database denies cross-case/ambient reads even if an app guard is missed.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Cross-case/role denial
- connection-pool context reset
- migrator versus runtime grants
- background worker scope
- previous app compatibility during rollout

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate B2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P02-03, P03-03, P04-03, P09-05, P10-02. Parallel opportunities: P13-01, P13-02, P13-03, P13-04

Objective acceptance criteria

- □ Move app runtime to a least-privilege non-DDL database role and enforce tenant/case membership policies with RLS/FORCE RLS before any PHI gate can open.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	PostgreSQL RLS, connection pooling, security architecture
Higher-capability review	Mandatory before merge

P13-06 — Harden B1/B2 backups, audit, alarms, and cohort controls

Field	Plan
Repository / base	`infra` (`main`)
Project item	P13-06
Gate / size / readiness	B2 / Exceptional / Dependencies unmet
Dependencies	P09-02, P13-02, P13-05
Blocking decisions/checkpoints	D-PHI-01, D-PHI-02
Source / salvage	Existing audit/backup/platform stack constructs

Objective

Add the infrastructure evidence and controls required for a named PII beta and later PHI review: backup/restore, data-event audit, least privilege, alarms, log retention, cohort disable, and emergency stop.

Scope

- RDS/S3 backup and restore evidence hooks

- CloudTrail S3 data events and access alarms
- Runtime/migrator/task-role least privilege
- Cohort disable and PHI gate kill switch
- Log retention/redaction contracts

Explicit non-goals

- No production/PHI activation
- No evidence claim without restore/tabletop

Expected files or modules

- `lib/stacks/platform-stack.ts`
- `lib/constructs/audit.ts`
- `lib/constructs/backup.ts`
- `test/security-controls.test.ts`
- `docs/runbooks/security-operations.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	Guarded, cost-aware controls with explicit dev defaults and documented operational owners.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- CDK policy assertions
- restore drill
- alarm path
- access-log redaction
- cohort/PHI gate disable
- role boundary

Local validation:

- `npm test`
- `npm run build`
- `Synthesize the affected CDK stack with billable/live changes disabled.`

Cloud validation: run only through Gate B2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Synthesize and review first; deploy only through the guarded dev workflow with billable/desired-count defaults unchanged, exact revision recorded, and explicit confirmation outside this planning task.

Rollback or forward repair

Return to the prior reviewed stack/task definition when compatible; otherwise keep resources fail-closed and forward-repair under the runbook. Never delete data-bearing resources as an automatic rollback.

Dependencies and parallel work

Hard dependencies: P09-02, P13-02, P13-05. Parallel opportunities: P13-01, P13-03, P13-04

Objective acceptance criteria

- □ Add the infrastructure evidence and controls required for a named PII beta and later PHI review: backup/restore, data-event audit, least privilege, alarms, log retention, cohort disable, and emergency stop.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Alan
Implementer	Alan (proposed; pending consolidated-plan sign-off)
Reviewer	Luke
Final approver	Luke
Required expertise	AWS security, backup/restore, CloudTrail/CloudWatch, IAM
Higher-capability review	Mandatory before merge

P13-07 — Publish B1 and B2 privacy, PHI, and operational go-no-go runbooks

Field	Plan
Repository / base	`.github` (`main`)
Project item	P13-07
Gate / size / readiness	B2 / Large / Blocked by decision
Dependencies	P13-04, P13-05, P13-06
Blocking decisions/checkpoints	D-PHI-01, D-PHI-02
Source / salvage	Natalyx compliance/company records by reference, all gate evidence

Objective

Create the programme evidence index and explicit human approval packets for PII beta (B1) and PHI readiness (B2), linking rather than duplicating controlled company/compliance records.

Scope

- Data classification and real-user boundary
- Access/logging/incident/backup/vendor/compliance evidence
- Named approvers and expiry/revalidation
- Rollback/stop response

Explicit non-goals

- No legal/compliance self-certification
- No PHI activation by merge

Expected files or modules

- `natalyx-repair-program/gates/B1-real-user-pii.md`
- `natalyx-repair-program/gates/B2-phi-go-no-go.md`
- `natalyx-repair-program/evidence-index.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Controlled go/no-go checklists and evidence links only.

Tests and validation

Objective tests:

- Evidence completeness review
- tabletop incident
- access review
- backup/restore evidence
- external security/privacy sign-off

Local validation:

- `git diff --check`
- `/Users/lukerhodes/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 scripts/validate_programme.py`

Cloud validation: run only through Gate B2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Merge only after document/sign-off checks; it changes no runtime. Re-sync the Project from the same canonical data.

Rollback or forward repair

Correct the canonical data and regenerate; do not rewrite implementation history.

Dependencies and parallel work

Hard dependencies: P13-04, P13-05, P13-06. Parallel opportunities: P13-01, P13-02, P13-03

Objective acceptance criteria

- □ Create the programme evidence index and explicit human approval packets for PII beta (B1) and PHI readiness (B2), linking rather than duplicating controlled company/compliance records.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The Project item has matching dependency, readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
DRI	Luke
Implementer	Joint (proposed; pending consolidated-plan sign-off)
Reviewer	External/TBD
Final approver	Luke
Required expertise	privacy/security governance, operations, compliance
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`.github`	programme/governance evidence and Project control plane
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain
`infra`	AWS resources, IAM/KMS/network/deployment/backup/audit contracts

Merge order inside this package: `P13-01` → `P13-02` → `P13-03` → `P13-05` → `P13-06` → `P13-04` → `P13-07`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Known revision; ALB/ECS/dependency health; Rollback; Synthetic A1/A2; PII controls; PHI go/no-go evidence	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Type	ID / risk	Owner	Affected PRs / latest point	Project item / treatment
Blocking/product/compliance decision	D-OPS-01 — Select the revision/schema pair to recover dev, rollback/forward-repair criteria and evidence owner.	Luke	P13-02; Before Gate A0 recovery action	D-OPS-01 — Use a tested immutable image and compatible schema; do not deploy from dirty or draft branches.
Blocking/product/compliance decision	D-PHI-01 — Approve data classification, vendors/BAAs, encryption/logging/access/incident/backup controls and prohibited data at each gate.	Luke	P09-01, P09-02, P09-05, P13-05, P13-06, P13-07; Before B1 for PII scope; before B2 for PHI	D-PHI-01 — B1 remains PII-only; B2 requires formal external security/privacy/compliance evidence.
Blocking/product/compliance decision	D-PHI-02 — Name security/privacy/compliance approvers, evidence recency, exception authority and revalidation cadence.	Luke	P13-06, P13-07; Before B1	D-PHI-02 — Luke is product go/no-go; independent qualified reviewers own their control attestations.
Implementation risk	Unknown running revision	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Rollback incompatible with schema	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	Synthetic success overclaimed	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan
Implementation risk	PHI enabled without organizational controls	Alan or named PR DRI	Package PRs; before associated gate	Mitigate through the explicit tests, rollout and forward-repair plan

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
No direct source	—	Build from approved contracts/evidence	Package PRs	New implementation	N/A

Package source inventory:

- Existing guarded AWS workflows and SSM contracts
- Current healthz
- Prior app/infra/intelligence AWS migration PR evidence

9. Ownership recommendation

Alan owns technical recovery/validation; Luke accepts gates and PHI boundary. External security, privacy, compliance, finance/payment, and clinical/legal review as applicable.

PR	DRI	Implementer	Reviewer	Final approver	External/high-capability review
P13-01	Alan	Alan (proposed)	Luke	Luke	Mandatory
P13-02	Alan	Alan (proposed)	Luke	Luke	Mandatory
P13-03	Alan	Alan (proposed)	Luke	Luke	Mandatory
P13-04	Alan	Alan (proposed)	Luke	Luke	Mandatory
P13-05	Alan	Alan (proposed)	Luke	Luke	Mandatory
P13-06	Alan	Alan (proposed)	Luke	Luke	Mandatory
P13-07	Luke	Joint (proposed)	External/TBD	Luke	Mandatory

Luke decides product/lifecycle/commercial/privacy/PHI/provider policy and accepts gates. Alan proposes implementation design, decomposition, migrations, code, tests, rollout and repair. This is a recommendation pending Alan's review/sign-off, not a compulsory assignment.

10. Definition of done

- ☐ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- ☐ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.

- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.